



Quality Report

Request 50004-000001

Report for Pipeline ILL1XG1G4_CNV_NextSeq_7 created: May 5, 2026

Gene panel HCS_v2_hg38: 84 genes | 1,350 target regions | 284,897 bp

Number of samples: 8

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Pipeline ILL1XG1G4_CNV_NextSeq_7

1.1 Dataset

1.1.1 Input files size overview

Table 1.1: This pipeline randomly sub-samples input files in case the maximally supported input file size is exceeded. The following table gives an overview of observed file sizes in MiB before and after the sub-sampling step. Minor variations in file size between input and retained files are expected due to differences in compression formats. All subsequent statistics are calculated based on the retained files.

Sample ID	Input1	Input2	Retained1	Retained2	Maximal Size
200042384	117	116	117	116	400
200042385	117	121	117	121	400
200042386	116	118	116	118	400
200042387	115	119	115	119	400
200042388	116	118	116	118	400
200042389	114	115	114	115	400
200042390	116	118	116	118	400
200042391	115	116	115	116	400

1.1.2 Read count overview

A total of 8 samples were sequenced, resulting in a total of 20,240,000 reads. Due to space limitations most figures are restricted to show only a subset of samples, however, a separate quality report is available for each sample. Information provided in tables will include all samples.

Table 1.2: Overview of datasets.

Sample ID	Number of reads	Number of Mapped reads	Number of reads with problematic mate	Percentage of Mapped reads
200042384-0-S22	2,530,000	2,473,552	41,262	97.77%
200042385-0-S23	2,530,000	2,516,120	26,700	99.45%
200042386-0-S11	2,530,000	2,515,478	27,704	99.43%
200042387-0-S10	2,530,000	2,509,702	29,174	99.20%
200042388-0-S5	2,530,000	2,513,845	30,577	99.36%
200042389-0-S1	2,530,000	2,511,066	29,952	99.25%
200042390-0-S2	2,530,000	2,509,406	32,318	99.19%
200042391-0-S3	2,530,000	2,511,050	30,442	99.25%
Total	20,240,000	20,060,219	248,129	99.11%

1.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 1.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 1.2 shows the distribution of average base call quality, while Figure 1.3 shows the distribution of read lengths.

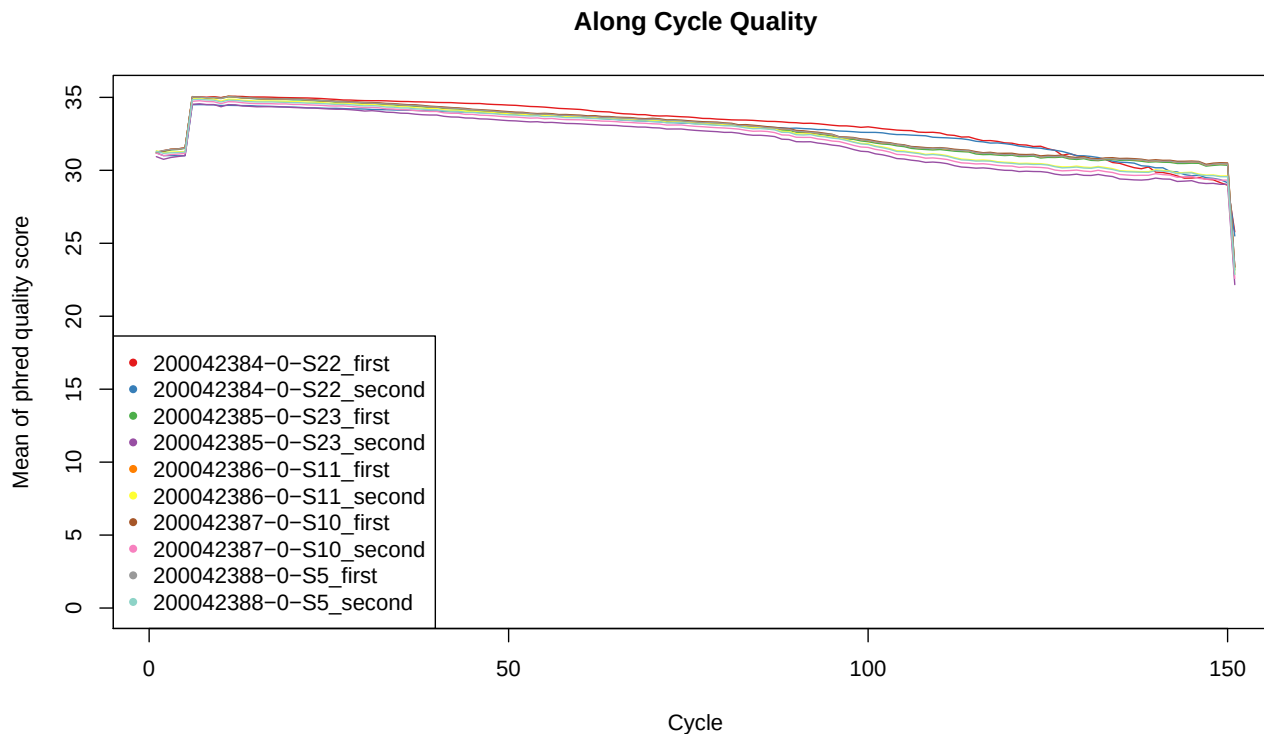


Figure 1.1: Mean base call quality by cycle.

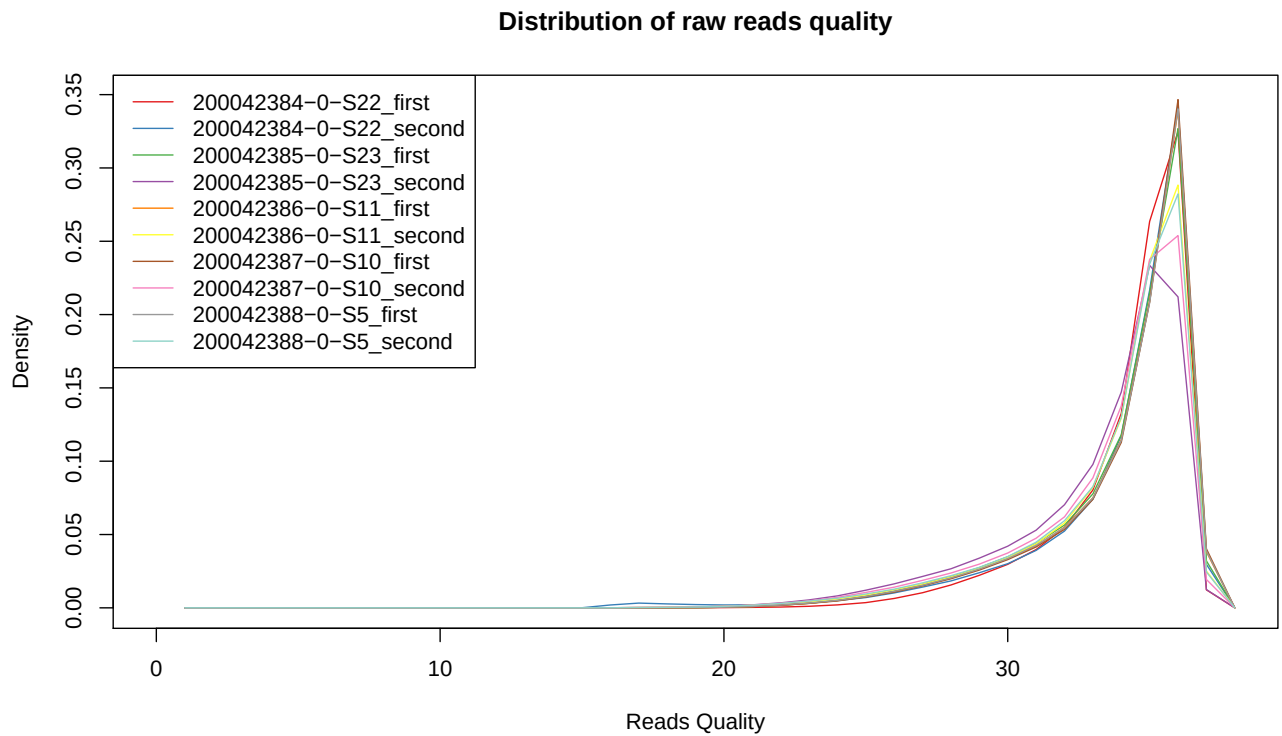


Figure 1.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

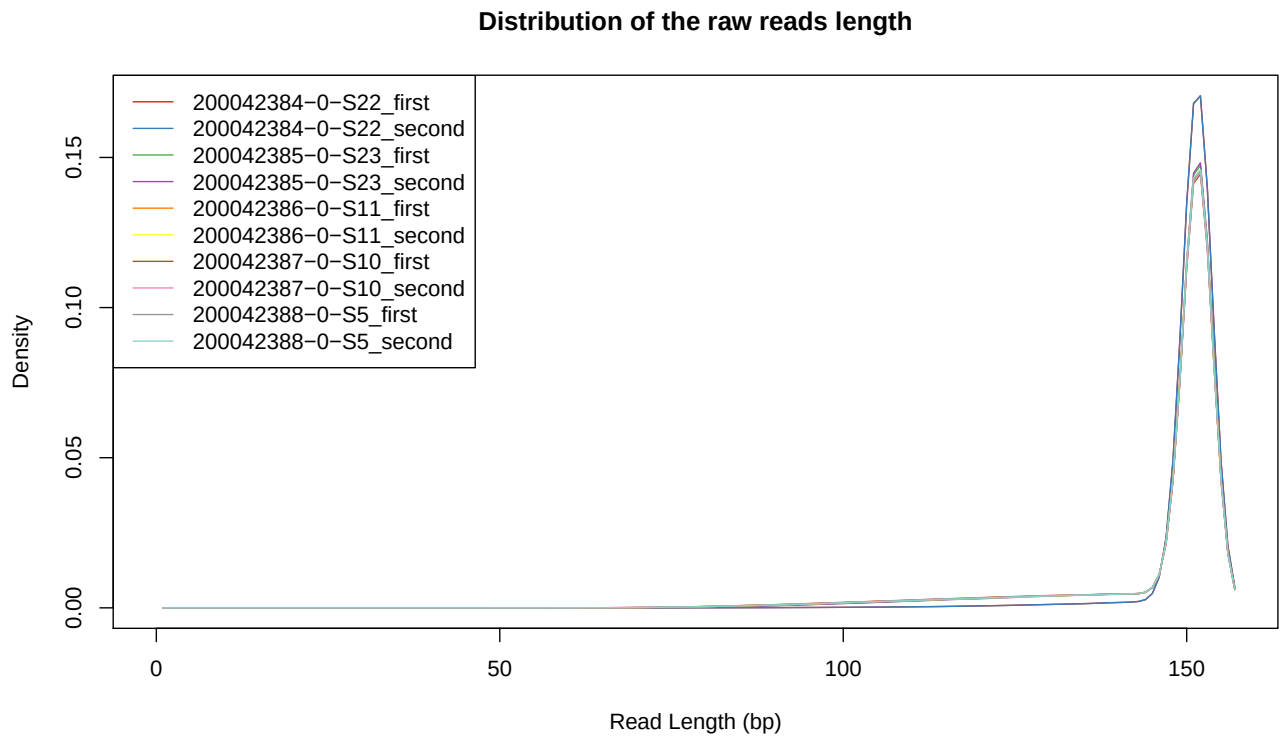
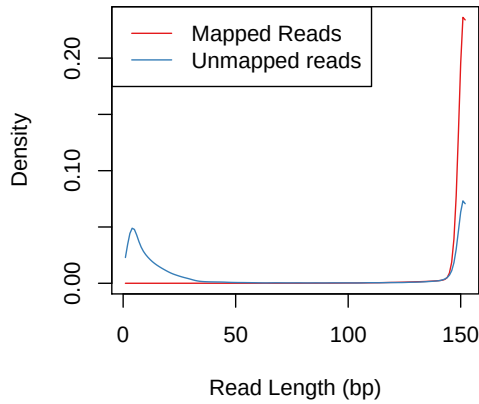


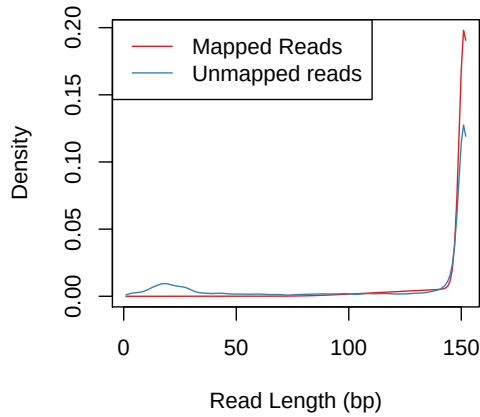
Figure 1.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

1.3 Alignment statistics

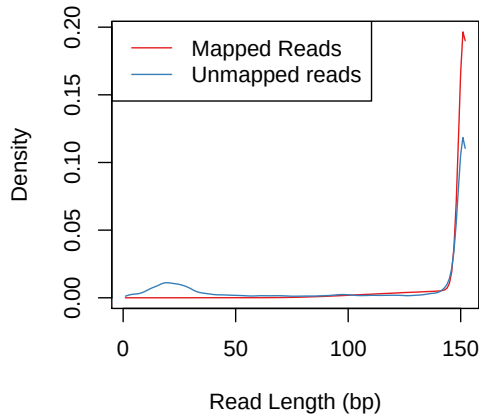
Length distribution of (un-)mapped reads for sample 200042384-0-S22



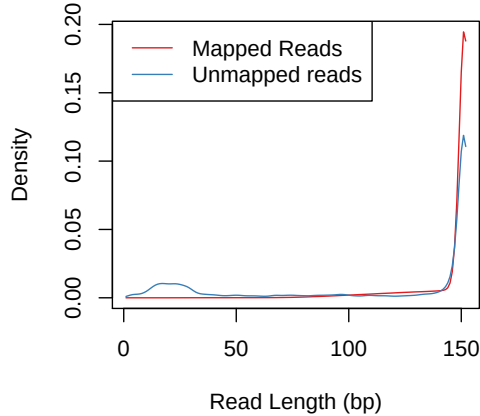
Length distribution of (un-)mapped reads for sample 200042385-0-S23



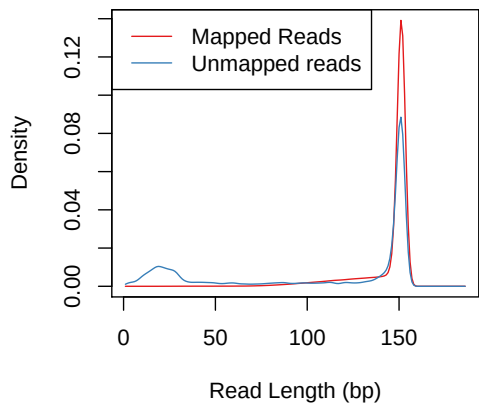
Length distribution of (un-)mapped reads for sample 200042386-0-S11



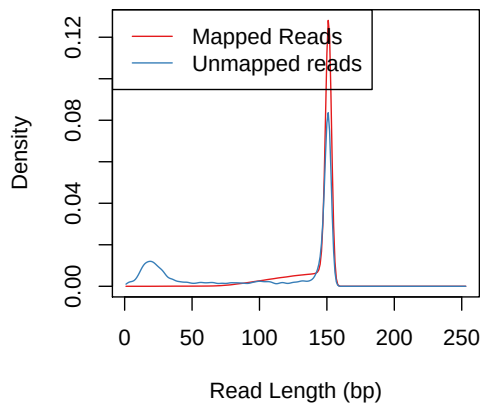
Length distribution of (un-)mapped reads for sample 200042387-0-S10



Length distribution of (un-)mapped reads for sample 200042388-0-S5



Length distribution of (un-)mapped reads for sample 200042389-0-S1



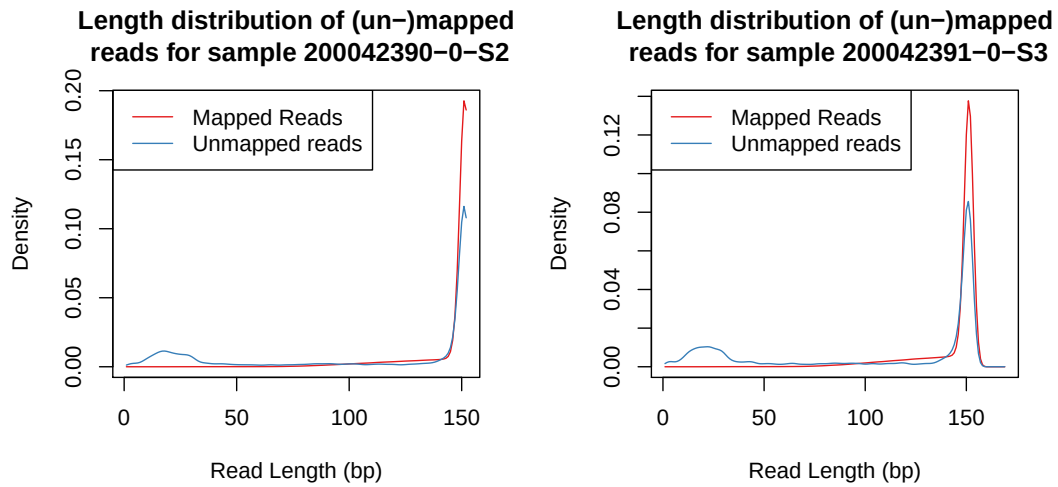


Figure 1.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

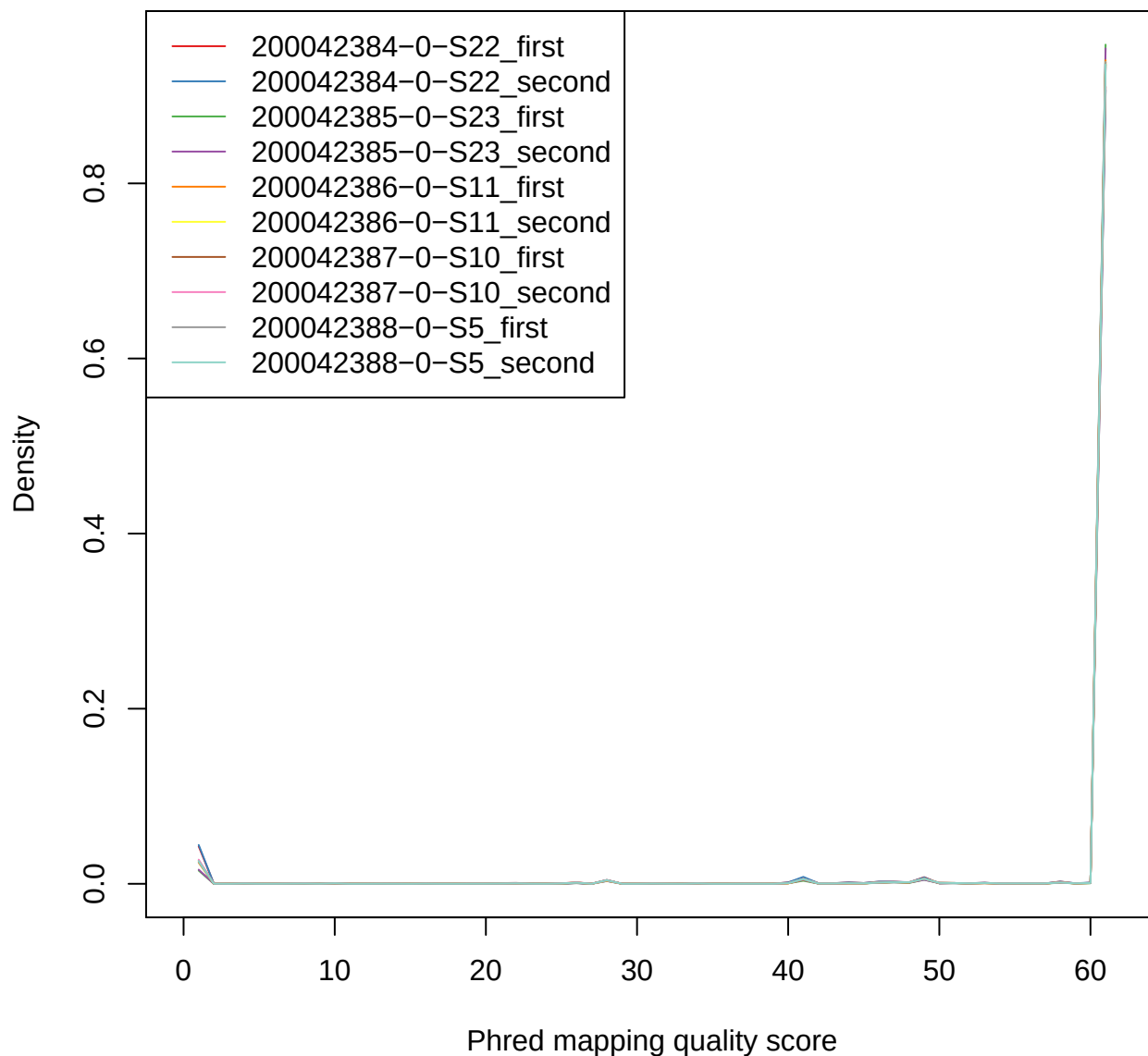


Figure 1.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

1.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

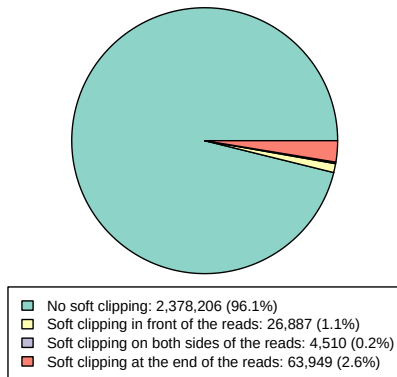
Table 1.3: Soft clipping

Sample ID	No soft clip	Front soft clip	Soft clip both ends	End soft clip
200042384-0-S22	2,378,206 (96.1%)	26,887 (1.1%)	4,510 (0.2%)	63,949 (2.6%)

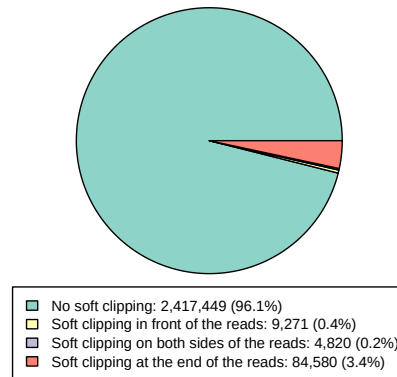
1. PIPELINE ILL1XG1G4_CNV_NEXTSEQ_7

200042385-0-S23	2,417,449 (96.1%)	9,271 (0.4%)	4,820 (0.2%)	84,580 (3.4%)
200042386-0-S11	2,423,162 (96.3%)	8,393 (0.3%)	4,088 (0.2%)	79,835 (3.2%)
200042387-0-S10	2,412,681 (96.1%)	10,024 (0.4%)	4,478 (0.2%)	82,519 (3.3%)
200042388-0-S5	2,416,250 (96.1%)	10,705 (0.4%)	4,360 (0.2%)	82,530 (3.3%)
200042389-0-S1	2,416,162 (96.2%)	9,602 (0.4%)	4,042 (0.2%)	81,260 (3.2%)
200042390-0-S2	2,411,019 (96.1%)	9,278 (0.4%)	4,616 (0.2%)	84,493 (3.4%)
200042391-0-S3	2,420,095 (96.4%)	8,362 (0.3%)	3,897 (0.2%)	78,696 (3.1%)

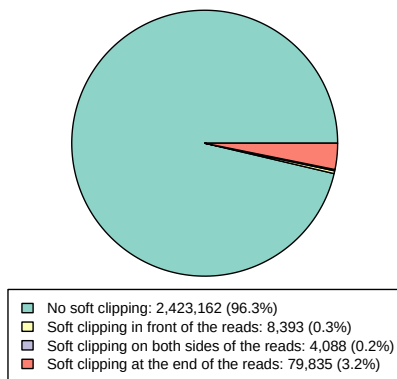
Soft clip statistics for sample 200042384-0-S22 (2,473,552 aligned reads)



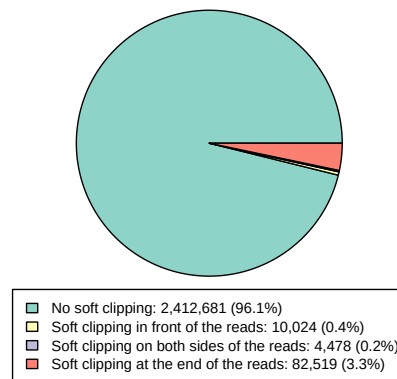
Soft clip statistics for sample 200042385-0-S23 (2,516,120 aligned reads)



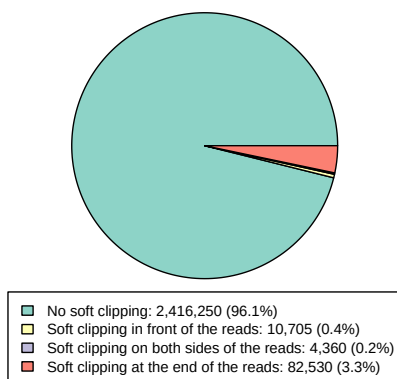
Soft clip statistics for sample 200042386-0-S11 (2,515,478 aligned reads)



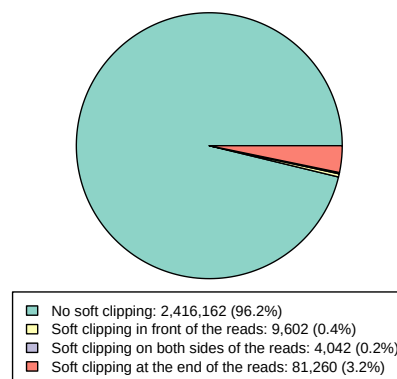
Soft clip statistics for sample 200042387-0-S10 (2,509,702 aligned reads)



Soft clip statistics for sample 200042388-0-S5 (2,513,845 aligned reads)



Soft clip statistics for sample 200042389-0-S1 (2,511,066 aligned reads)



**Soft clip statistics for sample
200042390-0-S2 (2,509,406 aligned reads)**

**Soft clip statistics for sample
200042391-0-S3 (2,511,050 aligned reads)**

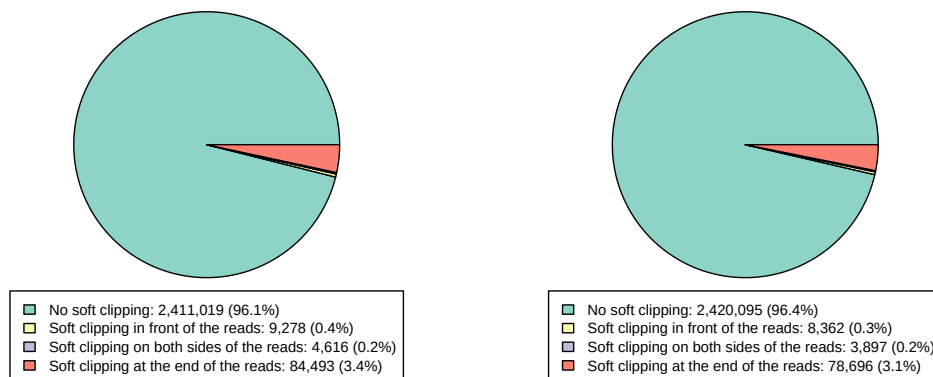
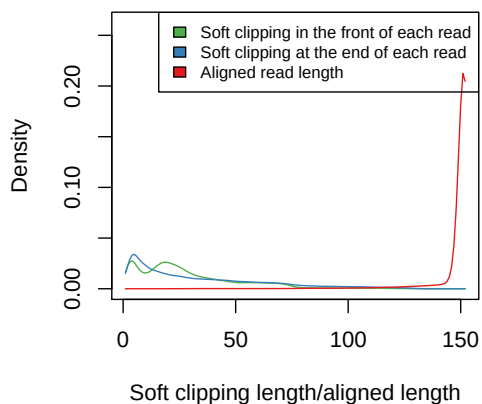
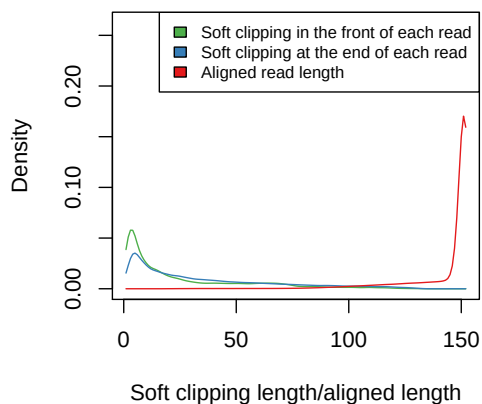
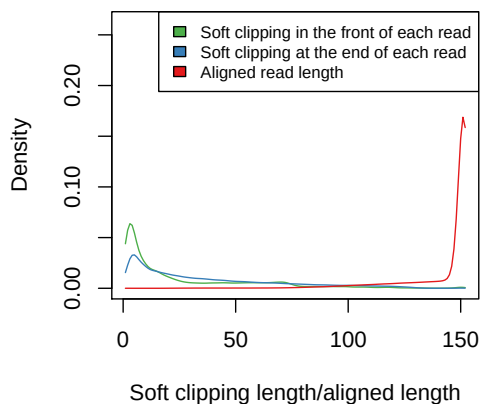
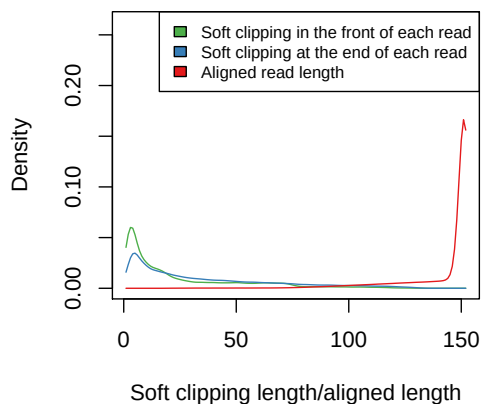
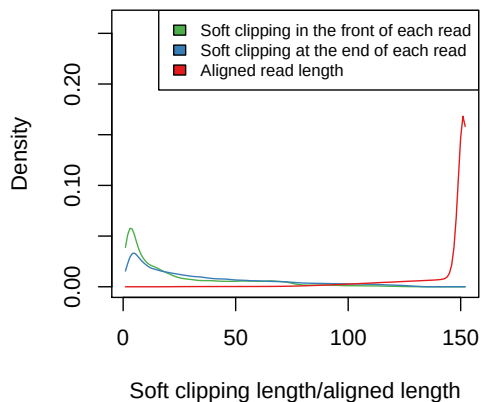
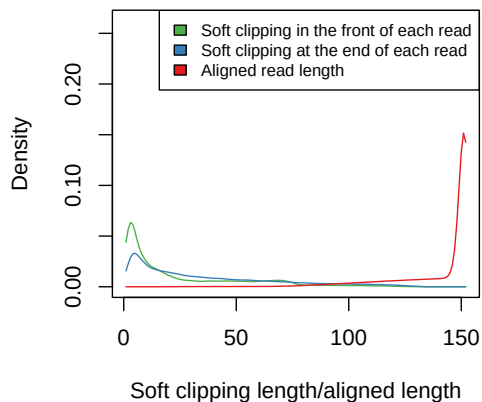


Figure 1.6: Pie charts illustrating the proportion of reads with soft clipping.

**Soft clipping length distribution
for sample 200042384-0-S22****Soft clipping length distribution
for sample 200042385-0-S23****Soft clipping length distribution
for sample 200042386-0-S11****Soft clipping length distribution
for sample 200042387-0-S10****Soft clipping length distribution
for sample 200042388-0-S5****Soft clipping length distribution
for sample 200042389-0-S1**

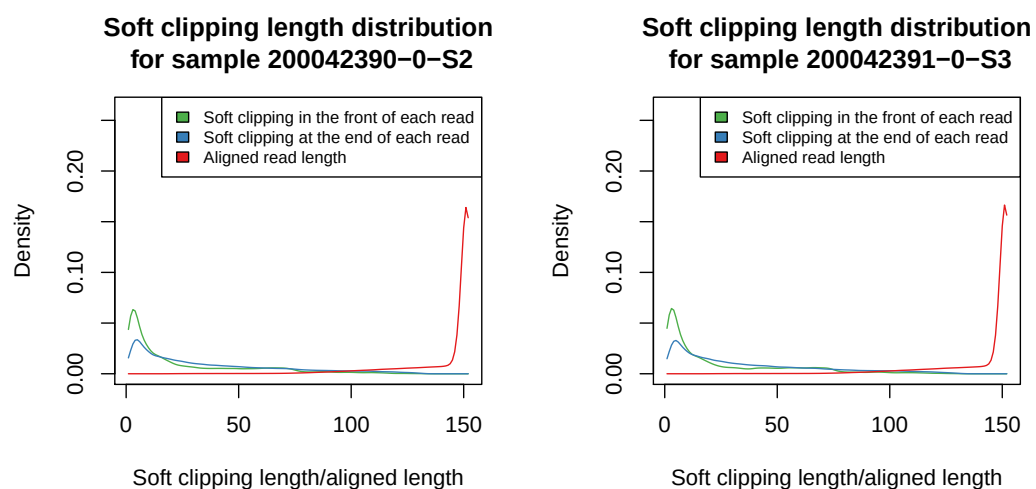


Figure 1.7: Density plots of soft clipping and aligned read length.

1.5 Sample correlation for HCS_v2_hg38 gene panel

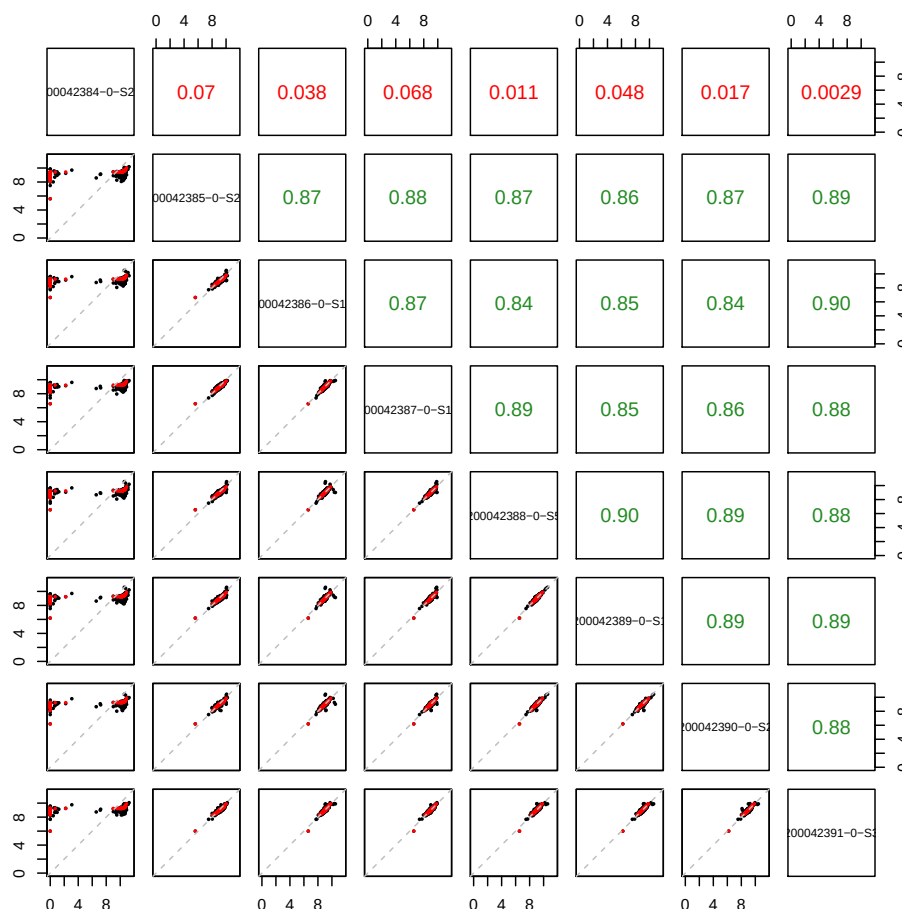


Figure 1.8: Pairwise correlation is presented in the upper panel as the Pearson correlation of the log transformed target region coverage values. The lower panel show the pairwise scatter plots of the log transformed coverage values.

1.6 Mapping statistics for HCS_v2_hg38 gene panel

Table 1.4: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (offTarget_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042384-O-S22	57.38%	10.71%	26.02%	5.88%
200042385-O-S23	80.47%	11.19%	3.48%	4.86%
200042386-O-S11	74.51%	12.01%	6.90%	6.57%
200042387-O-S10	73.96%	11.89%	7.42%	6.73%
200042388-O-S5	74.41%	11.95%	6.98%	6.66%
200042389-O-S1	75.36%	10.34%	7.32%	6.97%
200042390-O-S2	74.28%	11.47%	7.42%	6.83%
200042391-O-S3	74.19%	11.62%	7.48%	6.70%

Table 1.5: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042384-0-S22	29.23%	178790	53559	14985	4154	1497	1	0
200042385-0-S23	6.72%	72651	4926	358	17	5	0	0
200042386-0-S11	5.3%	58957	3164	181	13	2	1	0
200042387-0-S10	5.42%	59933	3316	206	11	1	0	0
200042388-0-S5	5.23%	57996	3134	192	15	0	0	0
200042389-0-S1	5.23%	57872	3155	204	11	2	0	0
200042390-0-S2	5.28%	58456	3204	178	11	2	0	0
200042391-0-S3	4.65%	51,972	2,614	141	4	0	0	0

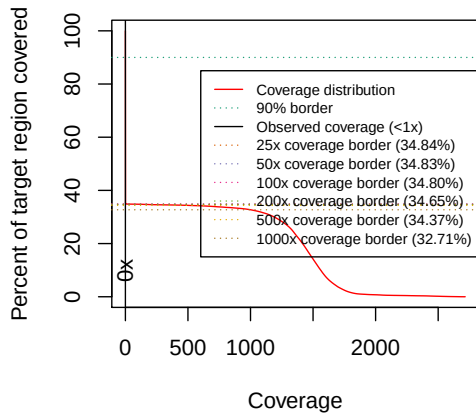
1.7 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

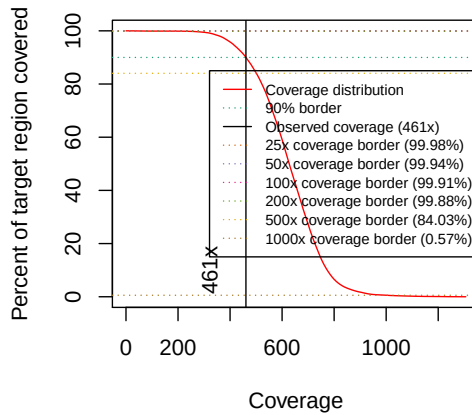
Table 1.6: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042384-0-S22	966
200042385-0-S23	2
200042386-0-S11	1
200042387-0-S10	1
200042388-0-S5	1
200042389-0-S1	2
200042390-0-S2	1
200042391-0-S3	1

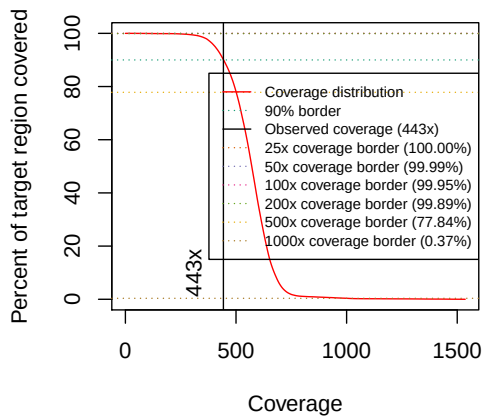
**Target region coverage distribution
for sample 200042384-0-S22**



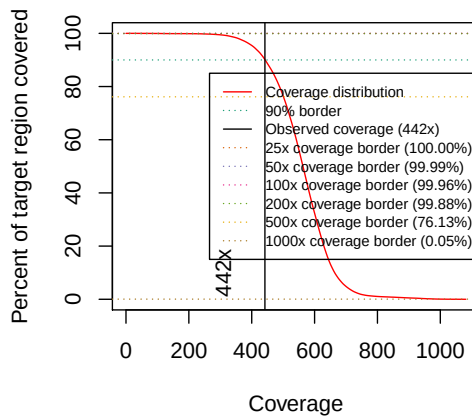
**Target region coverage distribution
for sample 200042385-0-S23**



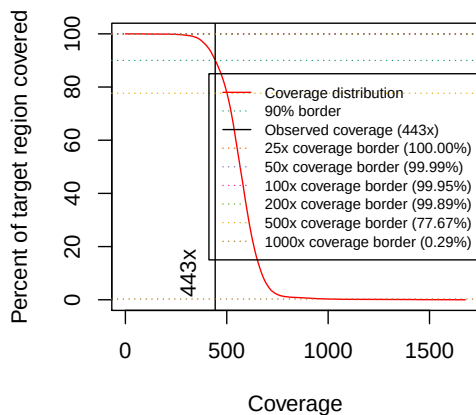
**Target region coverage distribution
for sample 200042386-0-S11**



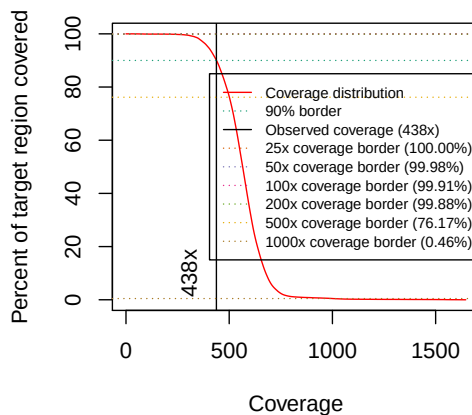
**Target region coverage distribution
for sample 200042387-0-S10**



**Target region coverage distribution
for sample 200042388-0-S5**



**Target region coverage distribution
for sample 200042389-0-S1**



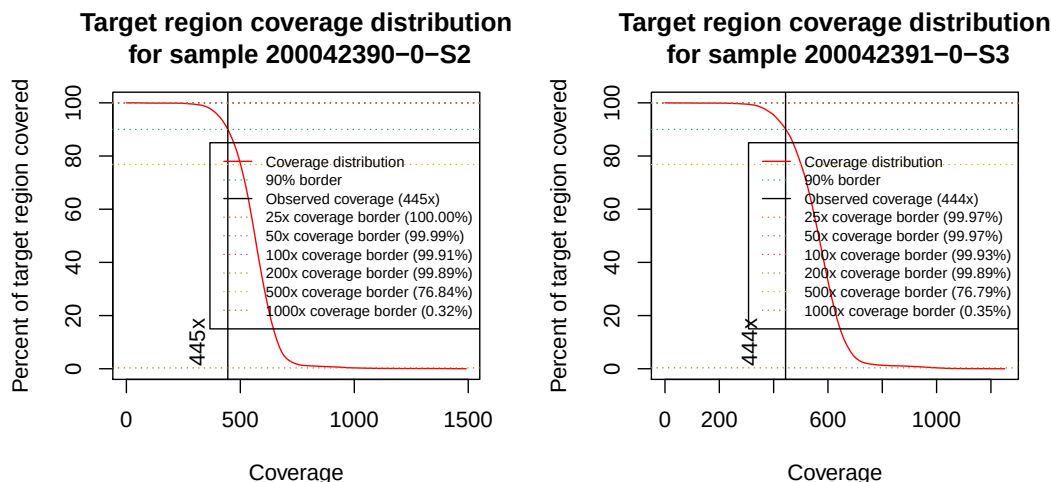


Figure 1.9: Target region coverage.

Table 1.7: Percentage of target regions with certain coverage and coverage heterogeneity across target region. Note, individual coverage uniformity plots can be found in the separate sample sections.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042384-0-S22	34.84%	34.83%	34.80%	34.65%	34.37%	32.71%	0×	35.29%
200042385-0-S23	99.98%	99.94%	99.91%	99.88%	84.03%	0.57%	461×	0.10%
200042386-0-S11	100.00%	99.99%	99.95%	99.89%	77.84%	0.37%	443×	0.05%
200042387-0-S10	100.00%	99.99%	99.96%	99.88%	76.13%	0.05%	442×	0.06%
200042388-0-S5	100.00%	99.99%	99.95%	99.89%	77.67%	0.29%	443×	0.07%
200042389-0-S1	100.00%	99.98%	99.91%	99.88%	76.17%	0.46%	438×	0.09%
200042390-0-S2	100.00%	99.99%	99.91%	99.89%	76.84%	0.32%	445×	0.09%
200042391-0-S3	99.97%	99.97%	99.93%	99.89%	76.79%	0.35%	444×	0.08%

Sample 200042384-0-S22

2.1 Dataset

For sample 200042384-0-S22 2,530,000 reads were sequenced, of these 2,473,552 were mapped corresponding to 97.77% of all reads.

2.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 2.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 2.2 shows the distribution of average base call quality, while Figure 2.3 shows the distribution of read lengths.

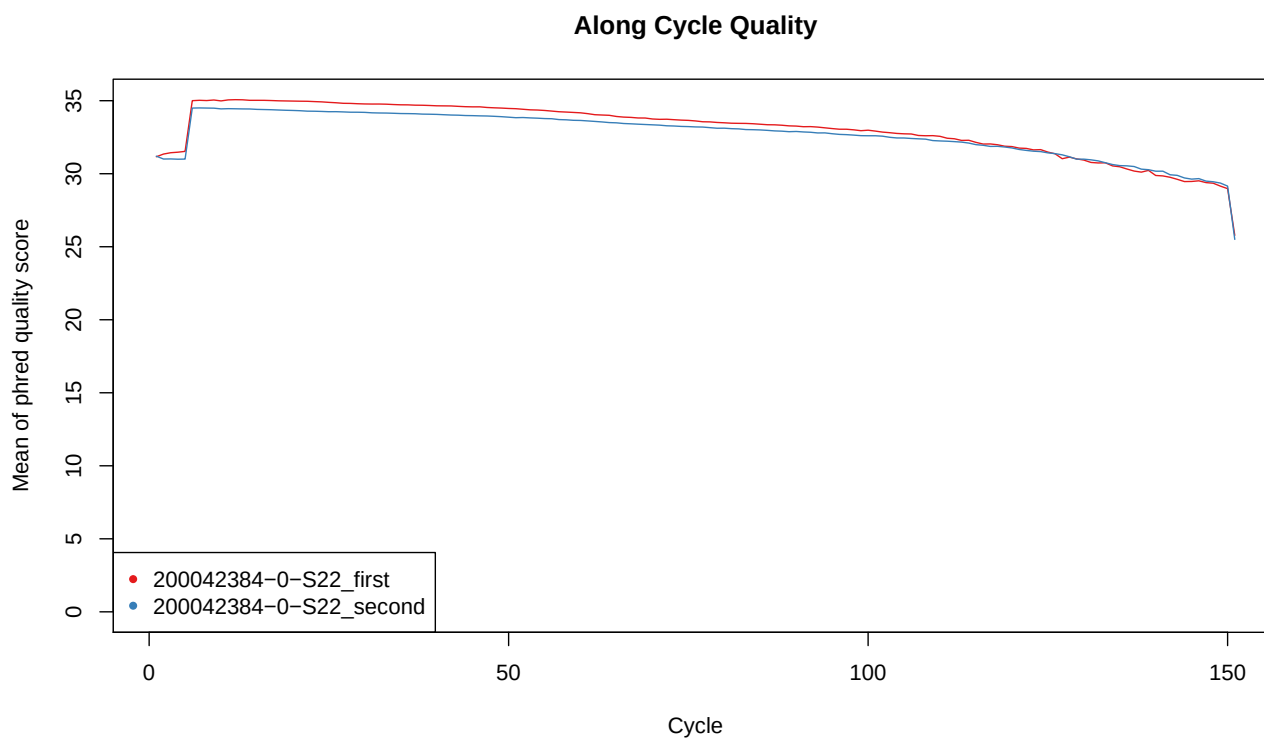


Figure 2.1: Mean base call quality by cycle.

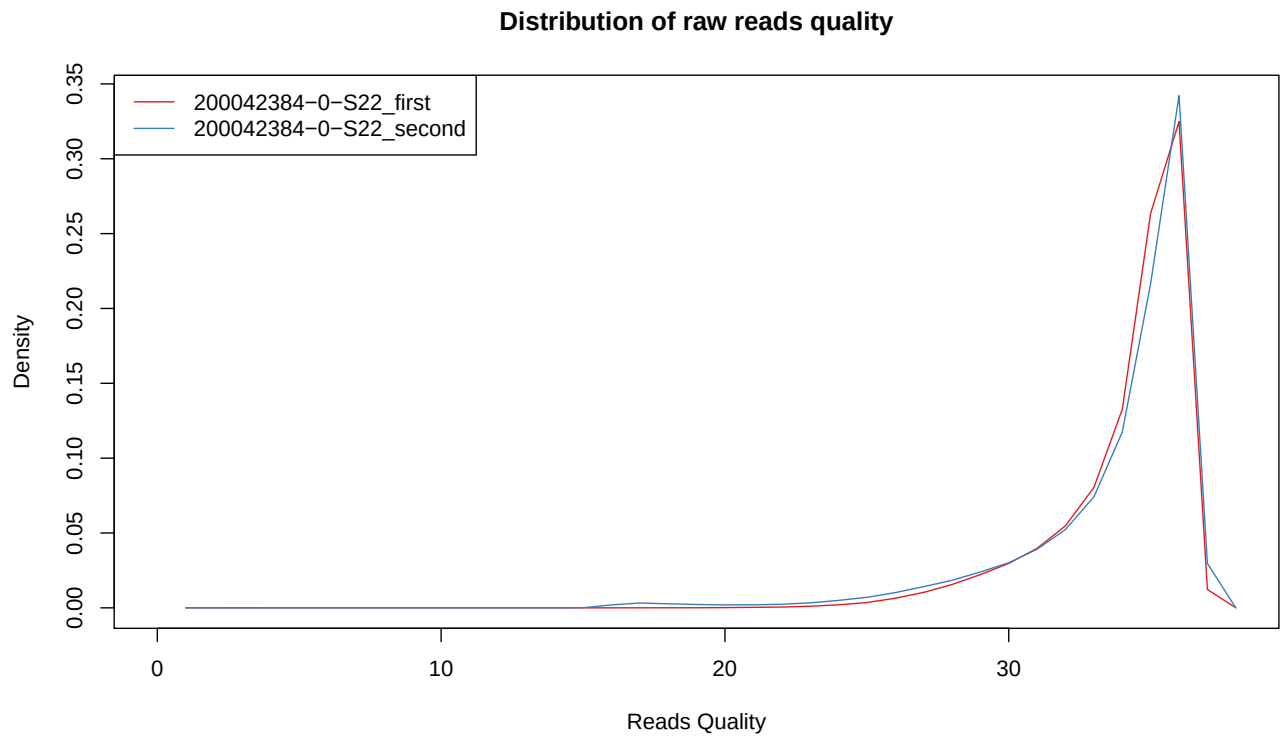


Figure 2.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

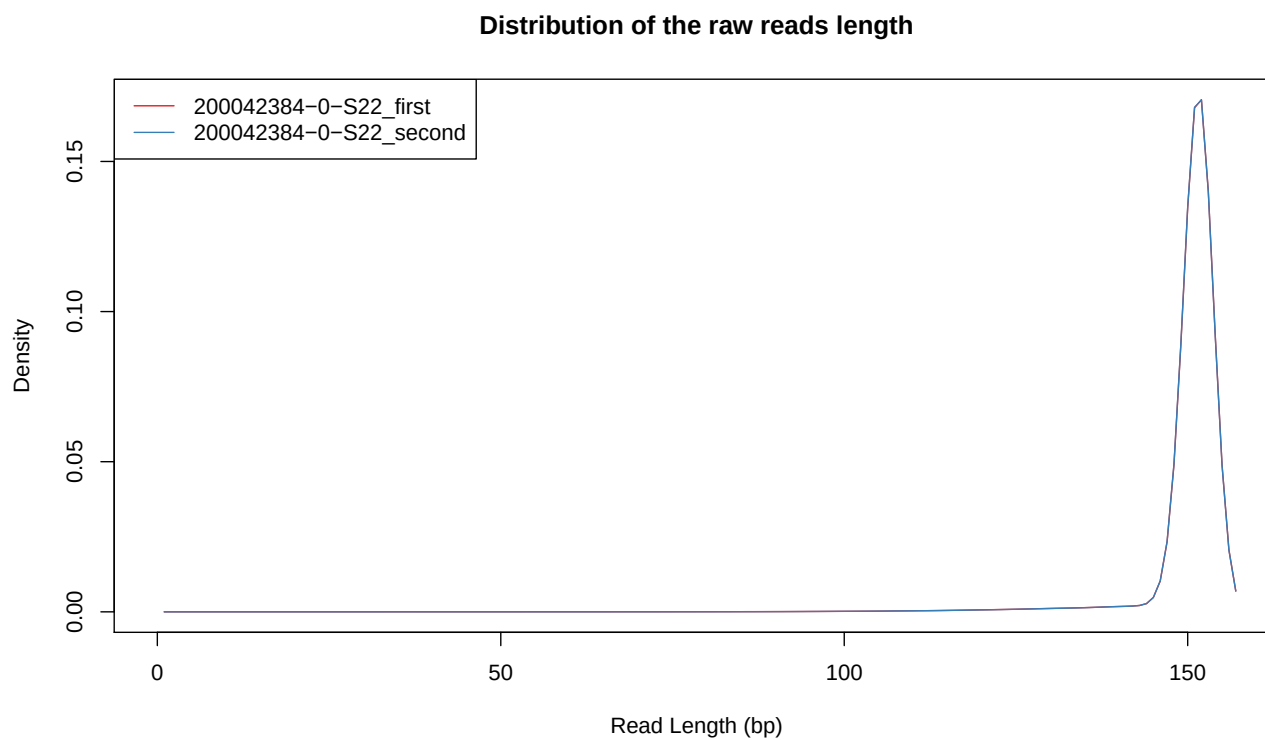


Figure 2.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

2.3 Alignment statistics

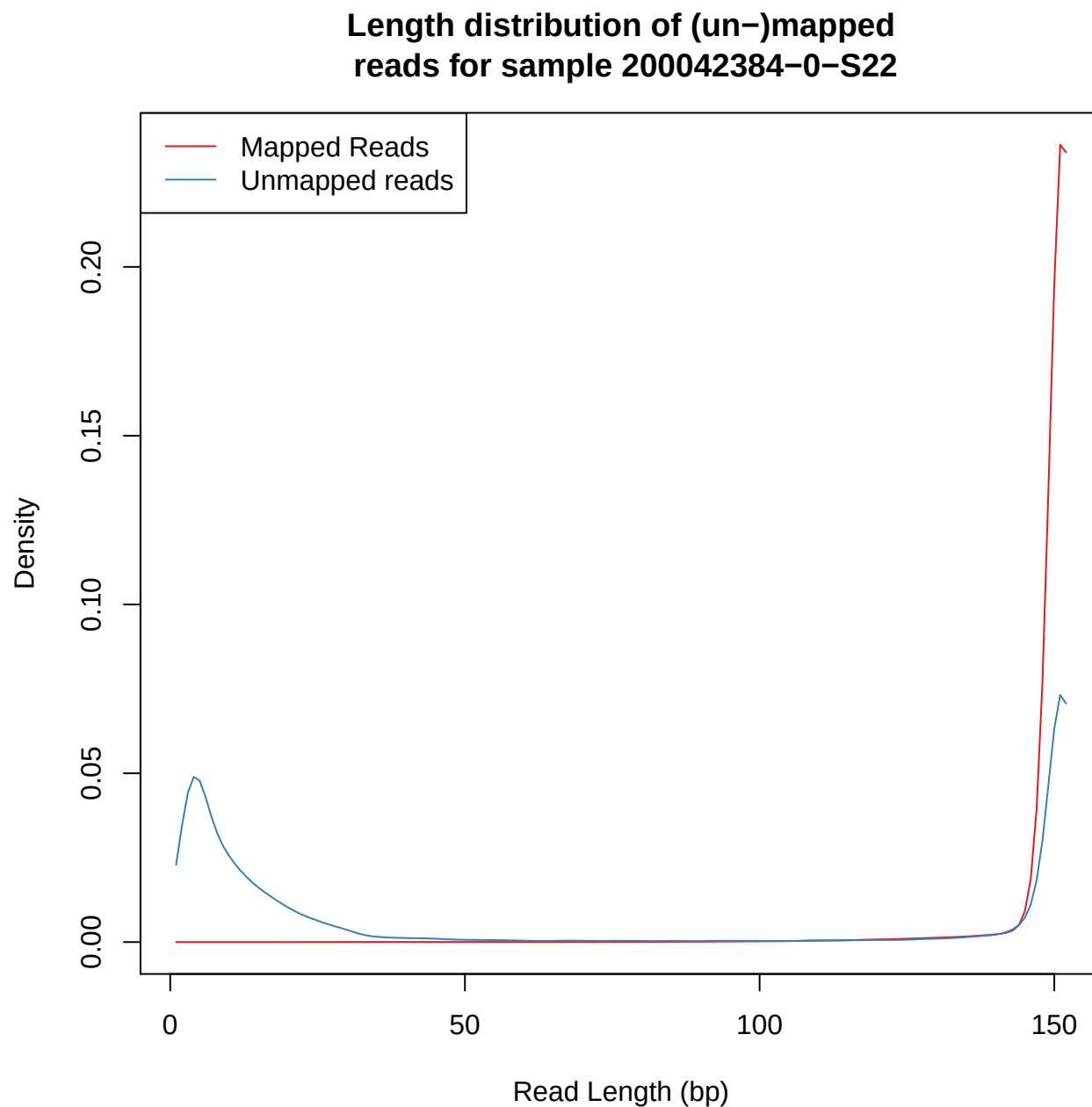


Figure 2.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

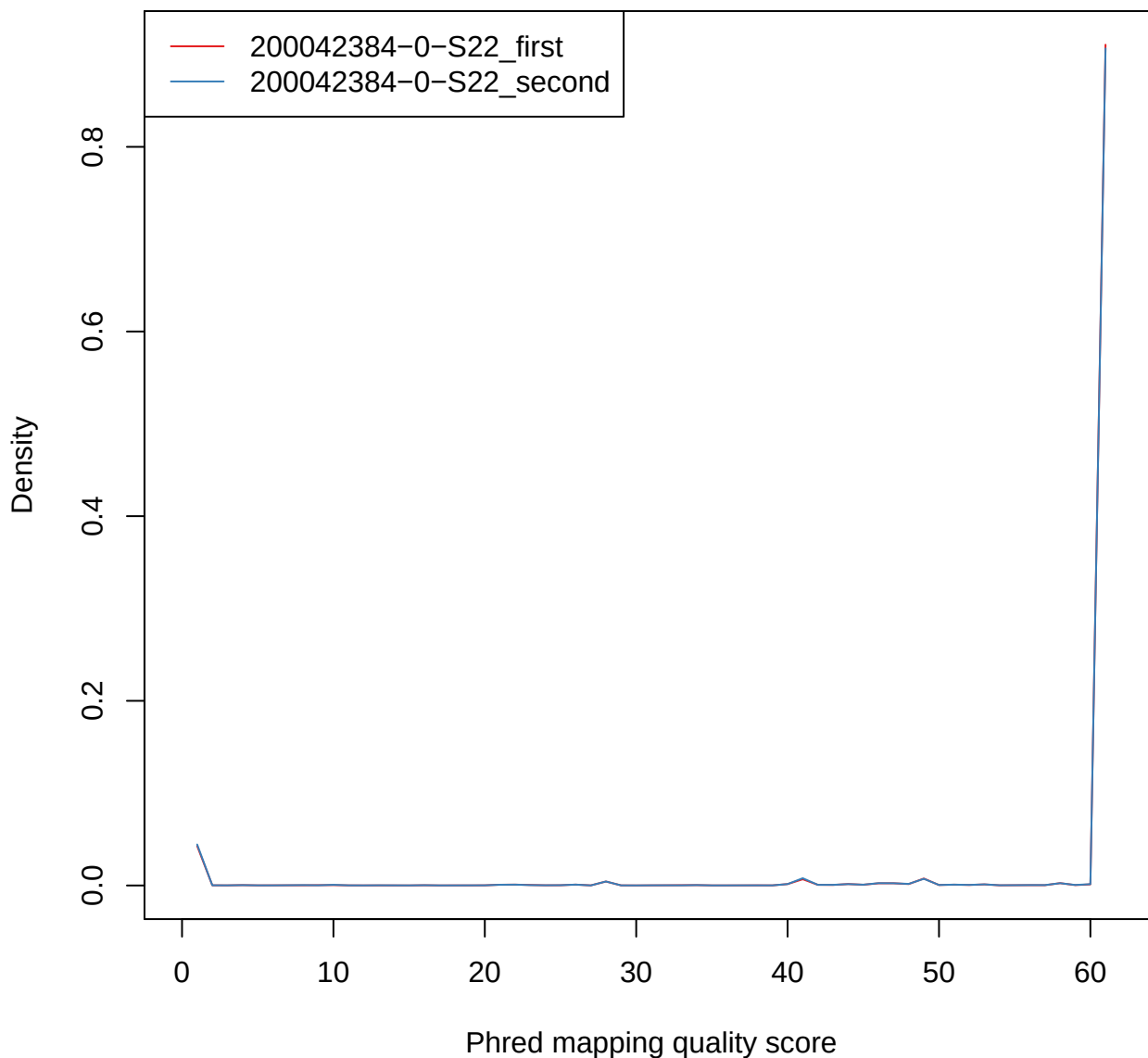


Figure 2.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

2.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042384-0-S22 (2,473,552 aligned reads)**

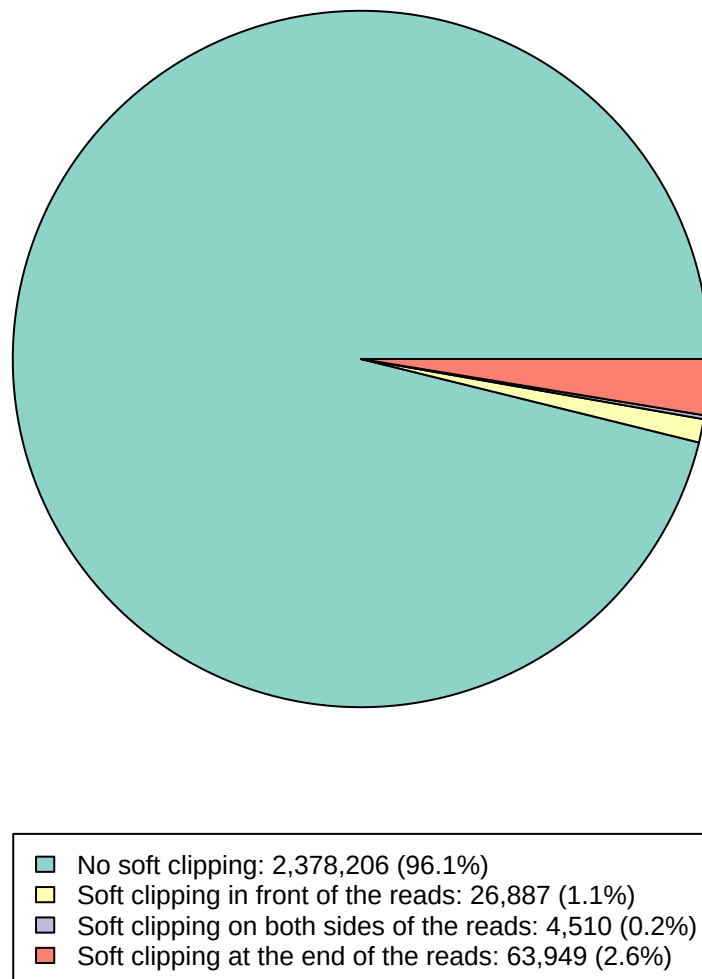


Figure 2.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042384-0-S22

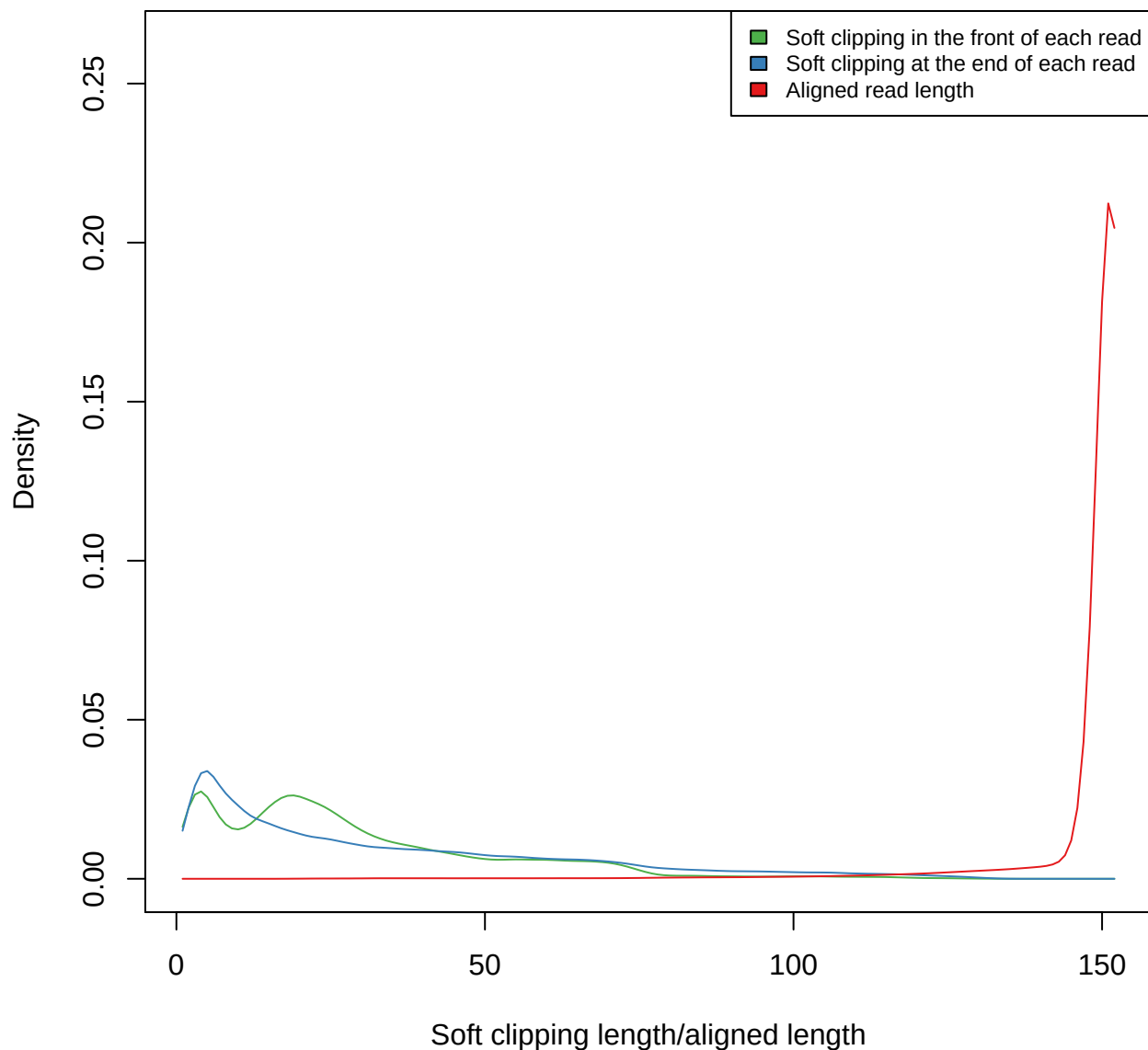


Figure 2.7: Density plots of soft clipping and aligned read length.

2.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 2.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042384-0-S22	57.38%	10.71%	26.02%	5.88%

Table 2.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042384-0-S22	29.23%	178790	53559	14985	4154	1497	1	0

2.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 2.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042384-0-S22	966

Too many low coverage regions were detected for reporting. For details please refer to the Warnings page in Sophia DDM®.

Target region coverage distribution for sample 200042384-0-S22

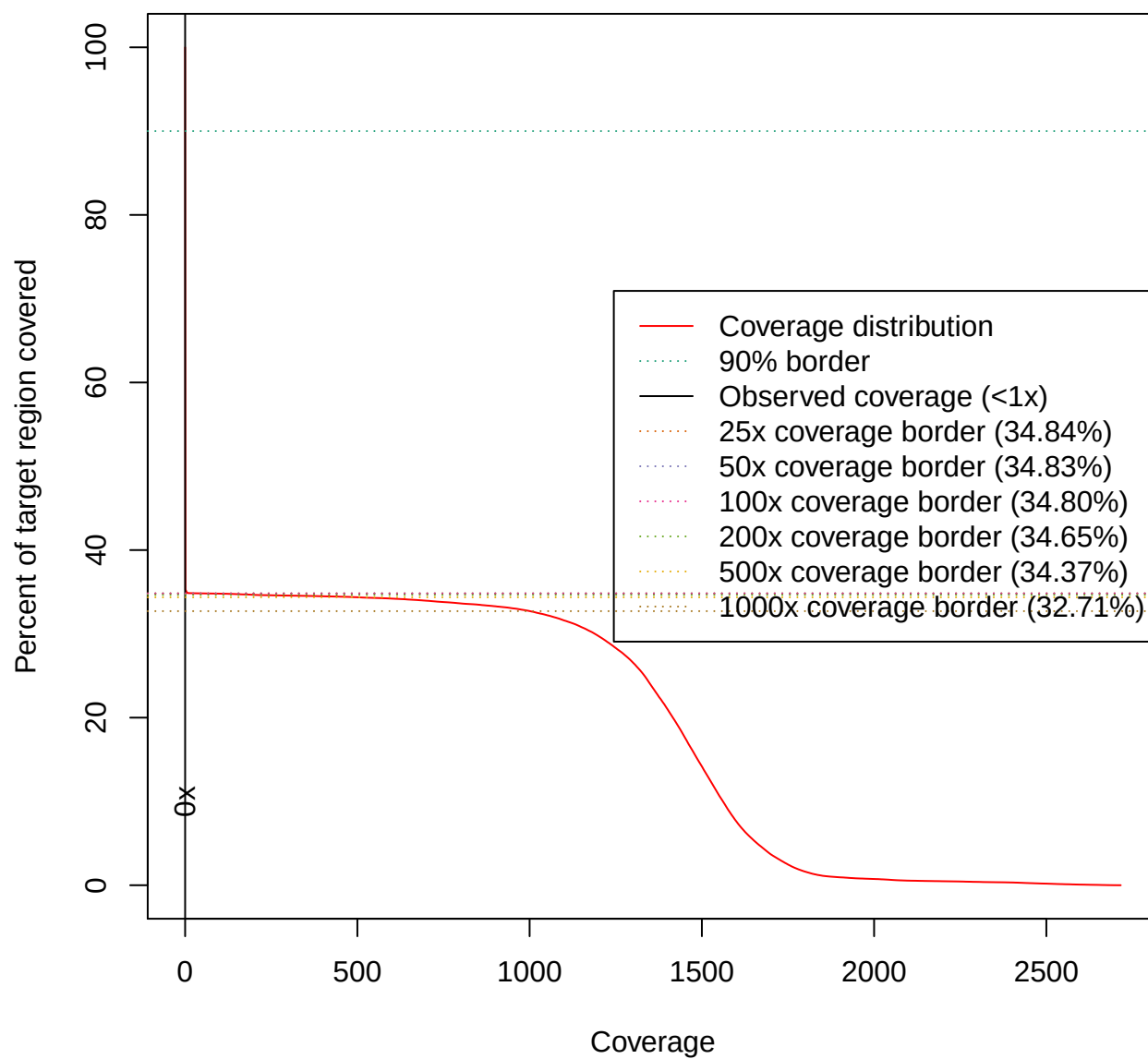


Figure 2.8: Target region coverage.

Table 2.4: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042384-0-S22	34.84%	34.83%	34.80%	34.65%	34.37%	32.71%	0×	35.29%

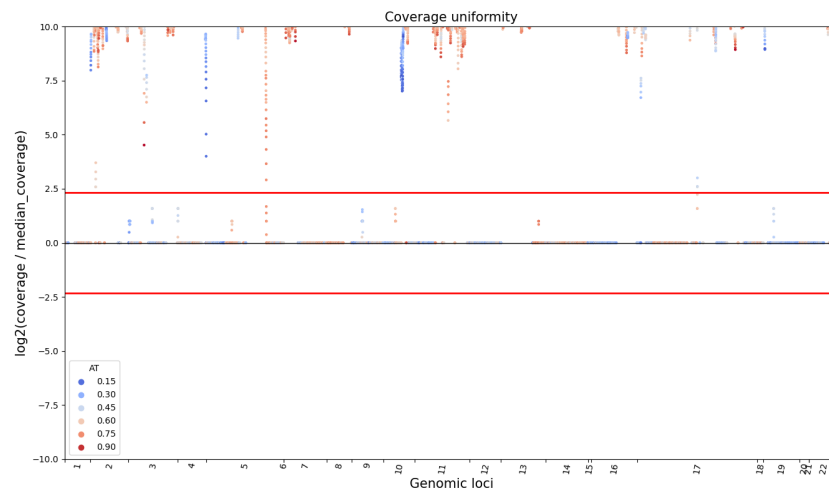


Figure 2.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.

Sample 200042385-0-S23

3.1 Dataset

For sample 200042385-0-S23 2,530,000 reads were sequenced, of these 2,516,120 were mapped corresponding to 99.45% of all reads.

3.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 3.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 3.2 shows the distribution of average base call quality, while Figure 3.3 shows the distribution of read lengths.

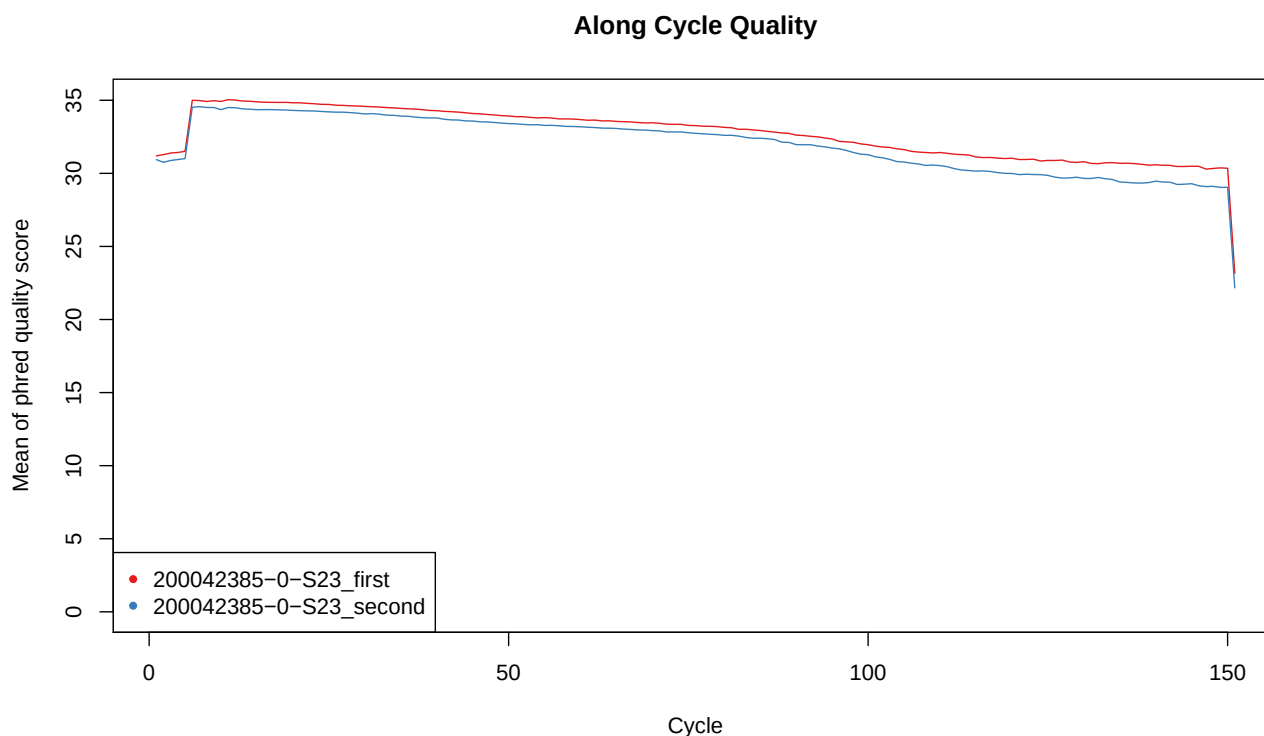


Figure 3.1: Mean base call quality by cycle.

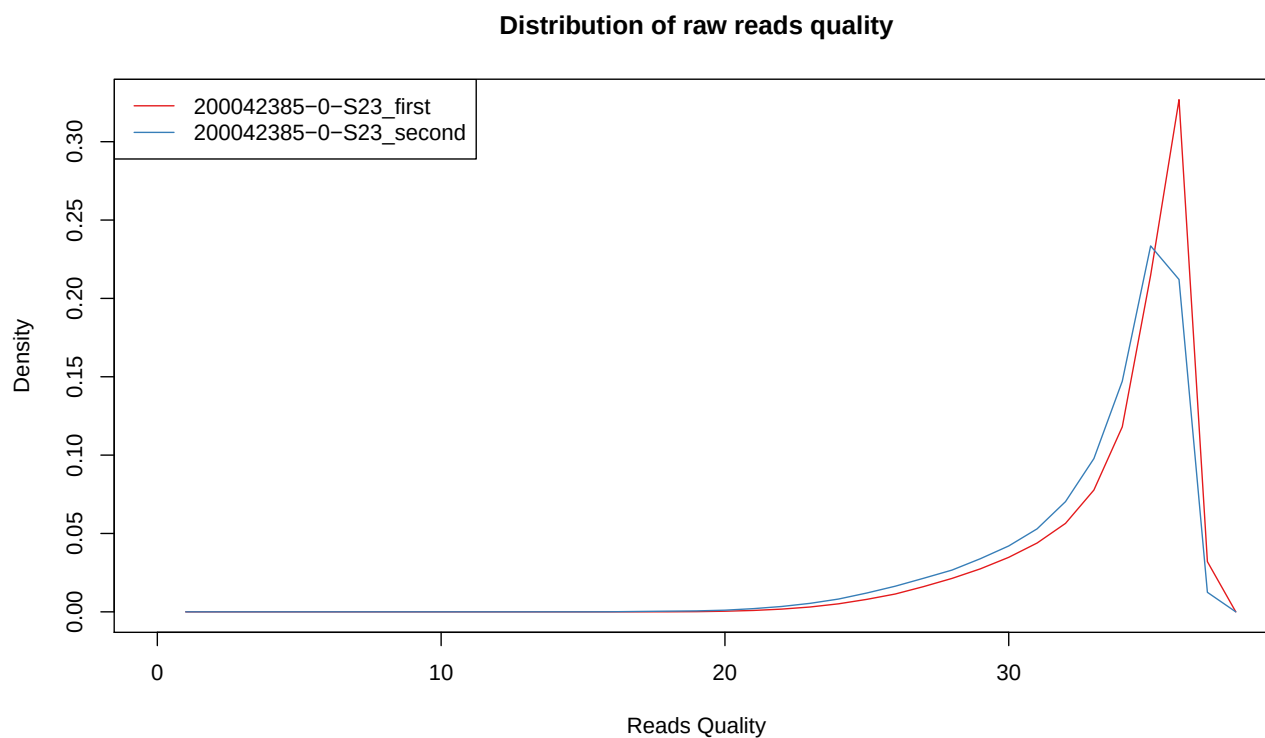


Figure 3.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

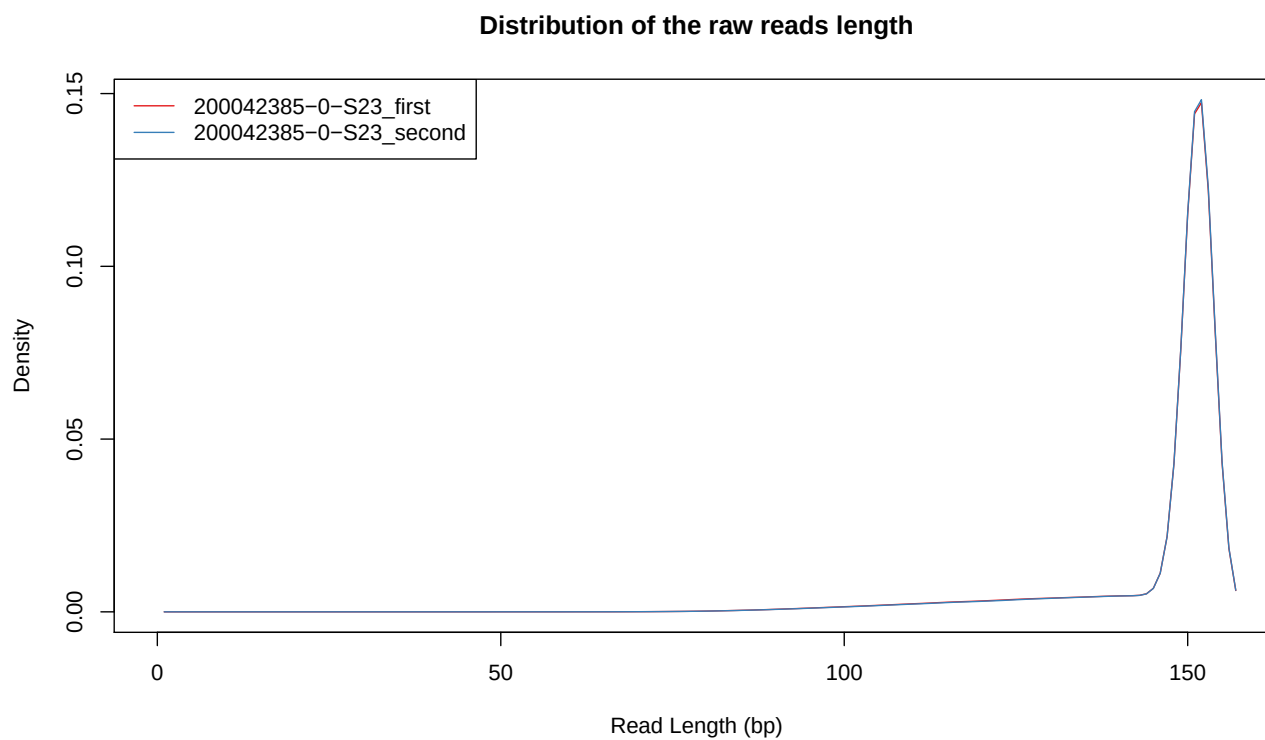


Figure 3.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

3.3 Alignment statistics

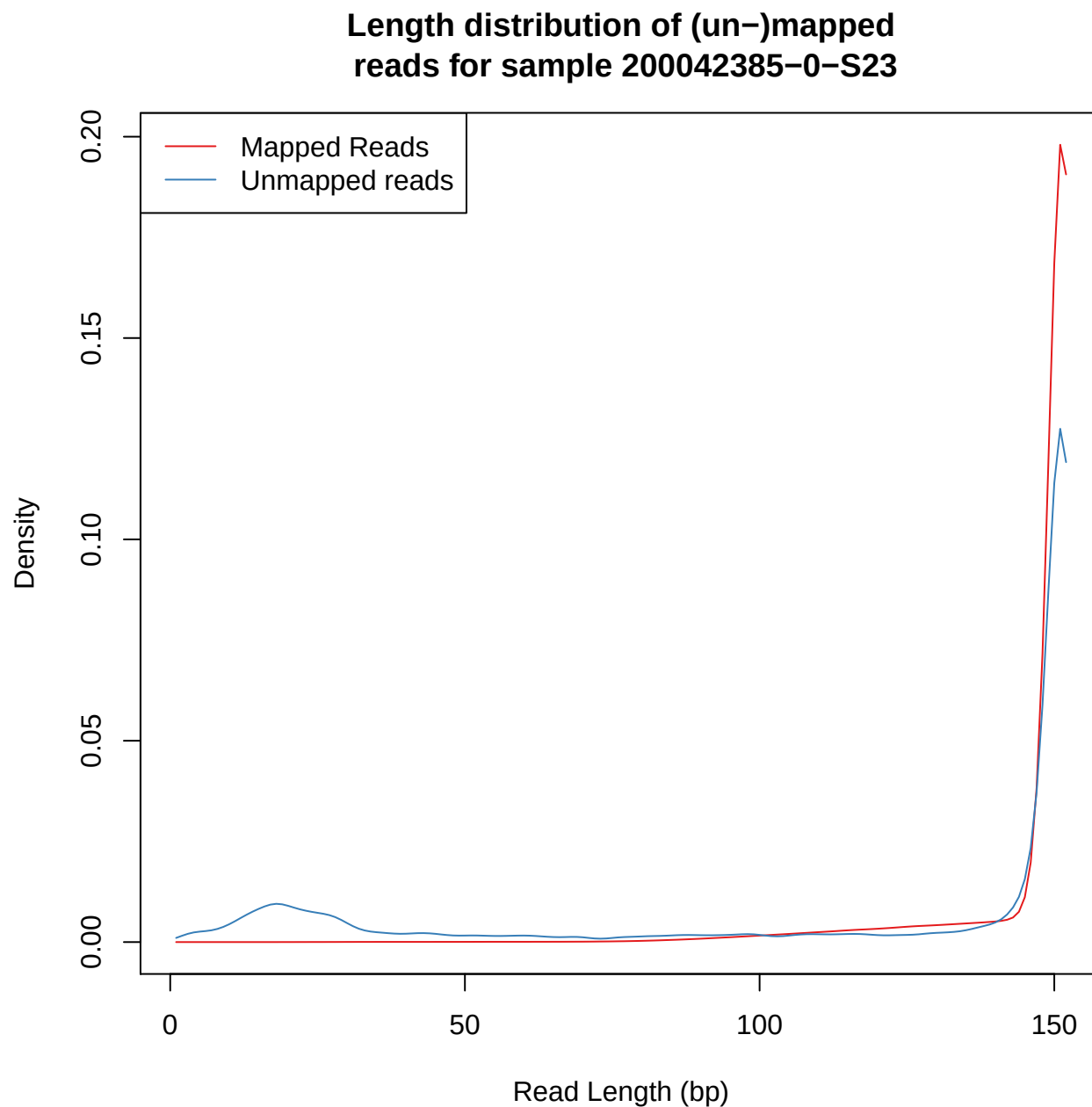


Figure 3.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

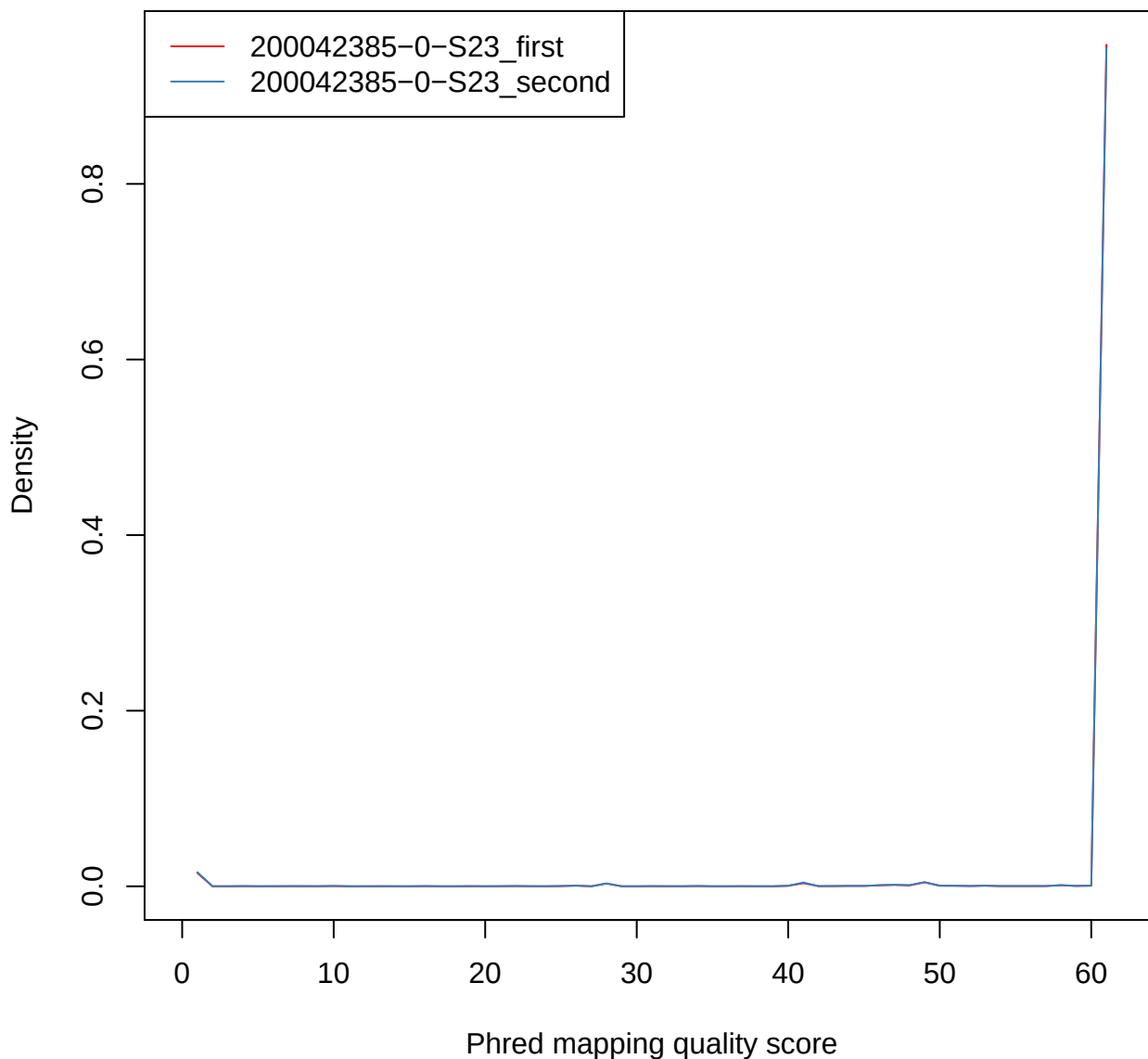


Figure 3.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

3.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042385-0-S23 (2,516,120 aligned reads)**

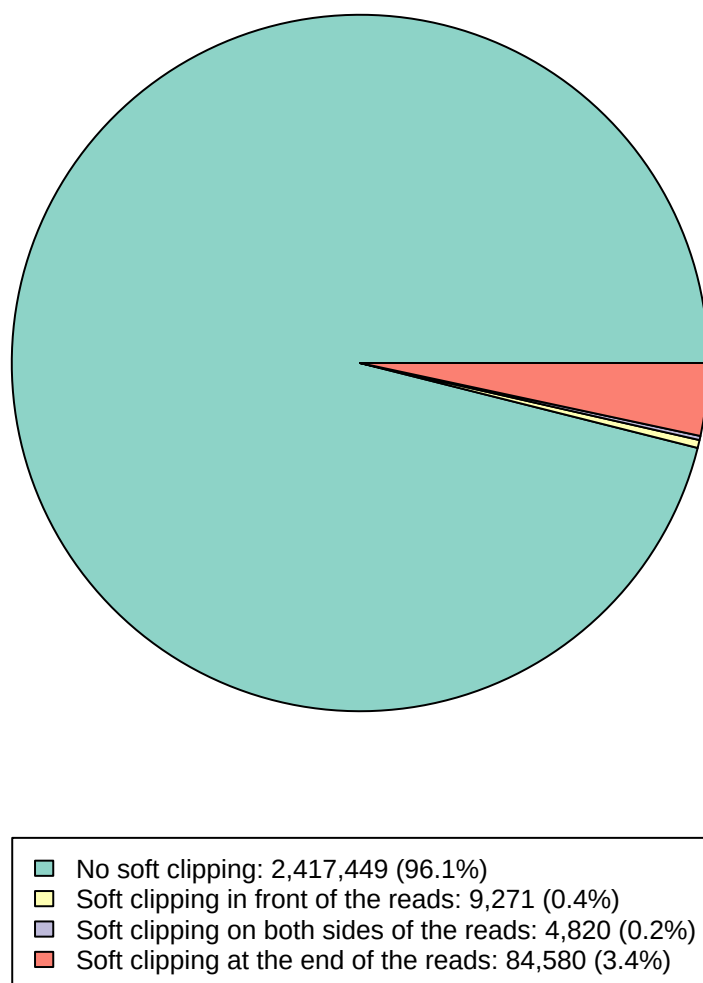


Figure 3.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042385-0-S23

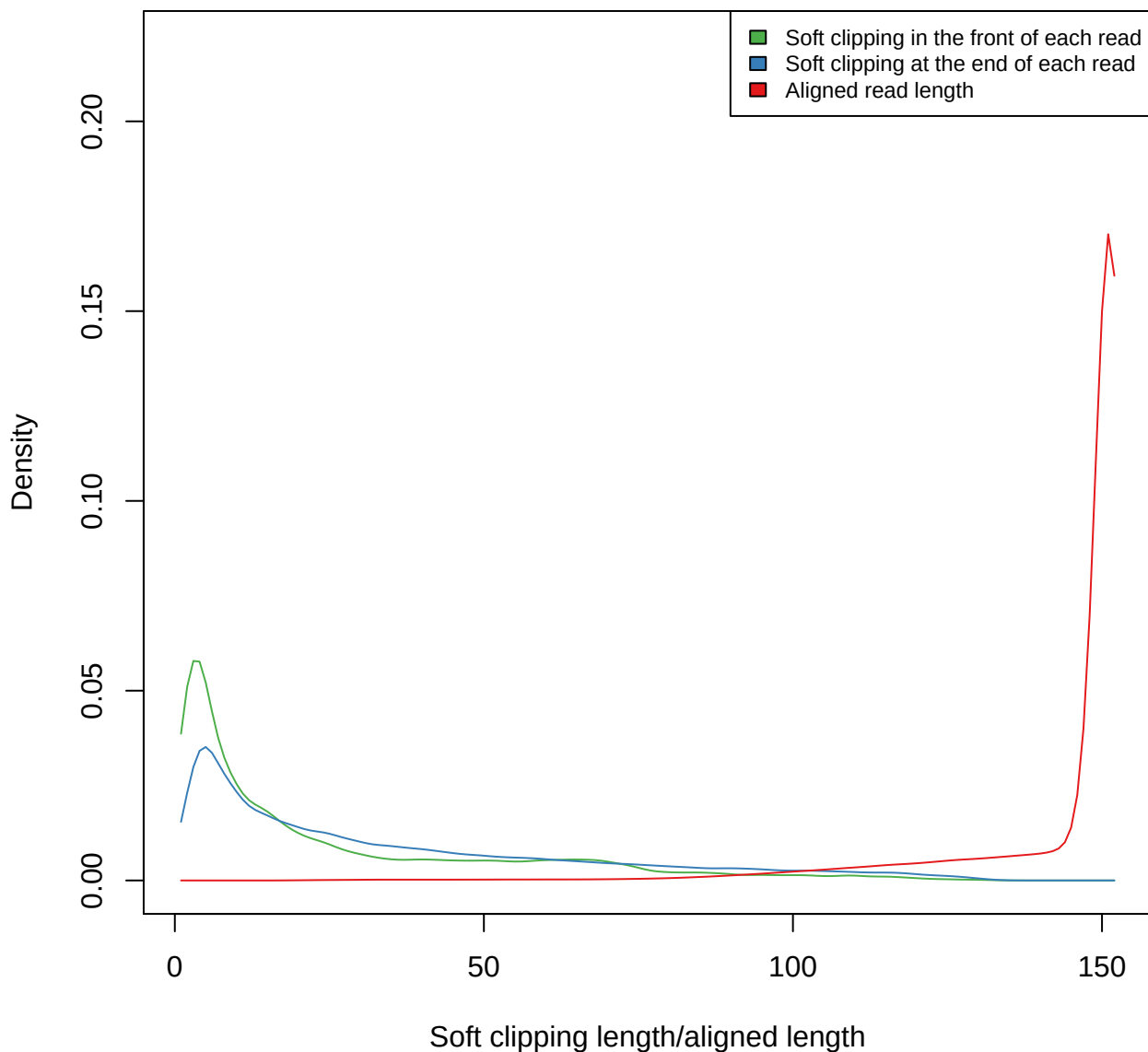


Figure 3.7: Density plots of soft clipping and aligned read length.

3.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 3.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042385-0-S23	80.47%	11.19%	3.48%	4.86%

Table 3.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042385-0-S23	6.72%	72651	4926	358	17	5	0	0

3.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 3.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042385-0-S23	2

Table 3.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	142	MSH2	NM_000251.3	8	c.1277-3297_1277-3156	33
2	47442499	16	MSH2	NM_000251.3	8	c.1277-3049_1277-3034	41

Target region coverage distribution for sample 200042385-0-S23

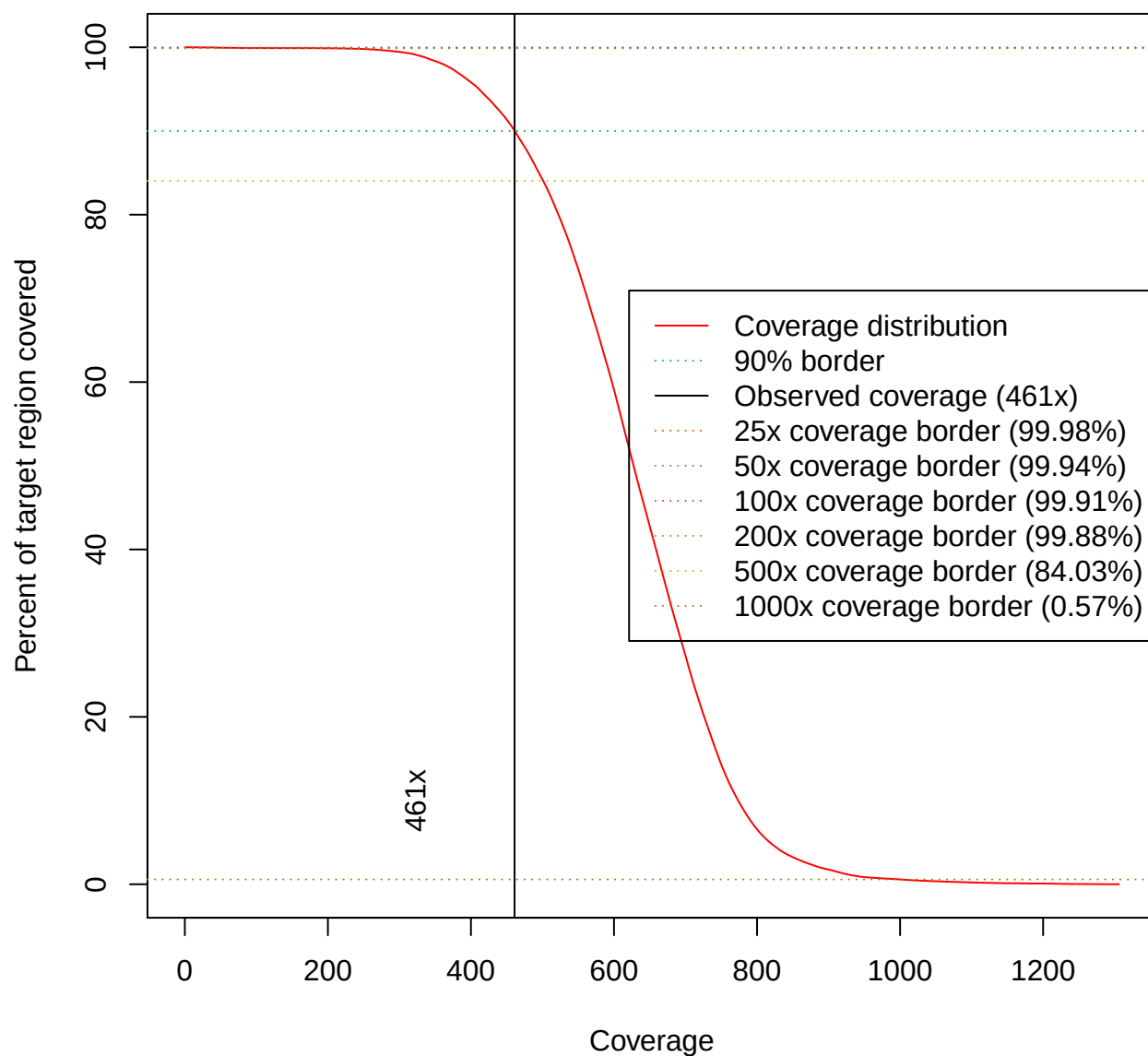


Figure 3.8: Target region coverage.

Table 3.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042385-0-S23	99.98%	99.94%	99.91%	99.88%	84.03%	0.57%	461×	0.10%

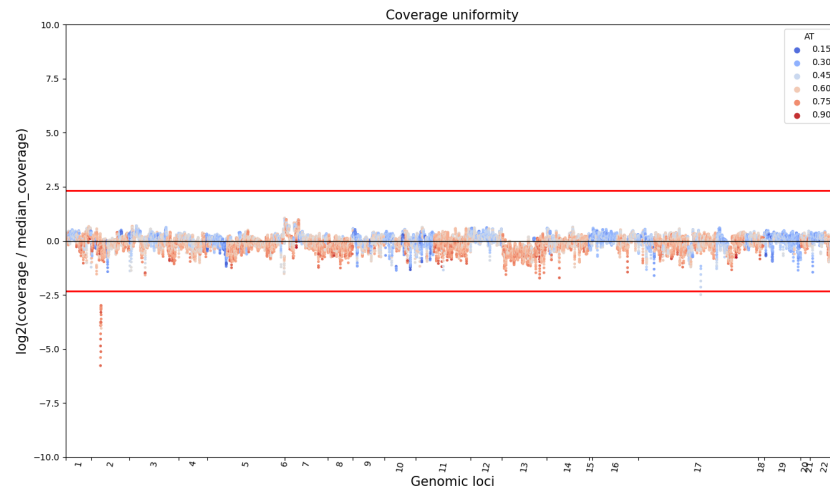


Figure 3.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.

Sample 200042386-0-S11

4.1 Dataset

For sample 200042386-0-S11 2,530,000 reads were sequenced, of these 2,515,478 were mapped corresponding to 99.43% of all reads.

4.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 4.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 4.2 shows the distribution of average base call quality, while Figure 4.3 shows the distribution of read lengths.

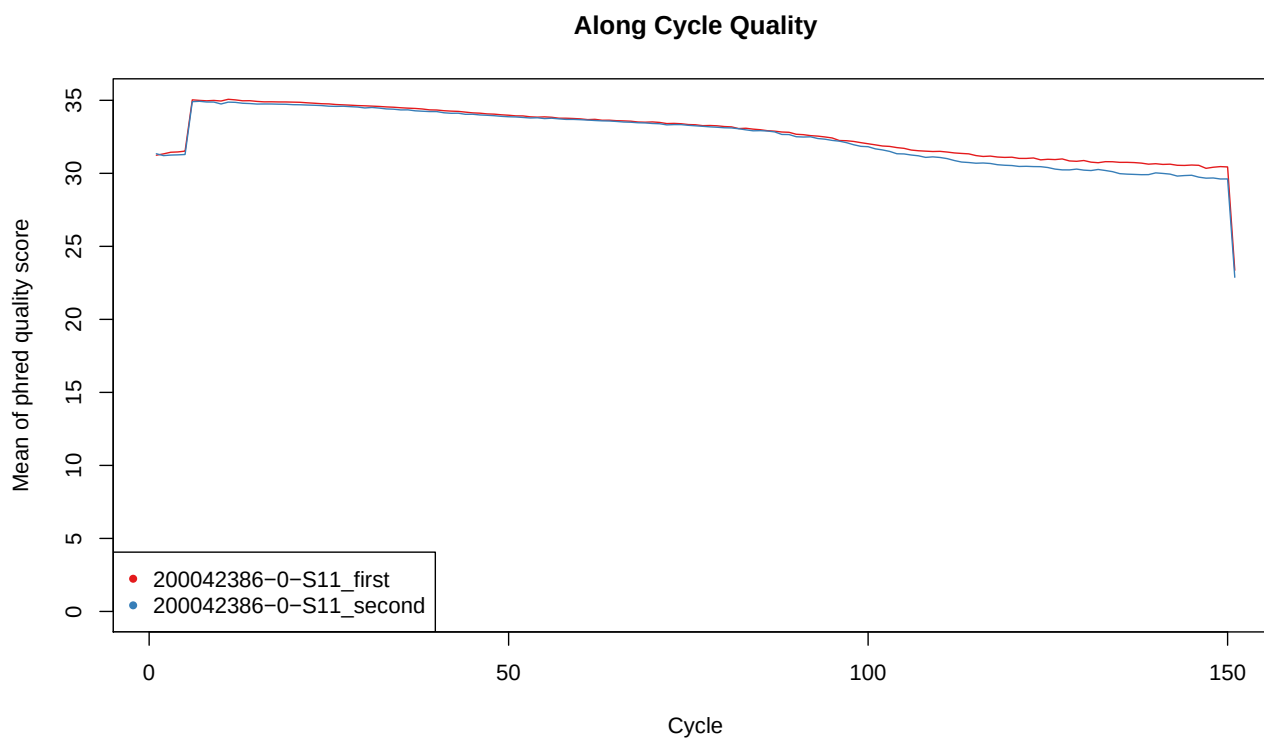


Figure 4.1: Mean base call quality by cycle.

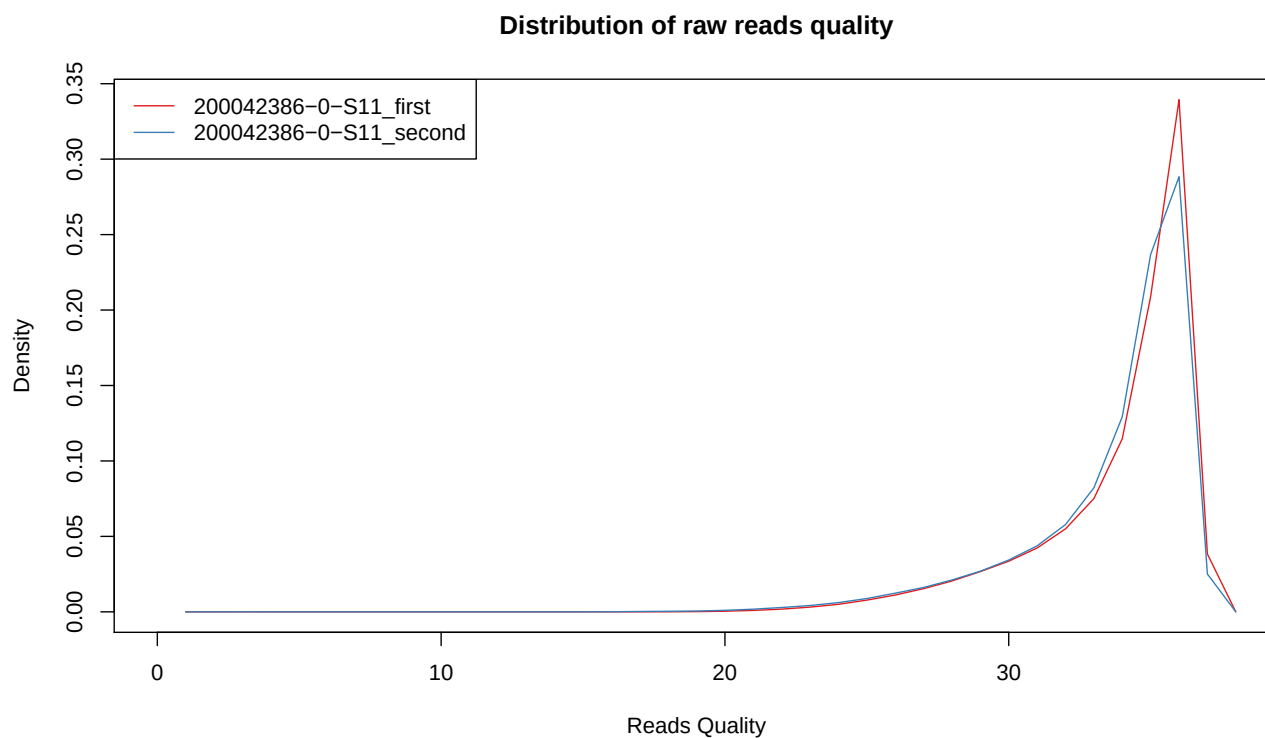


Figure 4.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

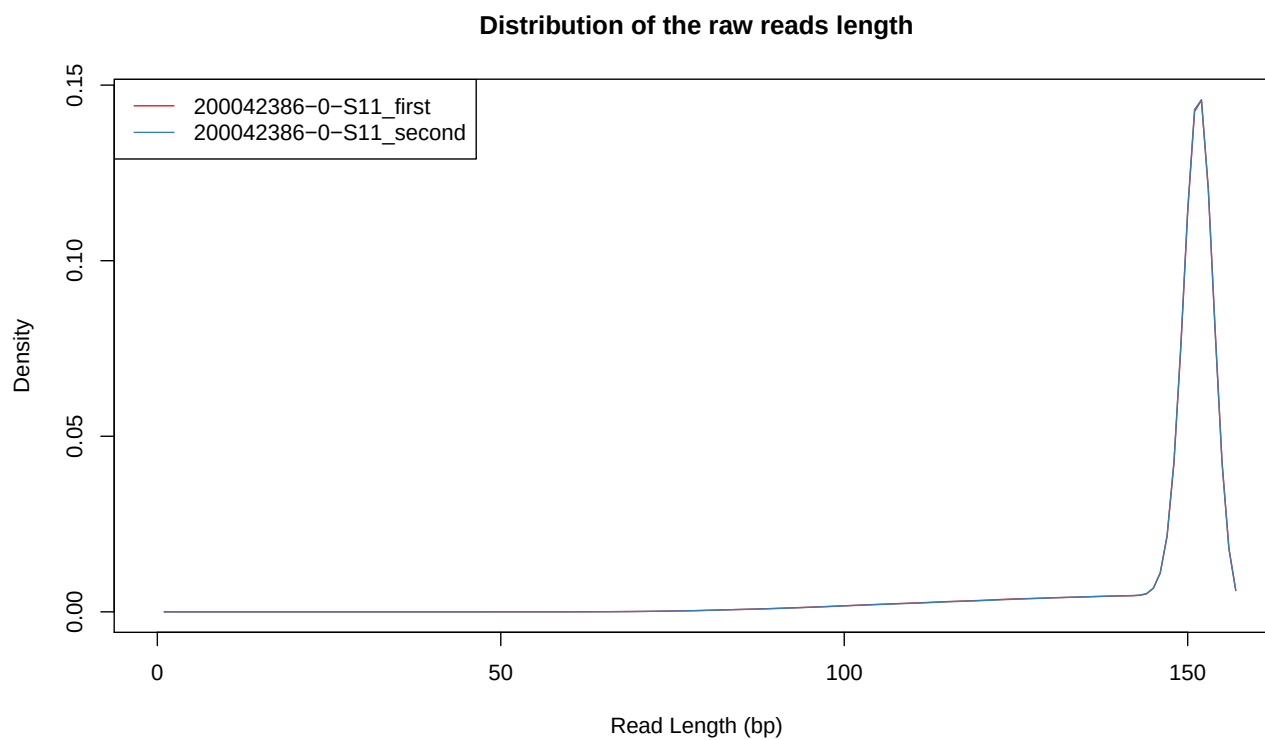


Figure 4.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

4.3 Alignment statistics

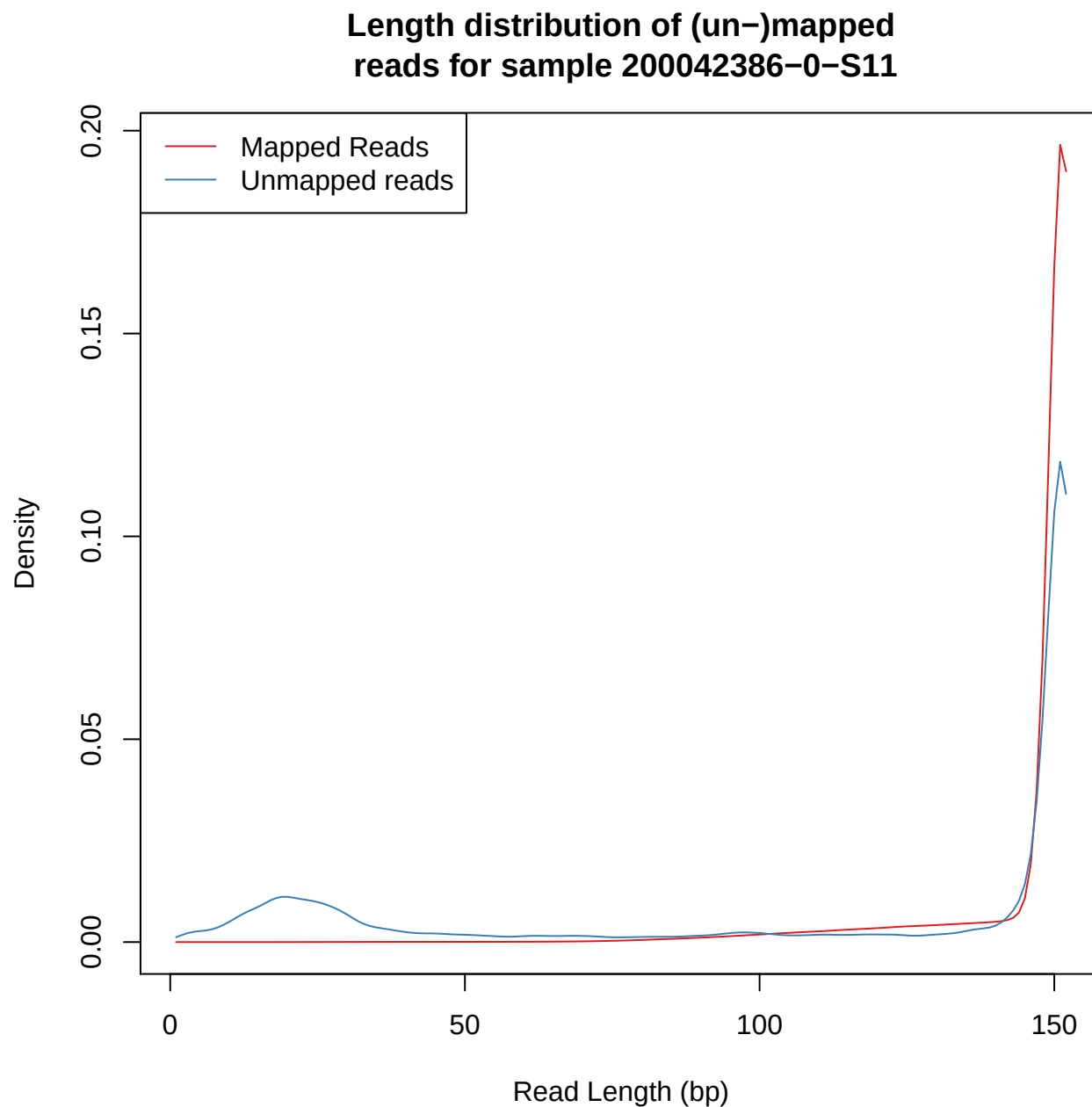


Figure 4.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

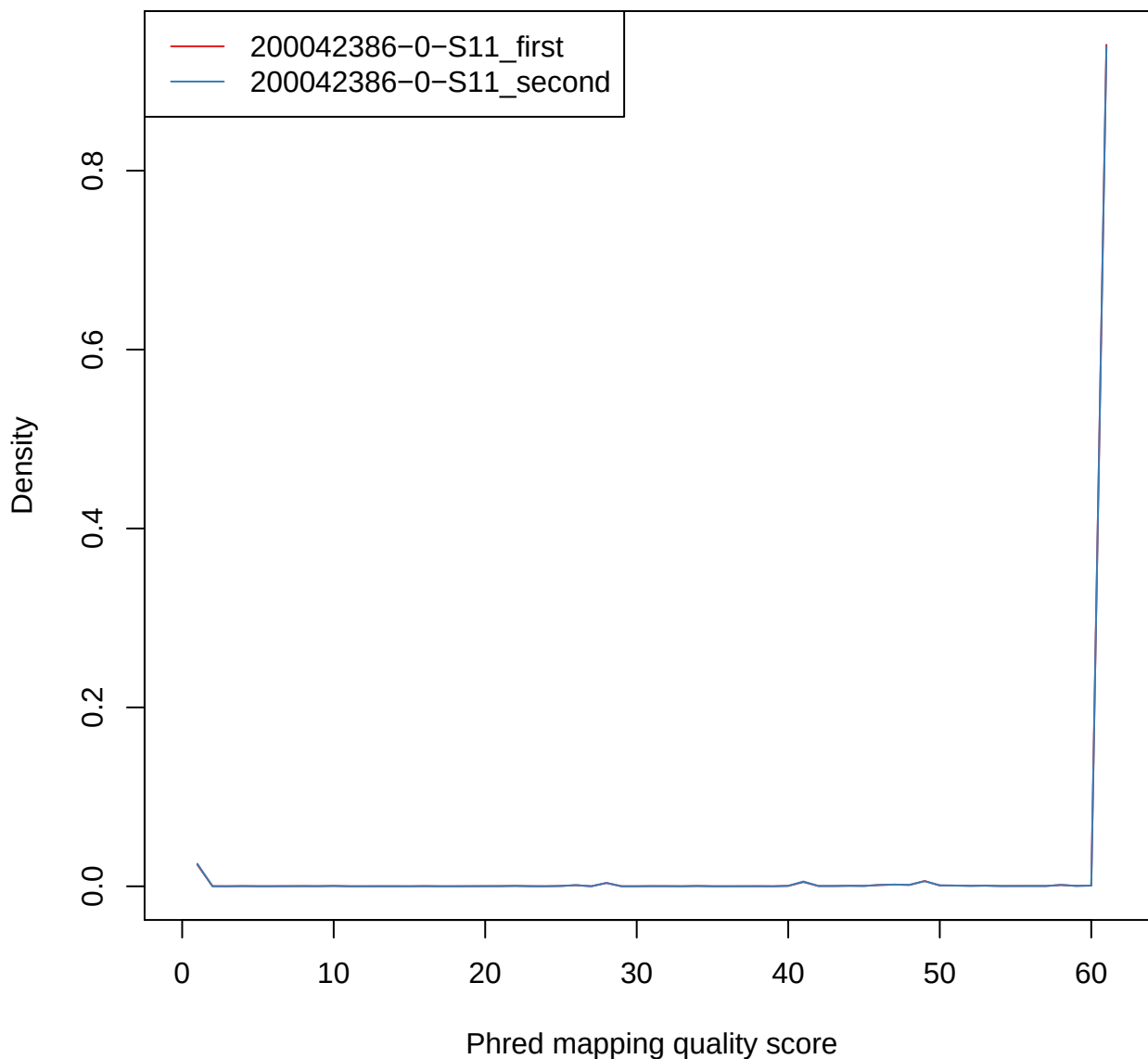


Figure 4.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

4.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042386-0-S11 (2,515,478 aligned reads)**

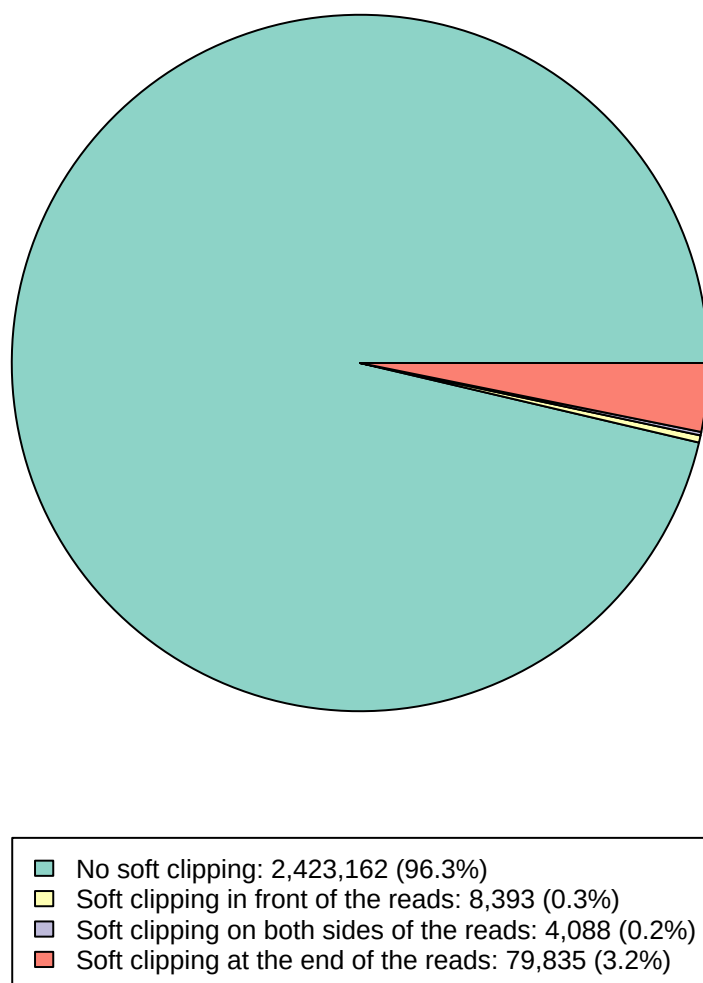


Figure 4.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042386-0-S11

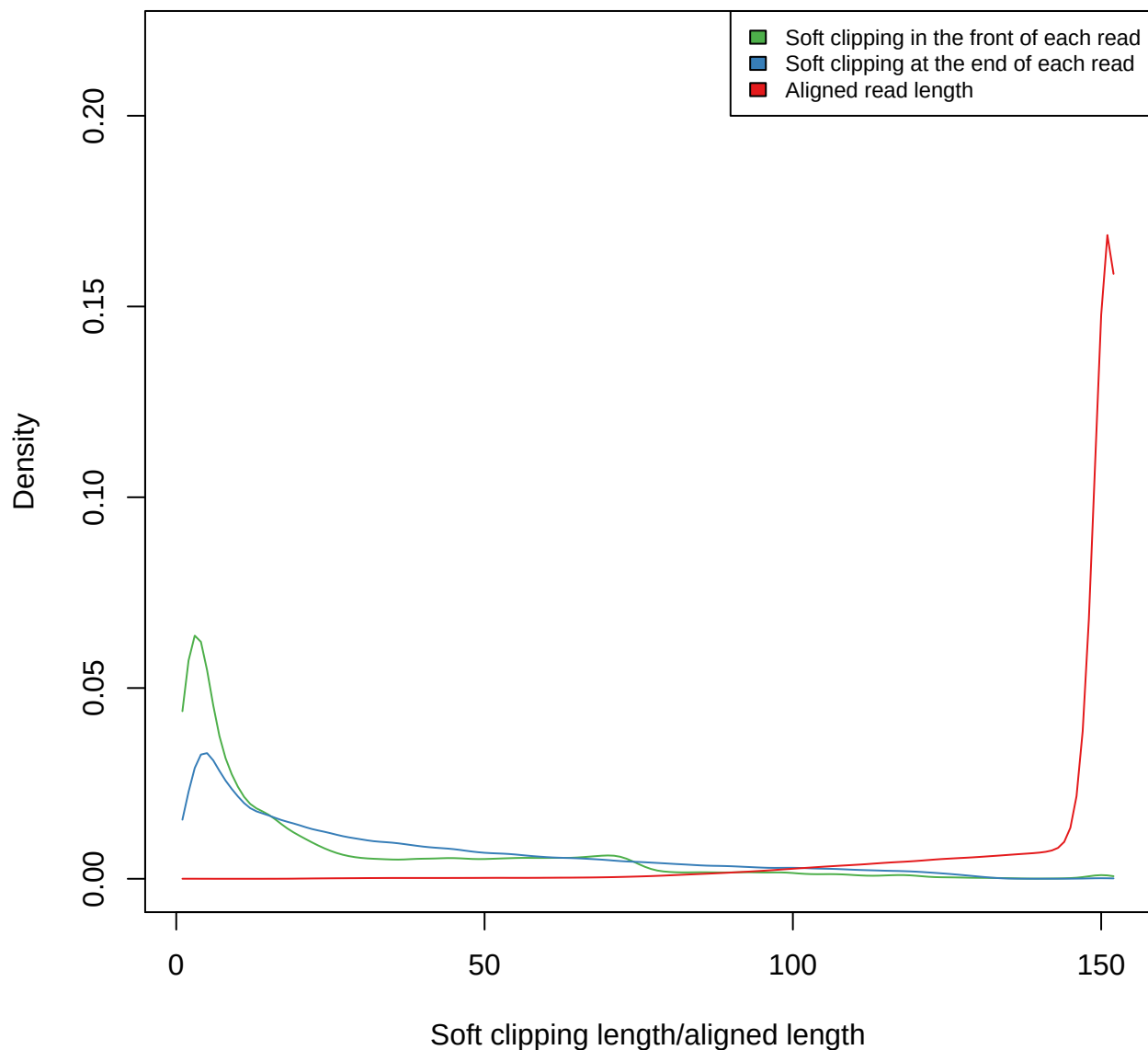


Figure 4.7: Density plots of soft clipping and aligned read length.

4.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 4.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042386-0-S11	74.51%	12.01%	6.90%	6.57%

Table 4.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042386-0-S11	5.3%	58957	3164	181	13	2	1	0

4.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 4.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042386-0-S11	1

Table 4.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	30	MSH2	NM_000251.3	8	c.1277-3297_1277-3268	36

Target region coverage distribution for sample 200042386-0-S11

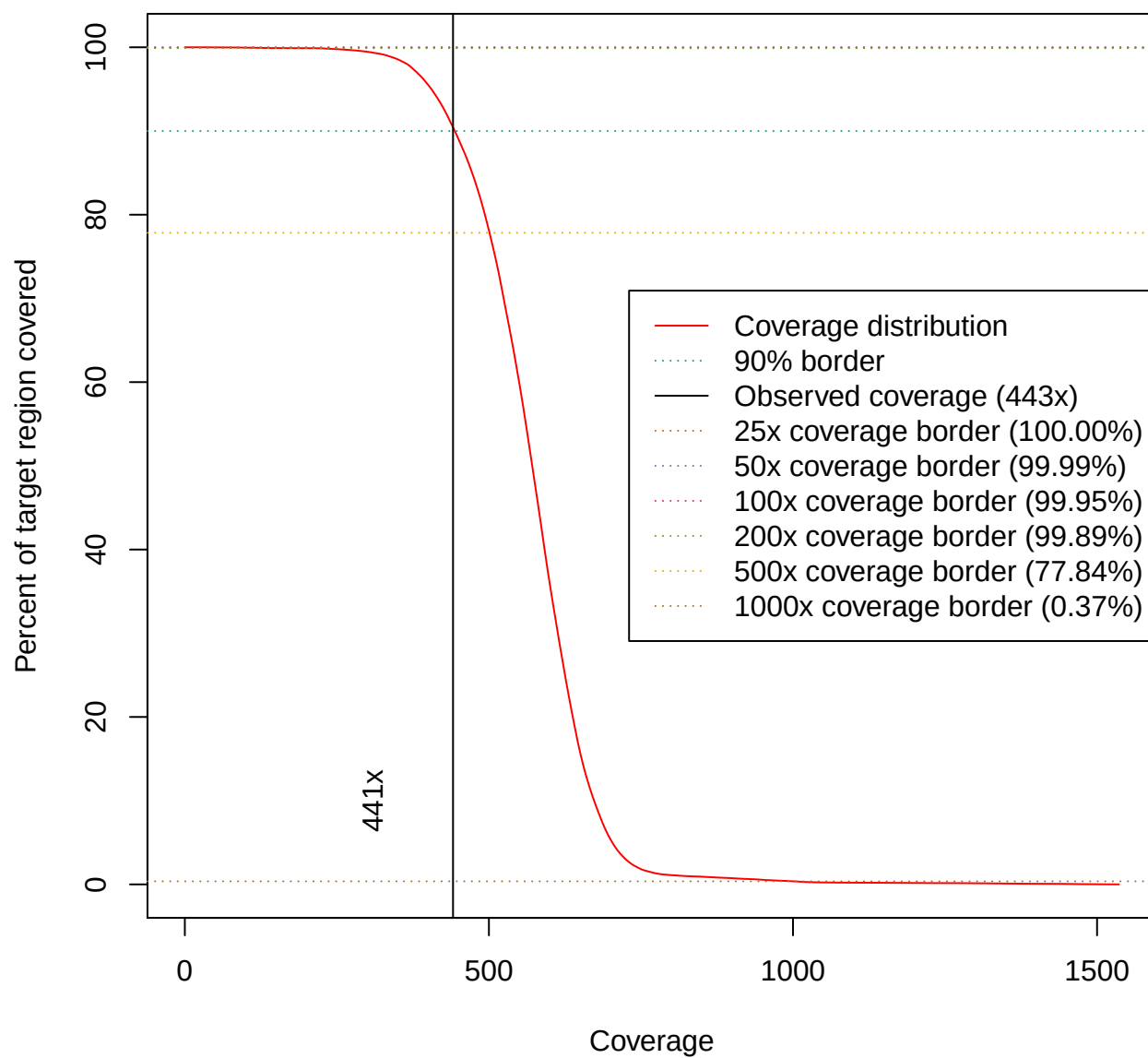


Figure 4.8: Target region coverage.

Table 4.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042386-0-S11	100.00%	99.99%	99.95%	99.89%	77.84%	0.37%	443×	0.05%

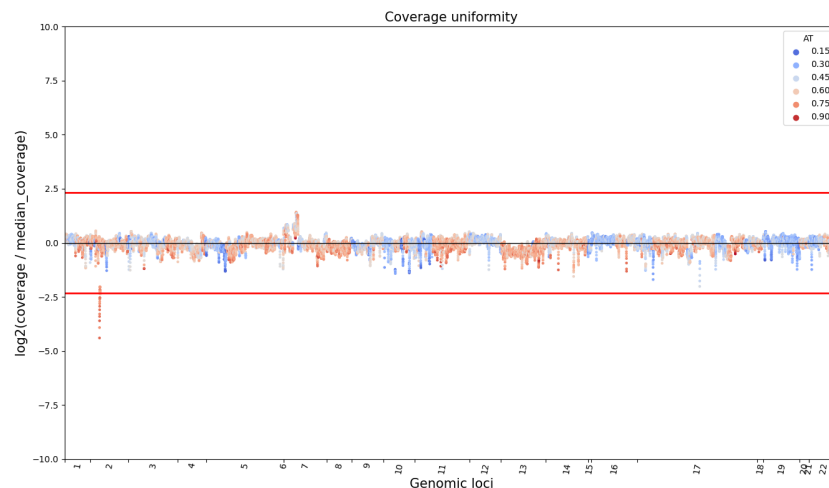


Figure 4.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.

Sample 200042387-0-S10

5.1 Dataset

For sample 200042387-0-S10 2,530,000 reads were sequenced, of these 2,509,702 were mapped corresponding to 99.2% of all reads.

5.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 5.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 5.2 shows the distribution of average base call quality, while Figure 5.3 shows the distribution of read lengths.

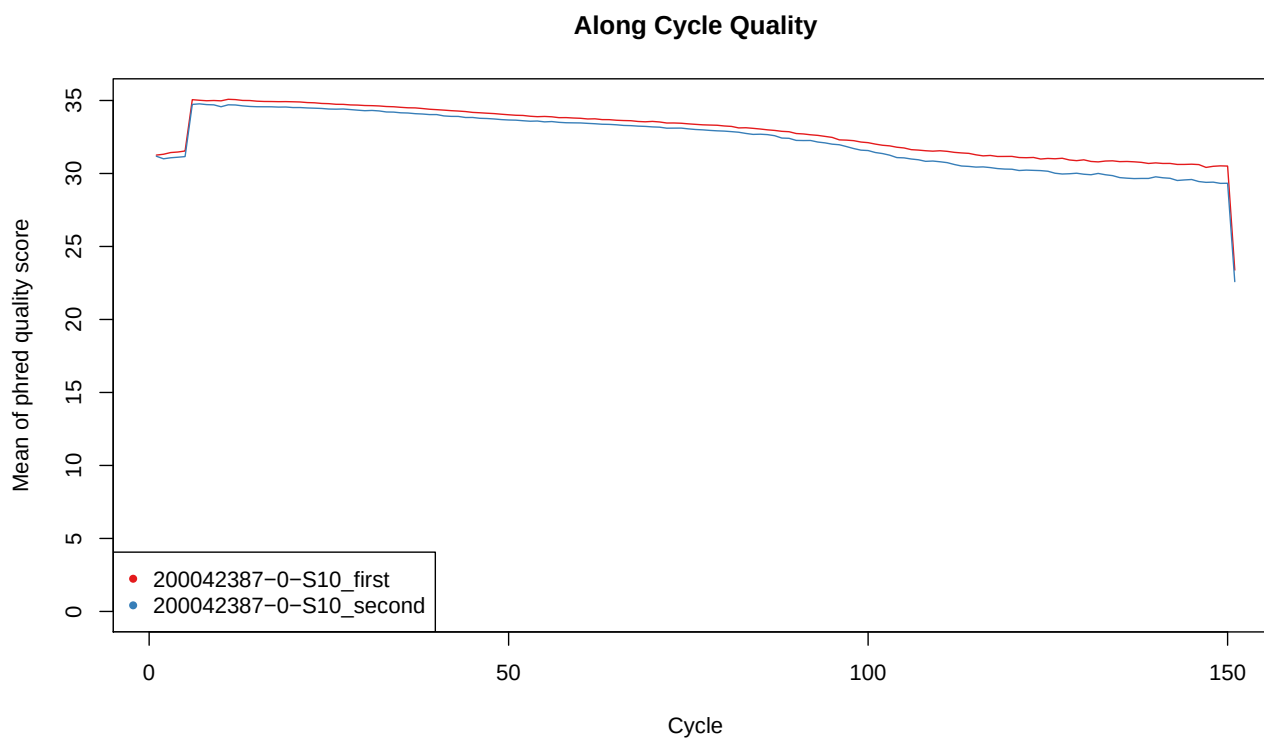


Figure 5.1: Mean base call quality by cycle.

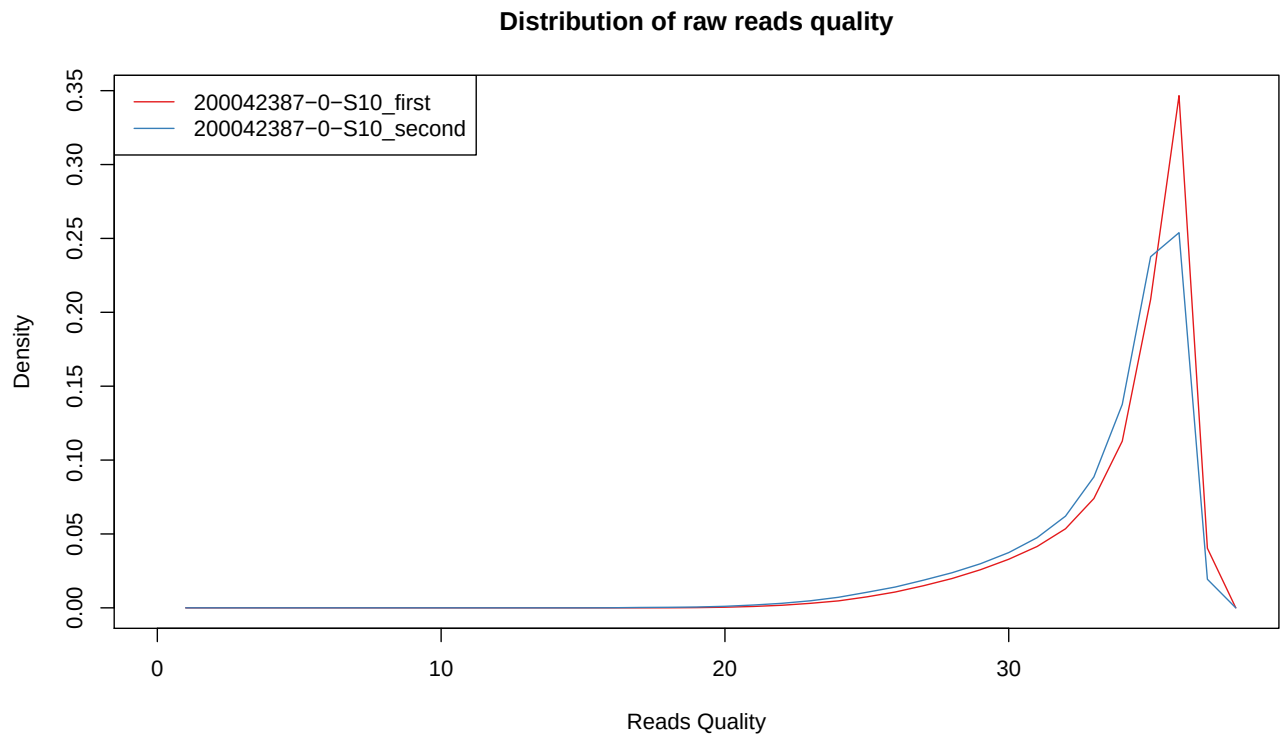


Figure 5.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

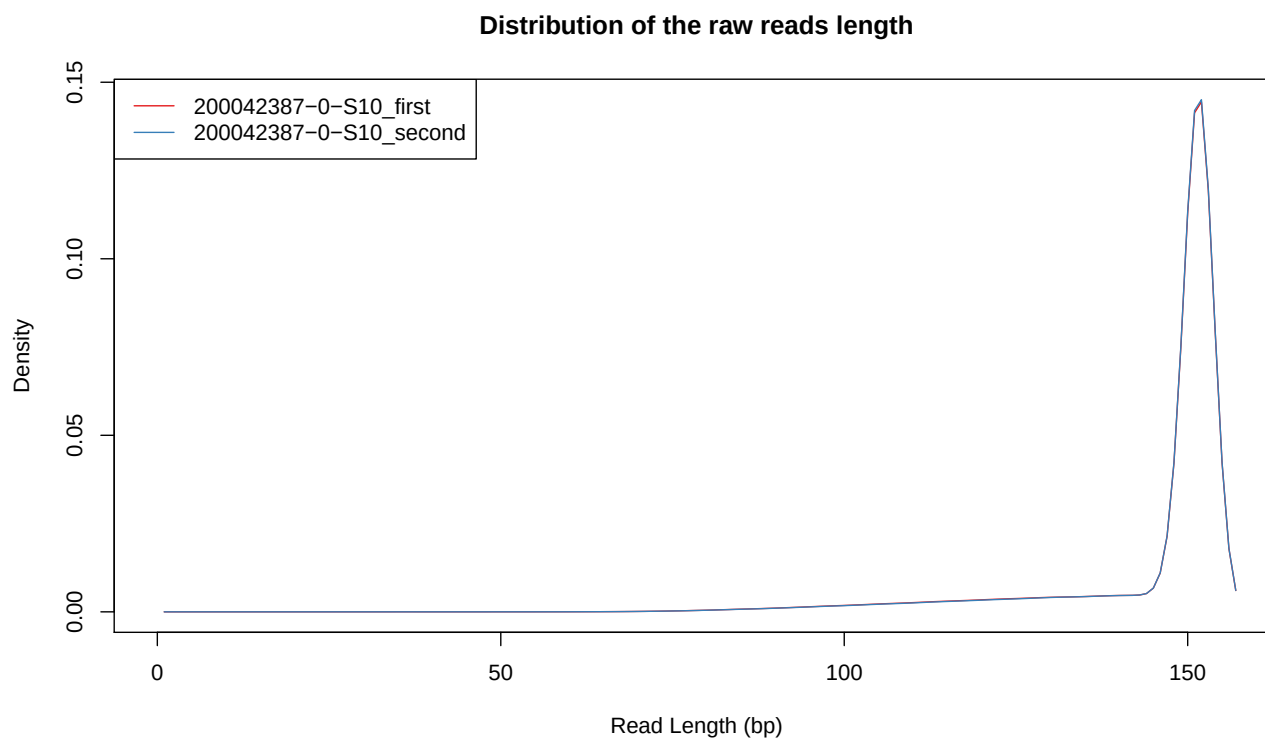


Figure 5.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

5.3 Alignment statistics

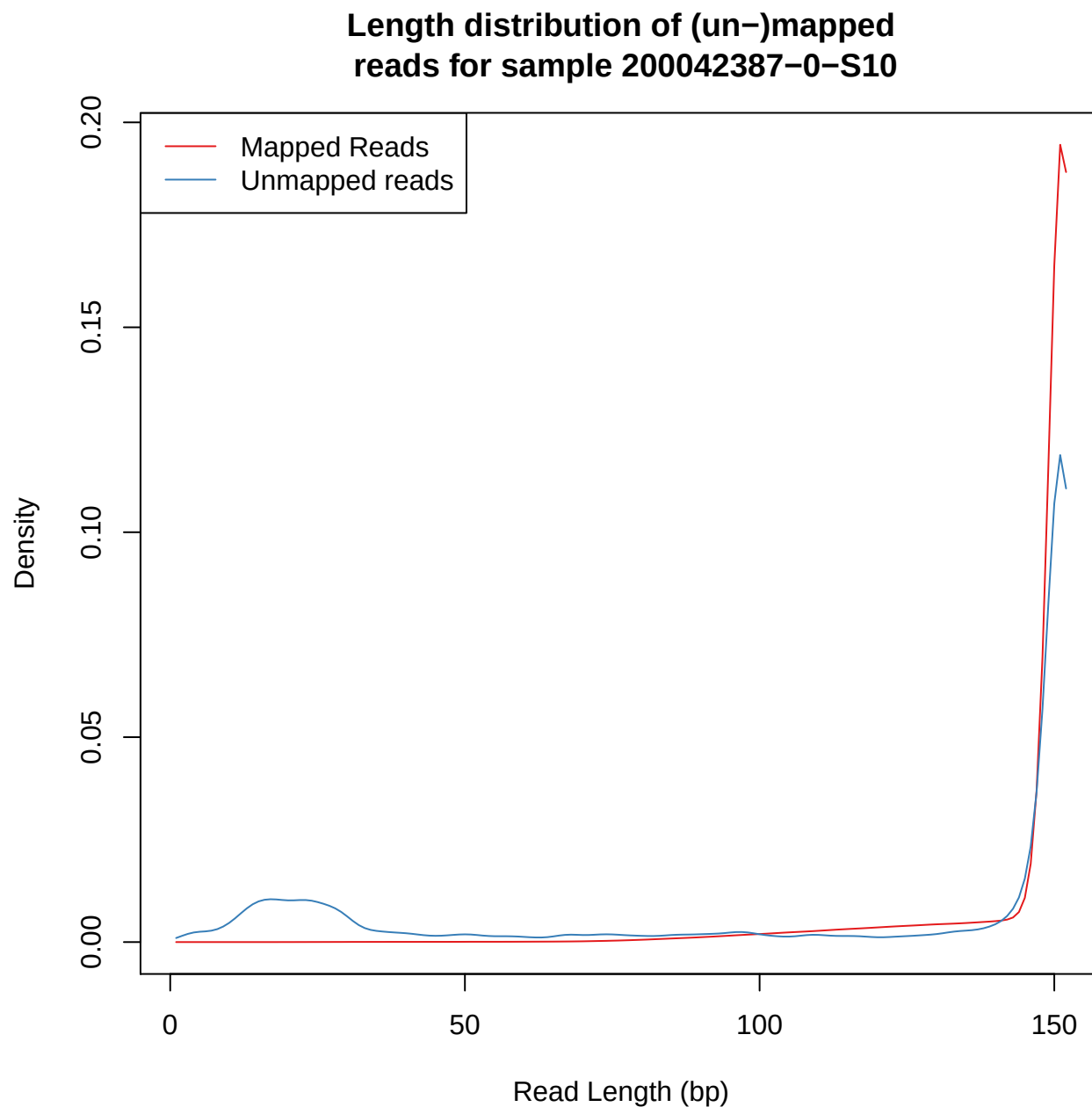


Figure 5.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

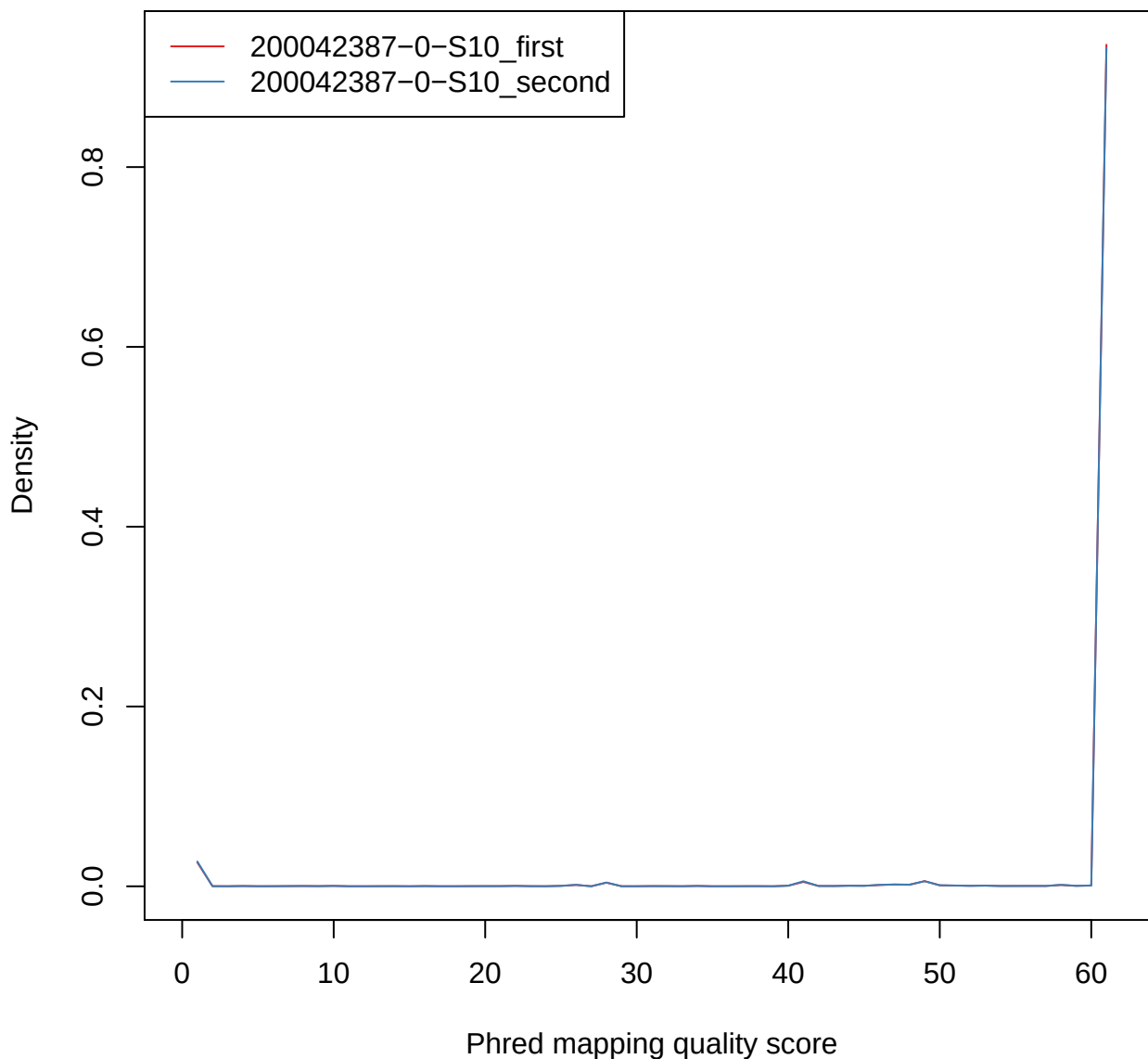


Figure 5.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

5.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042387-0-S10 (2,509,702 aligned reads)**

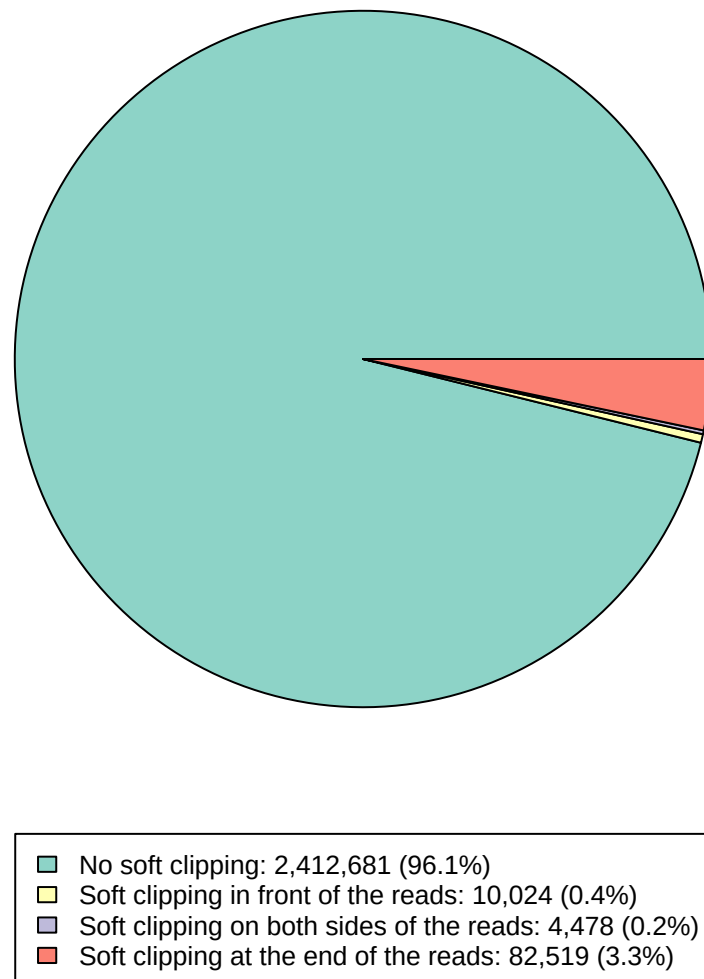


Figure 5.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042387-0-S10

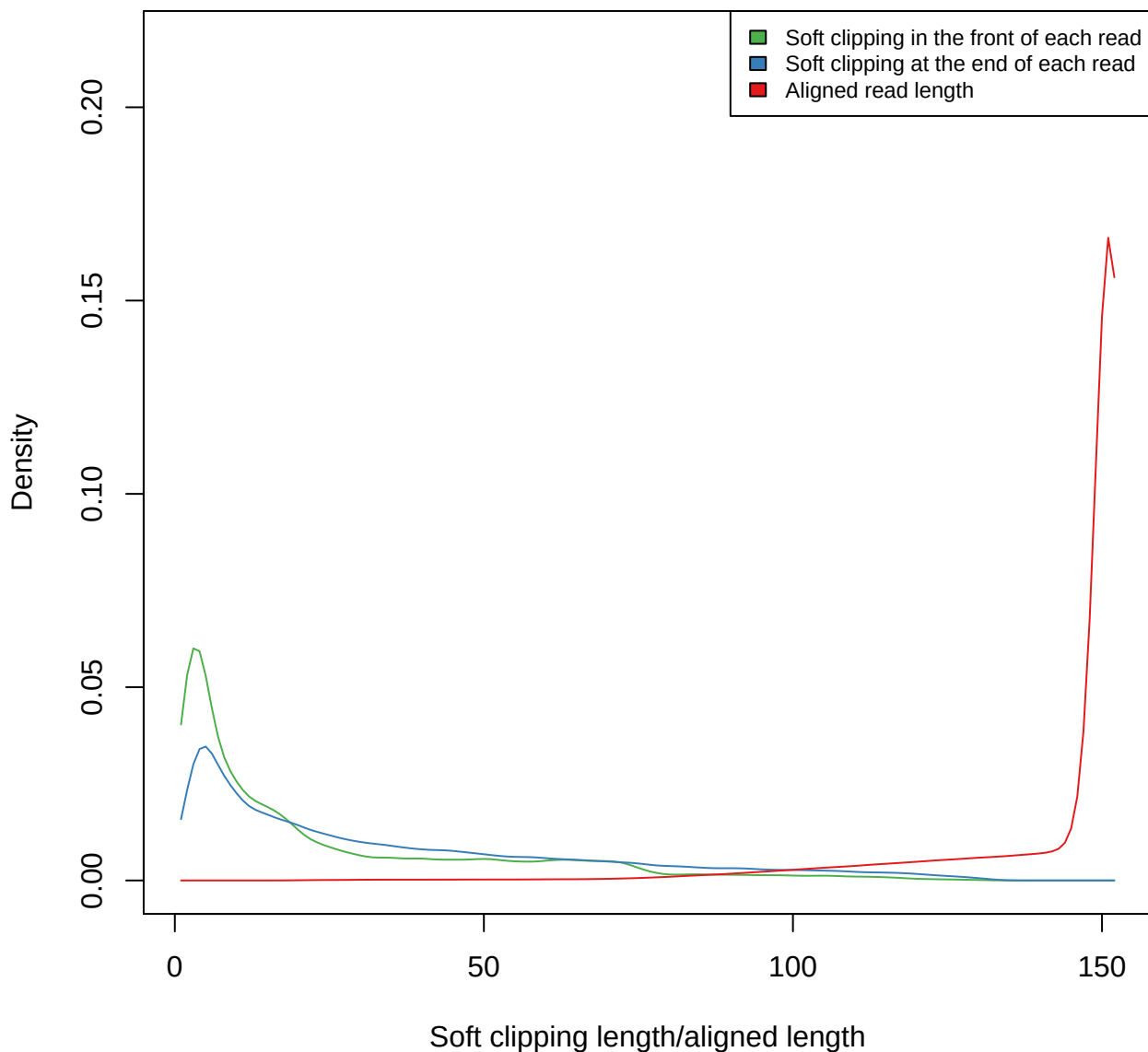


Figure 5.7: Density plots of soft clipping and aligned read length.

5.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 5.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042387-0-S10	73.96%	11.89%	7.42%	6.73%

Table 5.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042387-0-S10	5.42%	59933	3316	206	11	1	0	0

5.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 5.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042387-0-S10	1

Table 5.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	35	MSH2	NM_000251.3	8	c.1277-3297_1277-3263	37

Target region coverage distribution for sample 200042387-0-S10

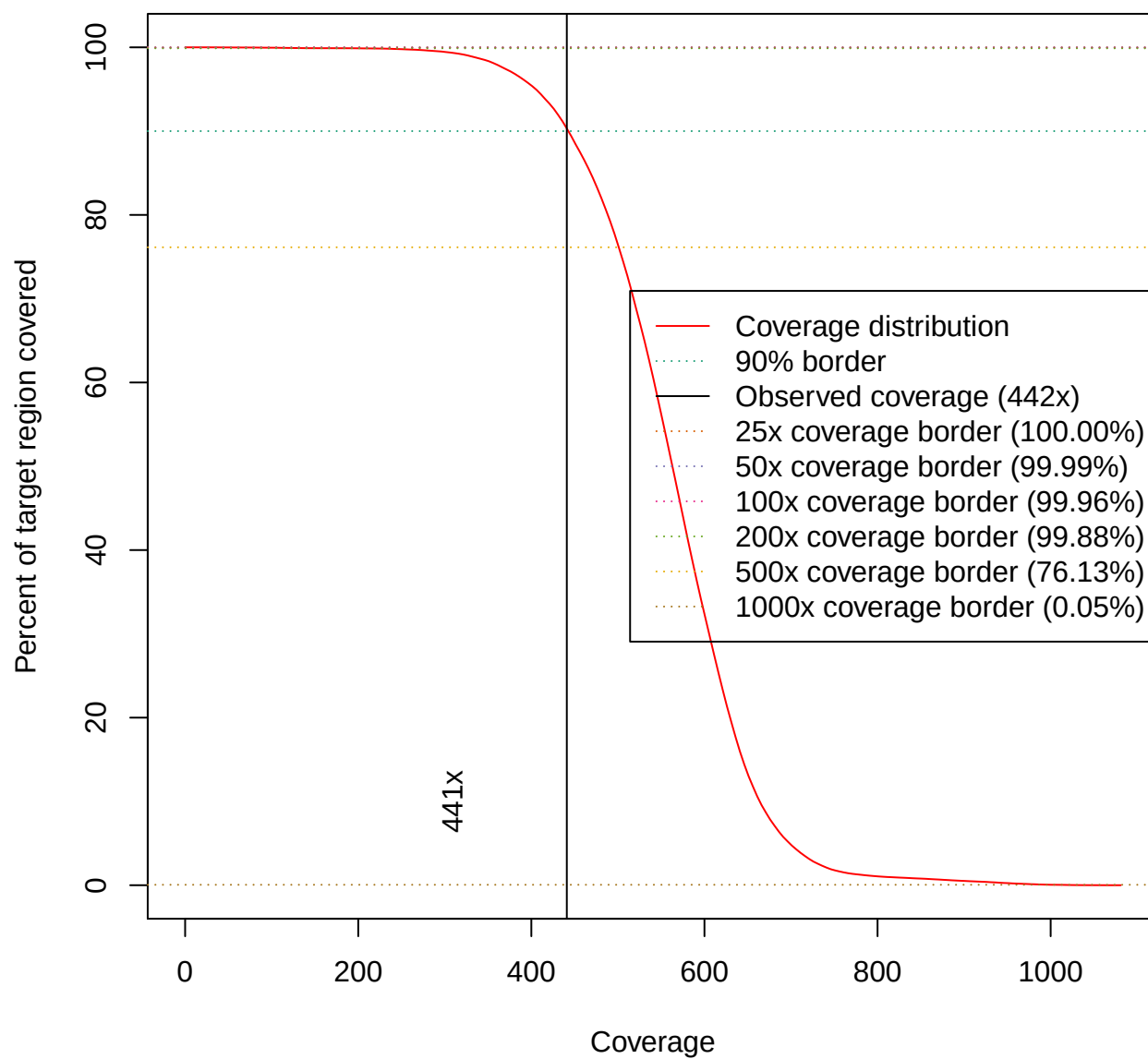


Figure 5.8: Target region coverage.

Table 5.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042387-0-S10	100.00%	99.99%	99.96%	99.88%	76.13%	0.05%	442×	0.06%

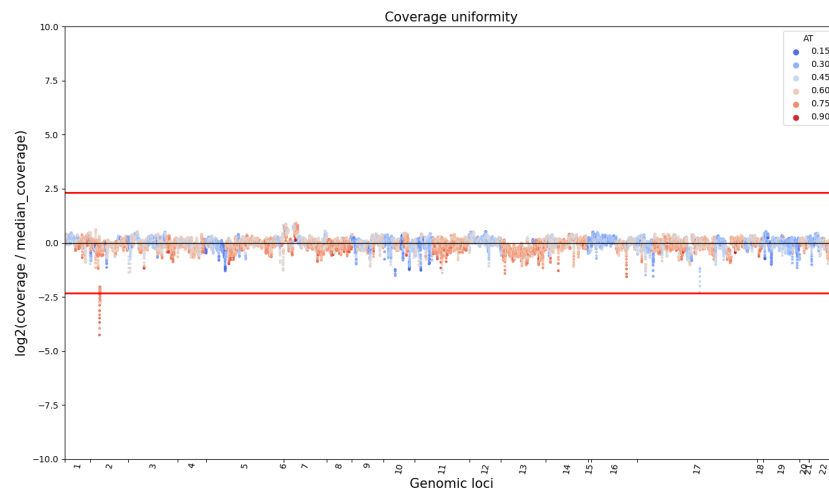


Figure 5.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.

Sample 200042388-0-S5

6.1 Dataset

For sample 200042388-0-S5 2,530,000 reads were sequenced, of these 2,513,845 were mapped corresponding to 99.36% of all reads.

6.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 6.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 6.2 shows the distribution of average base call quality, while Figure 6.3 shows the distribution of read lengths.

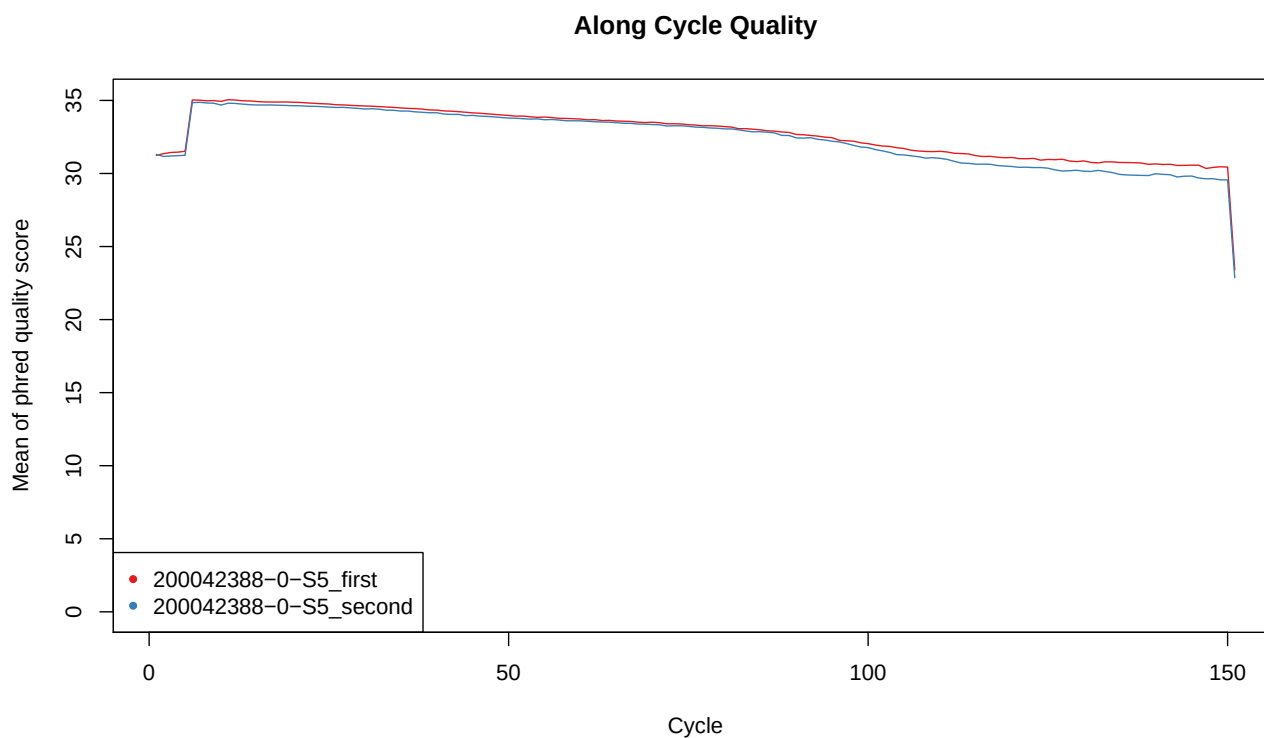


Figure 6.1: Mean base call quality by cycle.

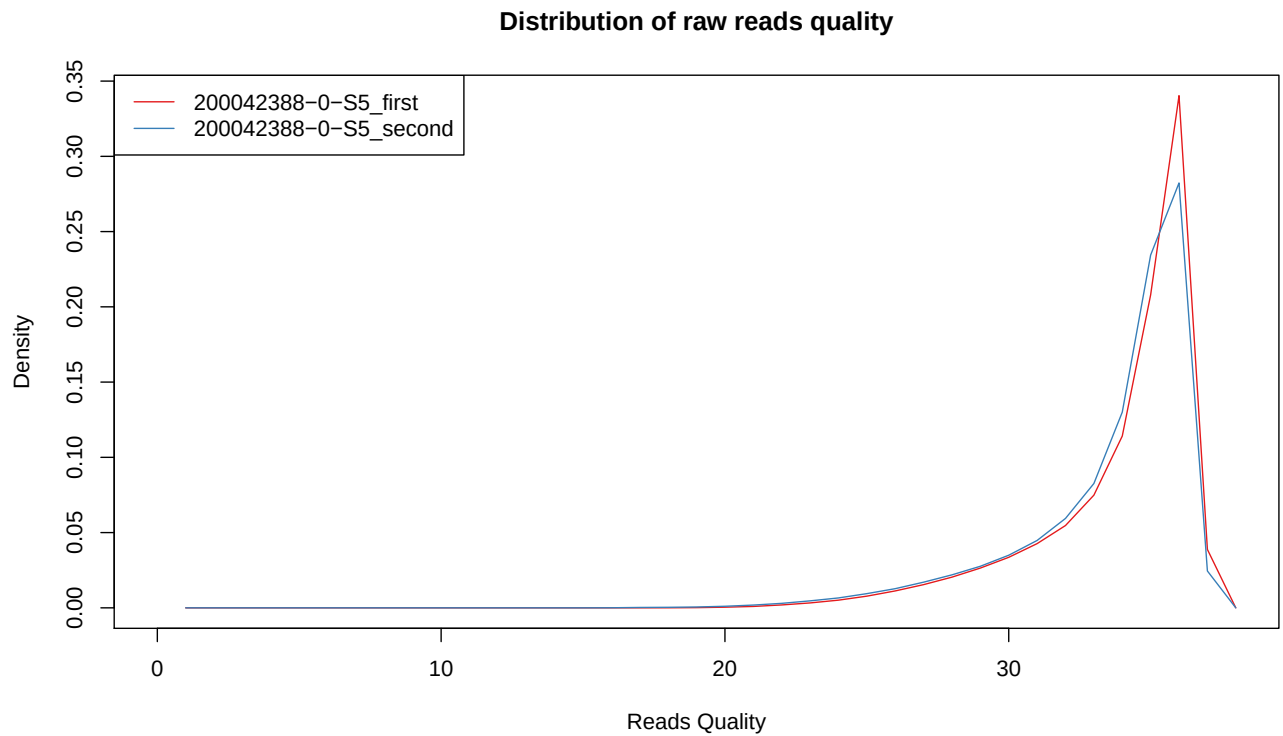


Figure 6.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

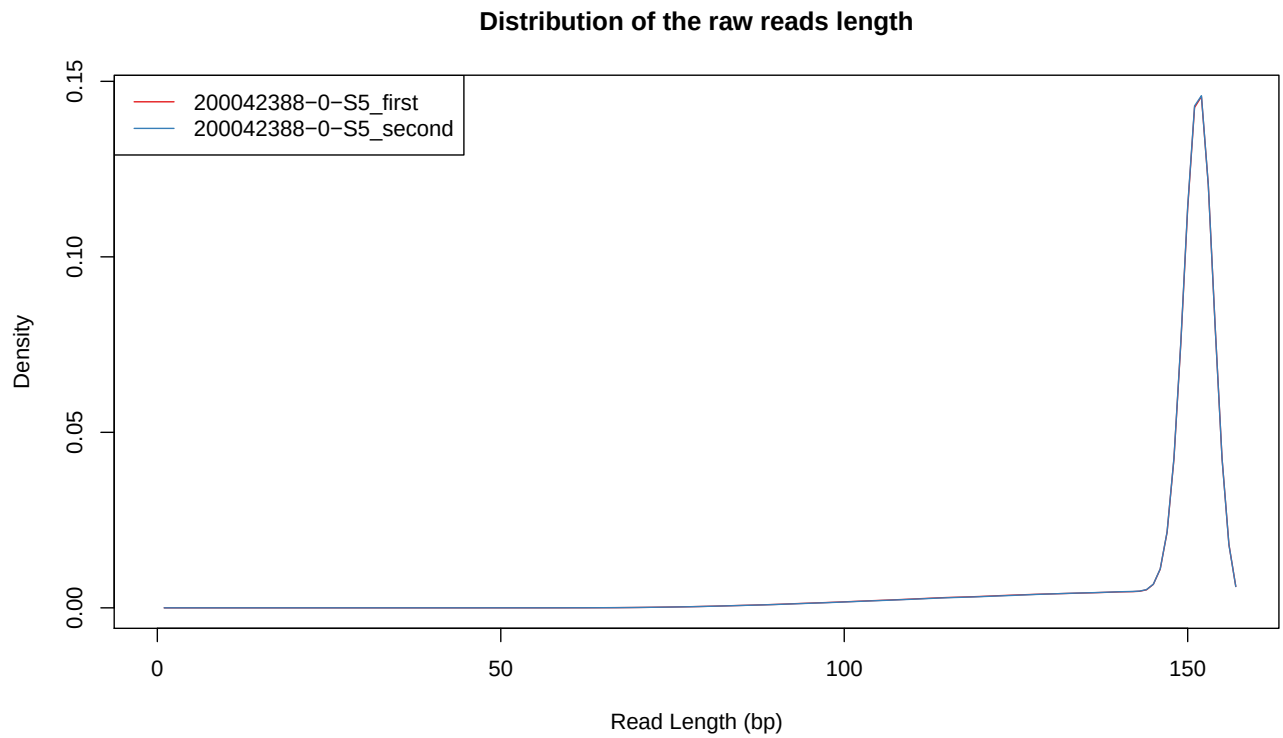


Figure 6.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

6.3 Alignment statistics

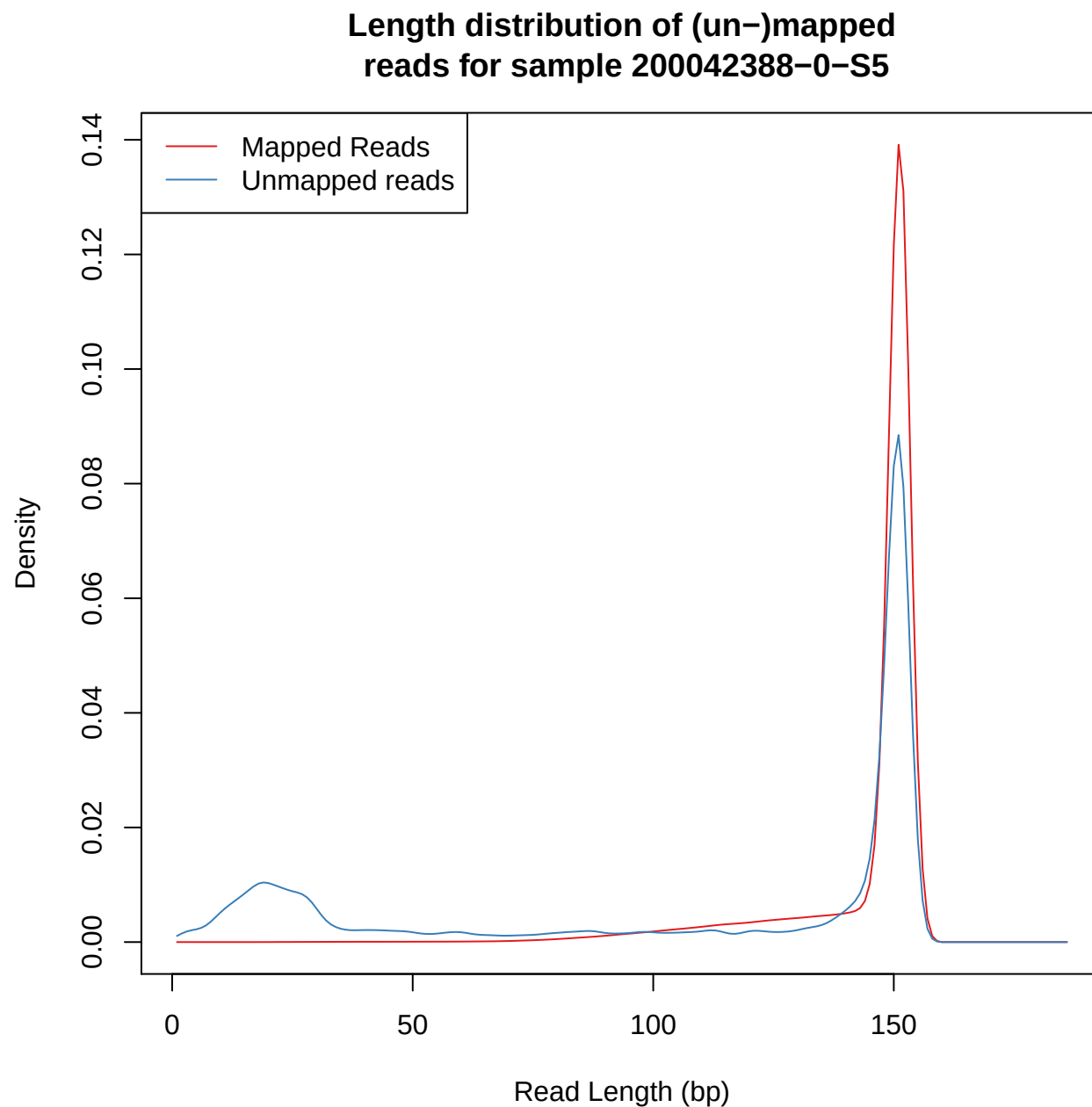


Figure 6.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

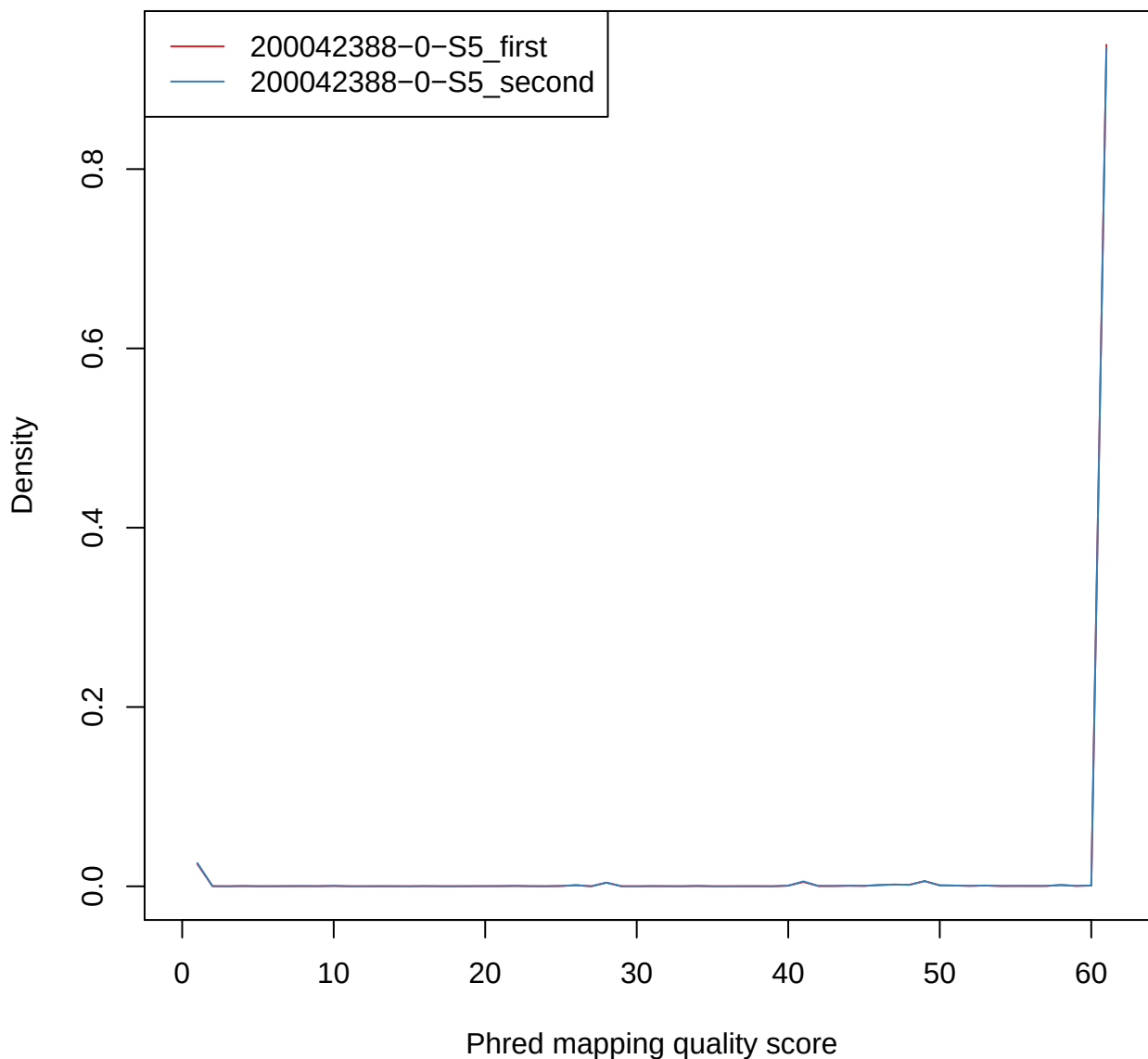


Figure 6.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

6.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042388-0-S5 (2,513,845 aligned reads)**

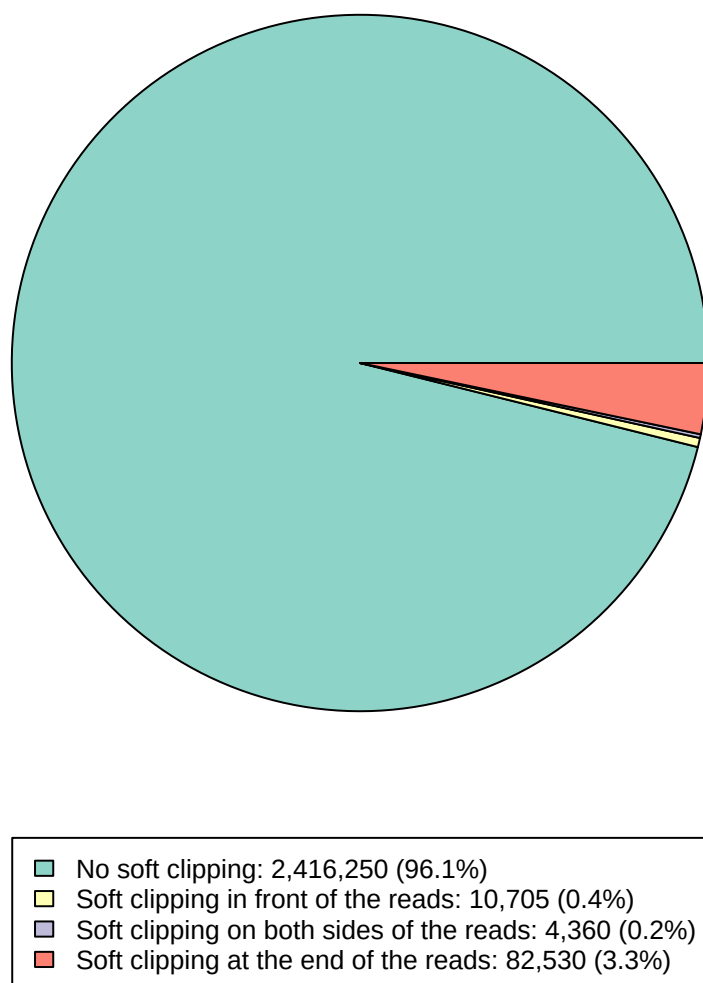


Figure 6.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042388-0-S5

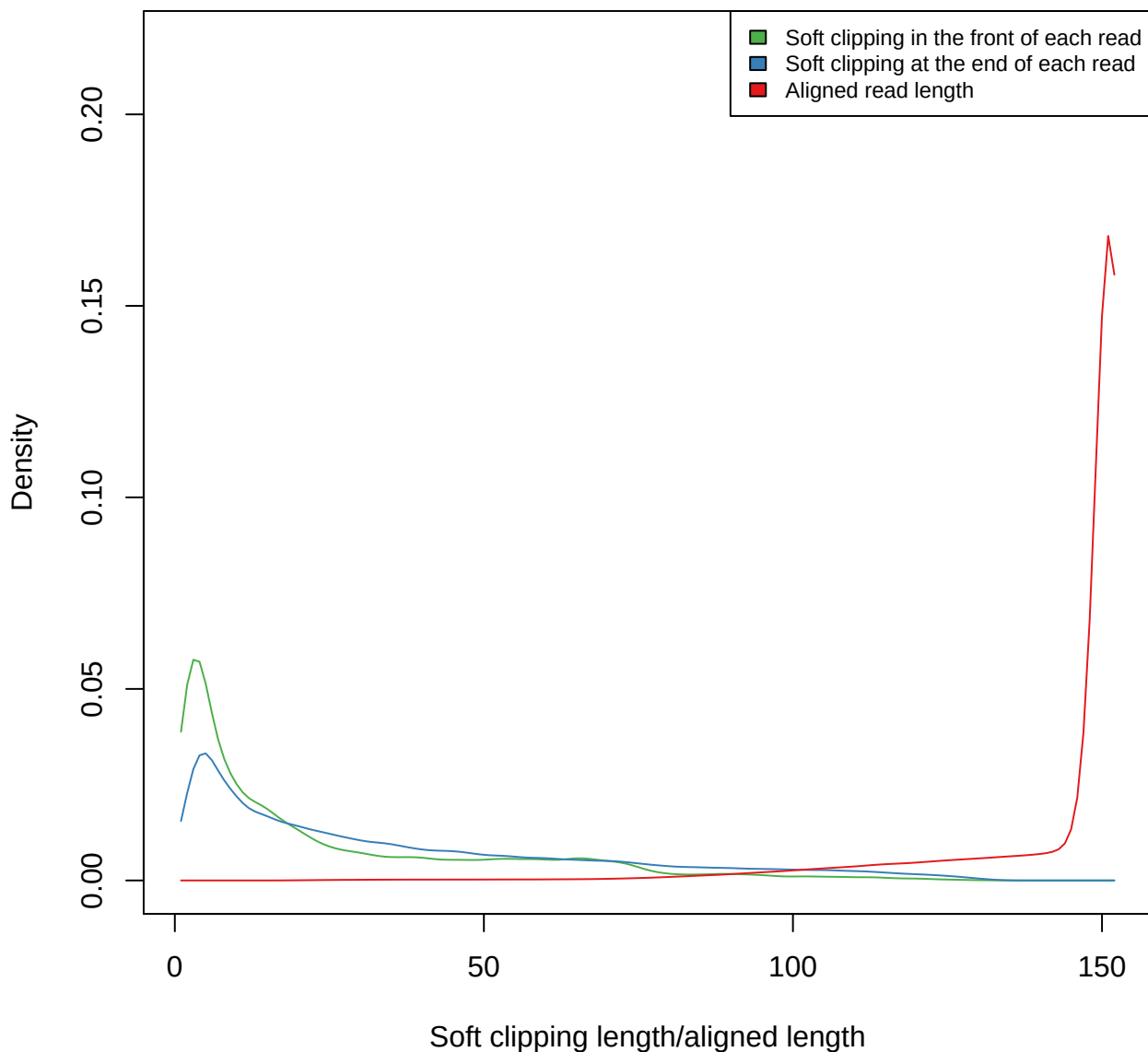


Figure 6.7: Density plots of soft clipping and aligned read length.

6.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 6.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042388-0-S5	74.41%	11.95%	6.98%	6.66%

Table 6.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042388-0-S5	5.23%	57996	3134	192	15	0	0	0

6.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 6.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042388-0-S5	1

Table 6.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	37	MSH2	NM_000251.3	8	c.1277-3297_1277-3261	43

Target region coverage distribution for sample 200042388-0-S5

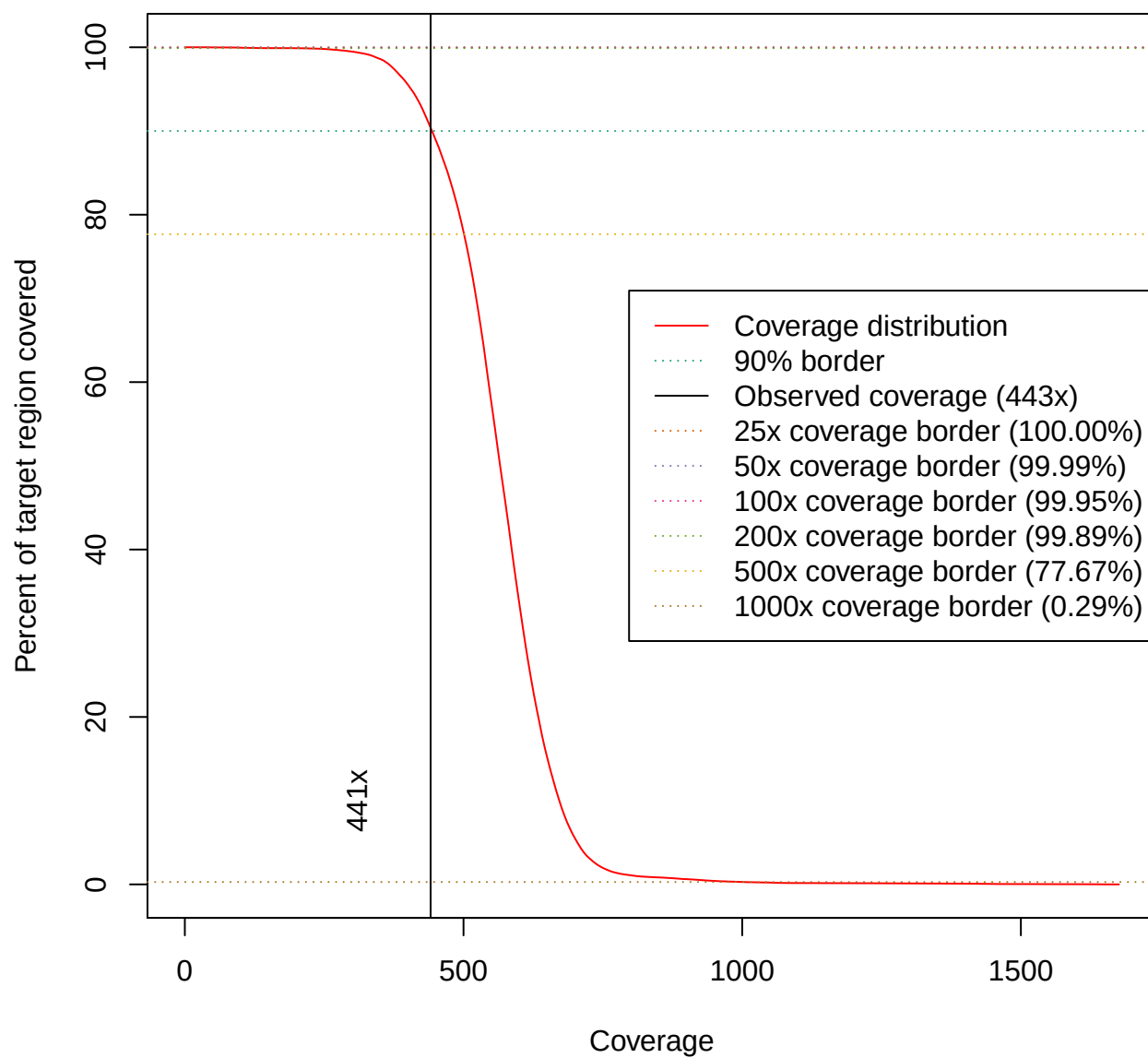


Figure 6.8: Target region coverage.

Table 6.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042388-0-S5	100.00%	99.99%	99.95%	99.89%	77.67%	0.29%	443×	0.07%

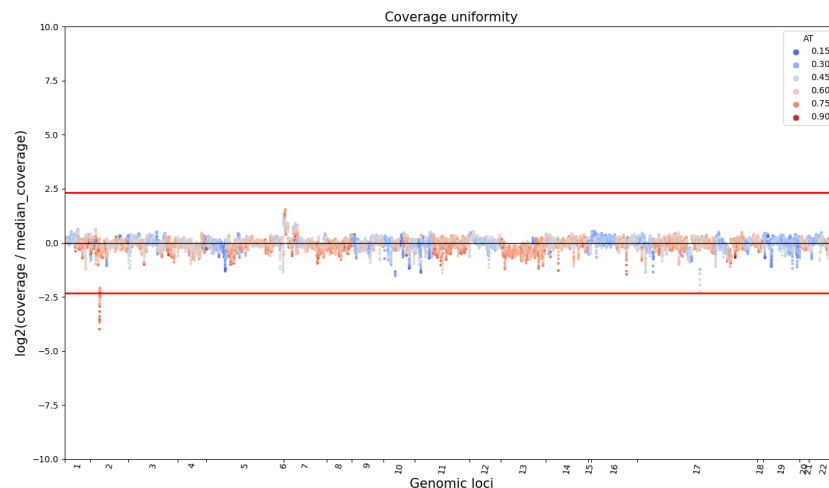


Figure 6.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.

Sample 200042389-0-S1

7.1 Dataset

For sample 200042389-0-S1 2,530,000 reads were sequenced, of these 2,511,066 were mapped corresponding to 99.25% of all reads.

7.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 7.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 7.2 shows the distribution of average base call quality, while Figure 7.3 shows the distribution of read lengths.

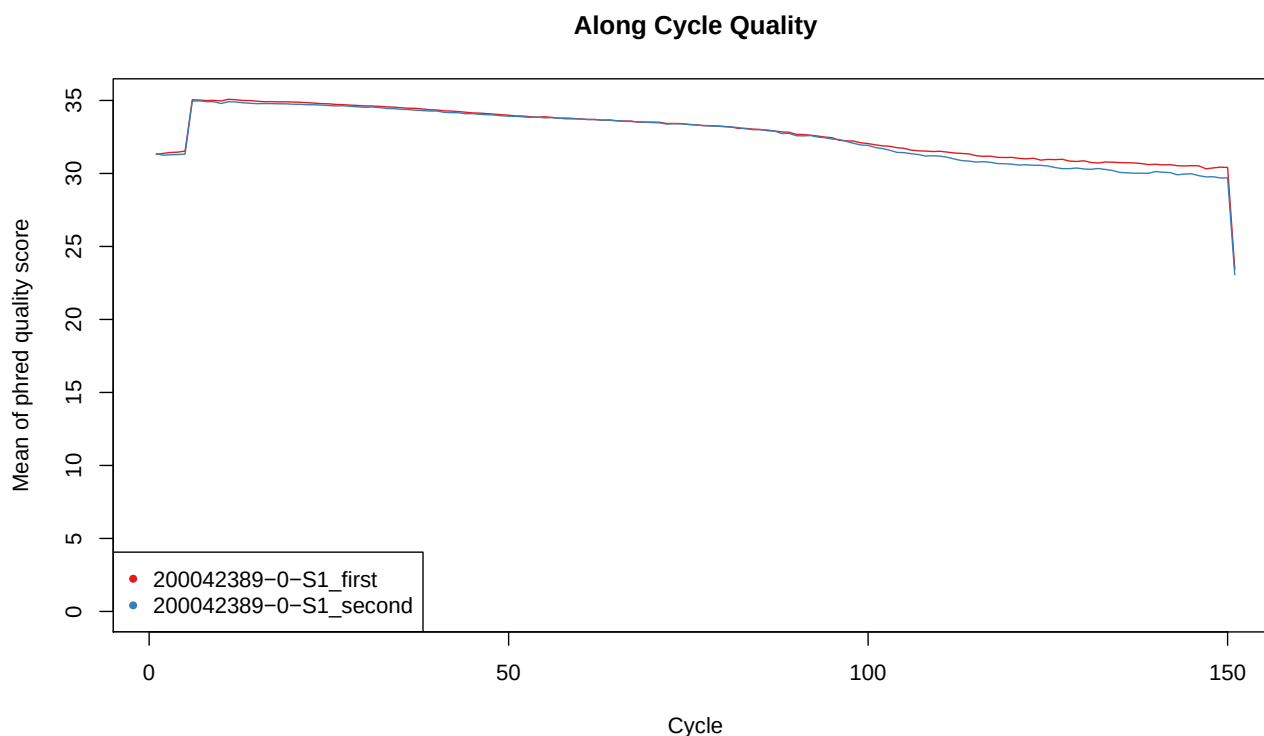


Figure 7.1: Mean base call quality by cycle.

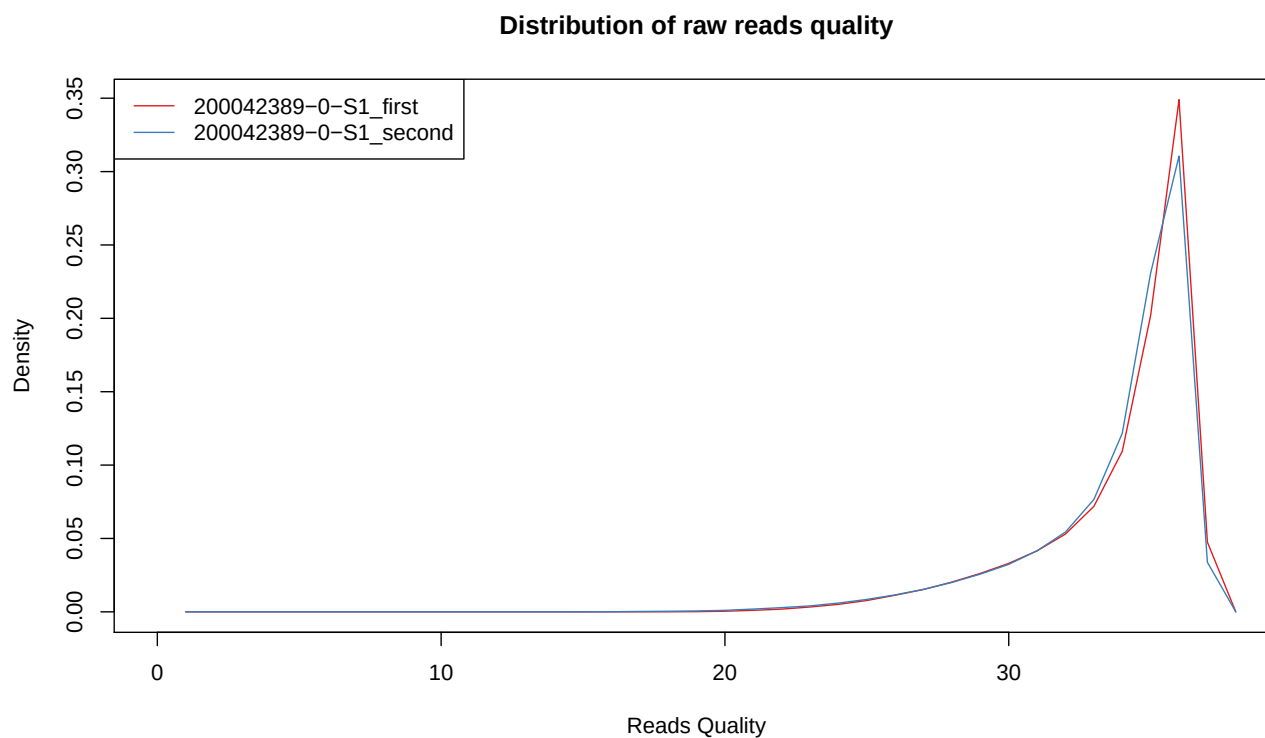


Figure 7.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

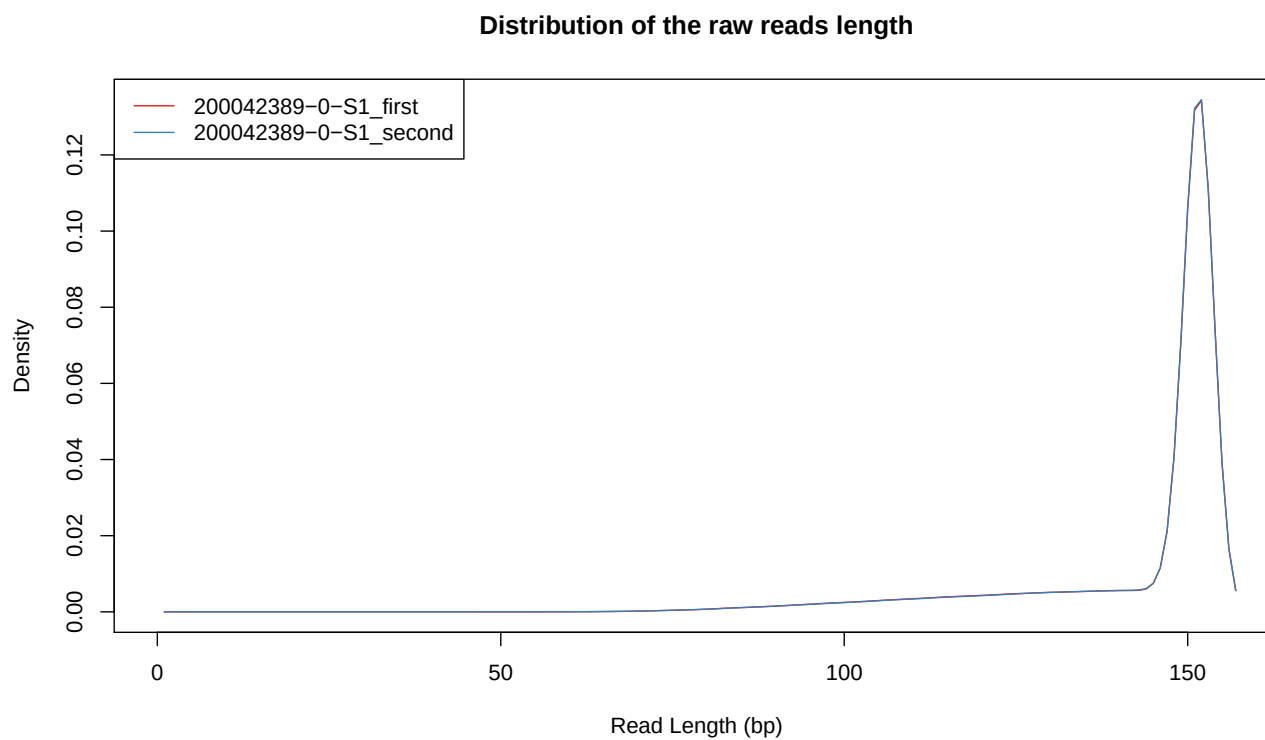


Figure 7.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

7.3 Alignment statistics

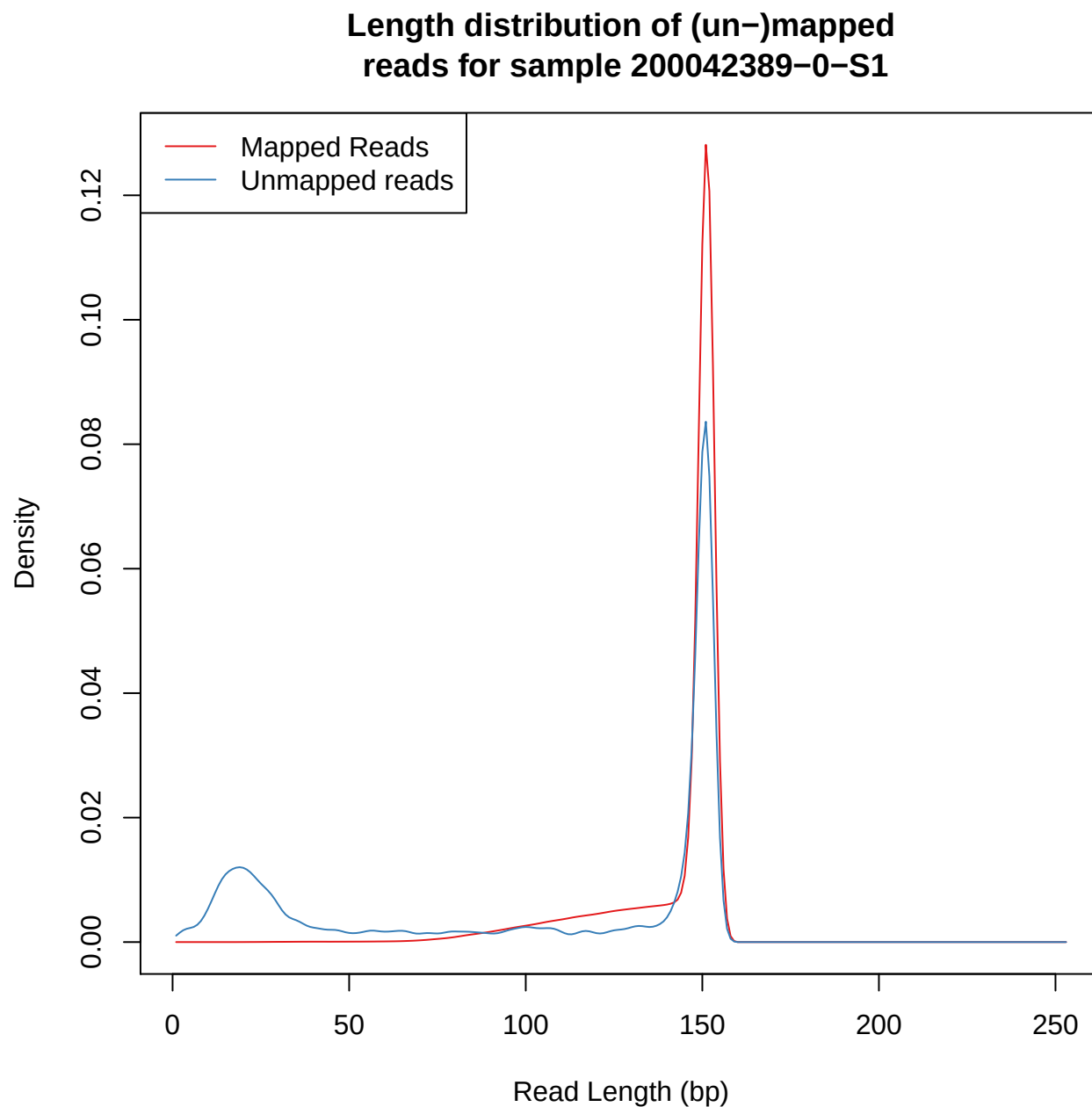


Figure 7.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

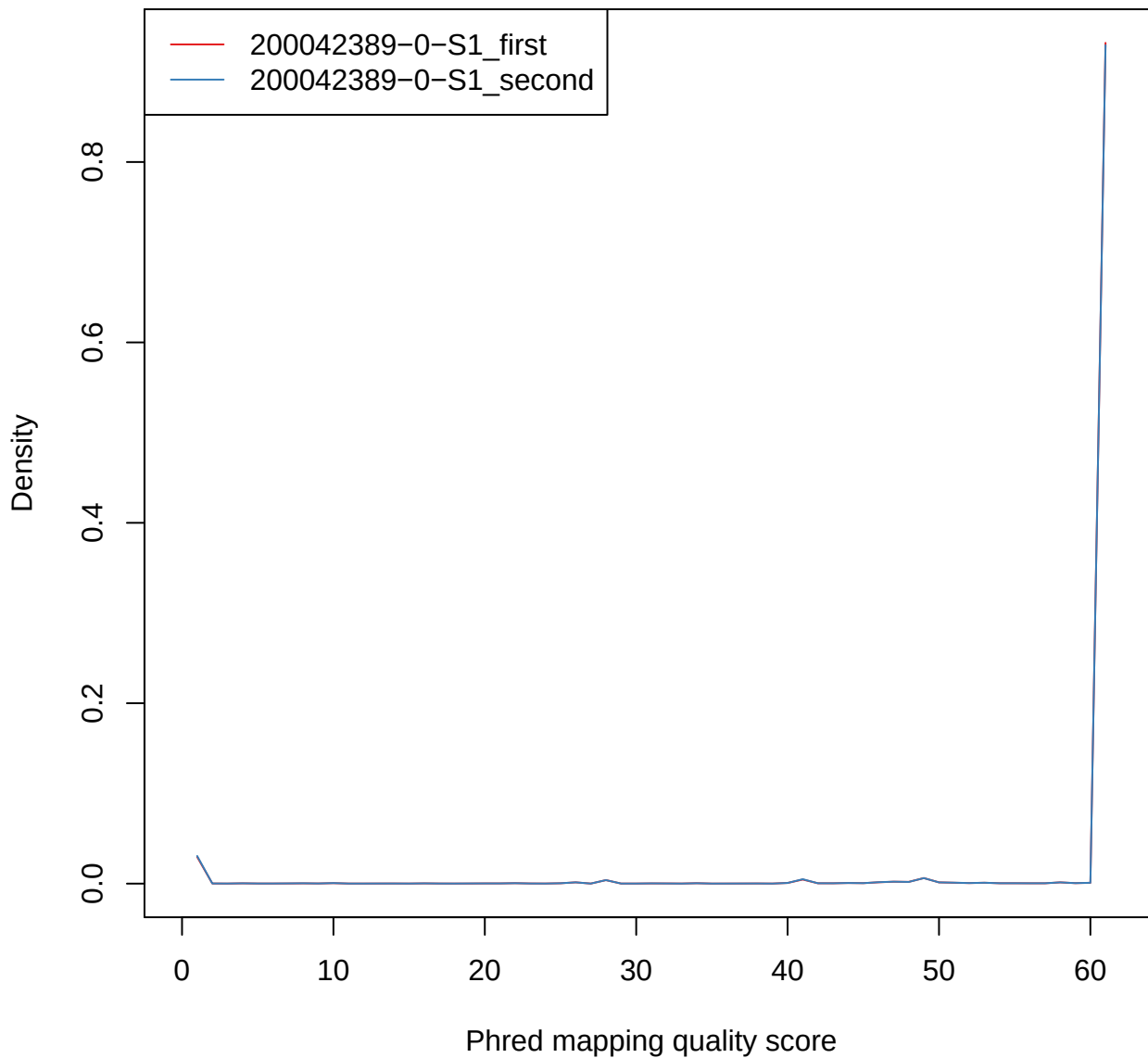


Figure 7.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

7.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042389-0-S1 (2,511,066 aligned reads)**

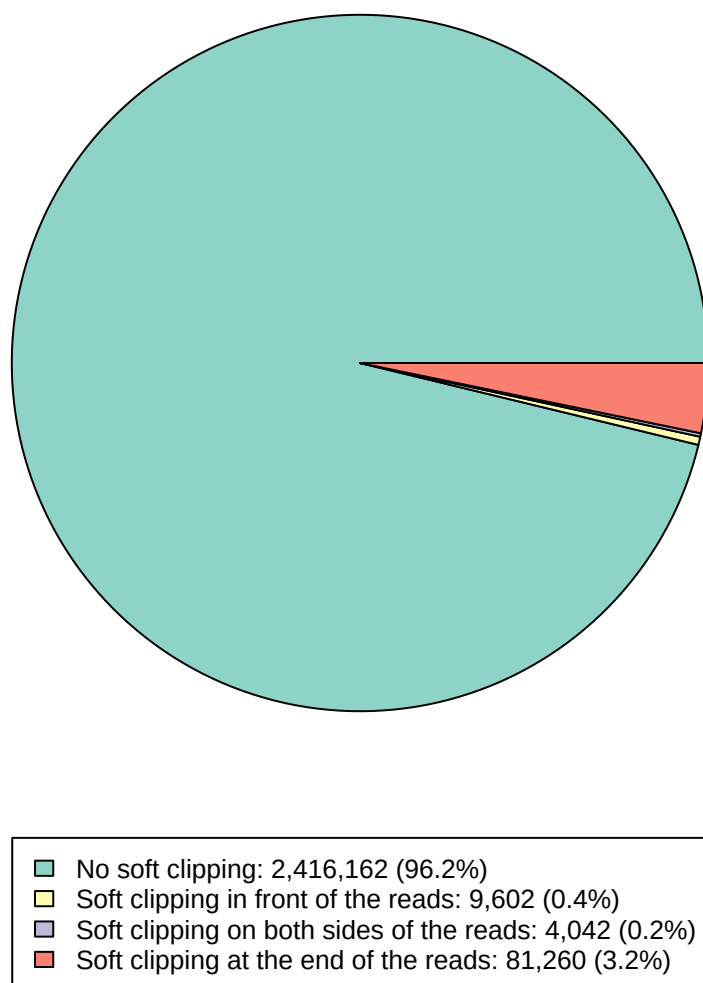


Figure 7.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042389-0-S1

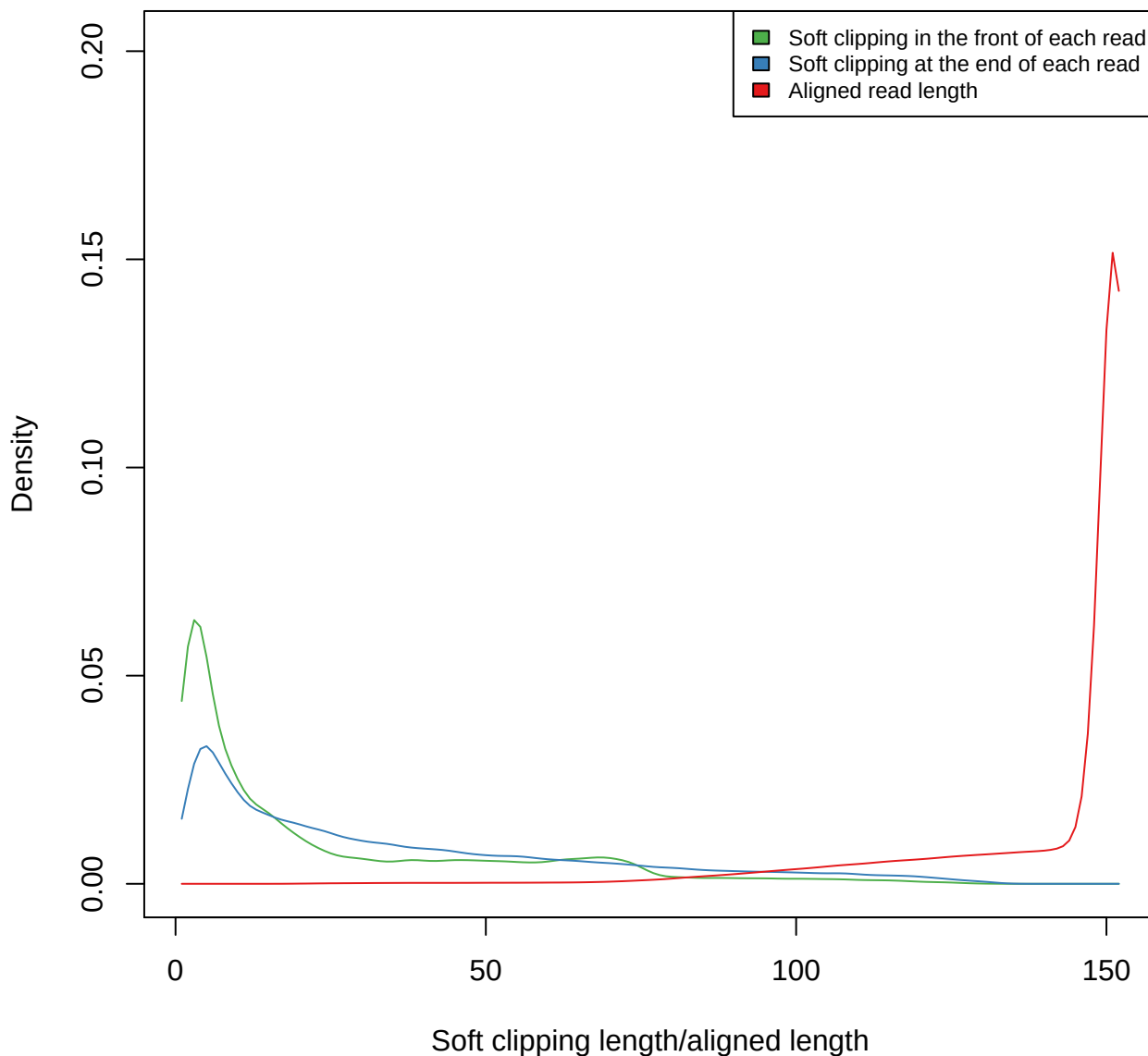


Figure 7.7: Density plots of soft clipping and aligned read length.

7.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 7.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042389-0-S1	75.36%	10.34%	7.32%	6.97%

Table 7.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042389-0-S1	5.23%	57872	3155	204	11	2	0	0

7.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 7.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042389-0-S1	2

Table 7.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	39	MSH2	NM_000251.3	8	c.1277-3297_1277-3259	40
2	47442513	2	MSH2	NM_000251.3	8	c.1277-3035_1277-3034	49

Target region coverage distribution for sample 200042389-0-S1

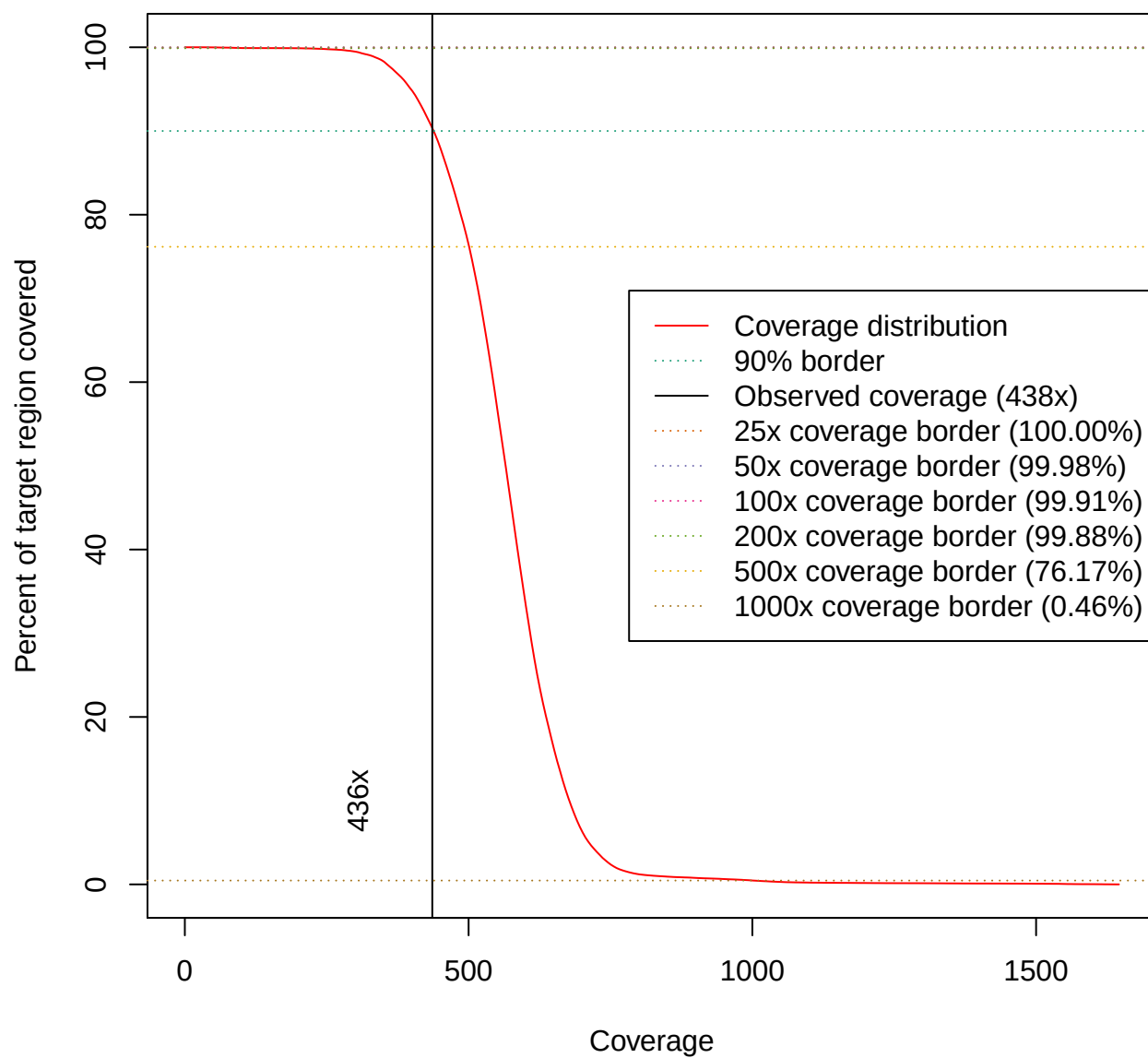


Figure 7.8: Target region coverage.

Table 7.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042389-0-S1	100.00%	99.98%	99.91%	99.88%	76.17%	0.46%	438×	0.09%

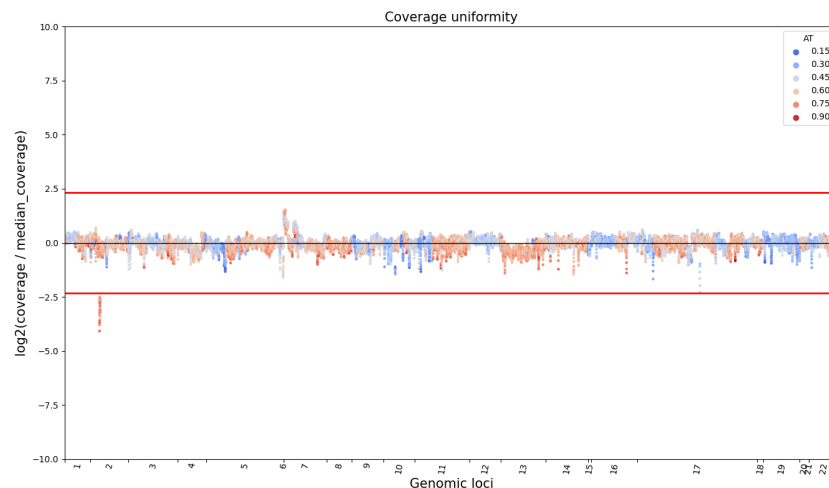


Figure 7.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.

Sample 200042390-0-S2

8.1 Dataset

For sample 200042390-0-S2 2,530,000 reads were sequenced, of these 2,509,406 were mapped corresponding to 99.19% of all reads.

8.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 8.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 8.2 shows the distribution of average base call quality, while Figure 8.3 shows the distribution of read lengths.

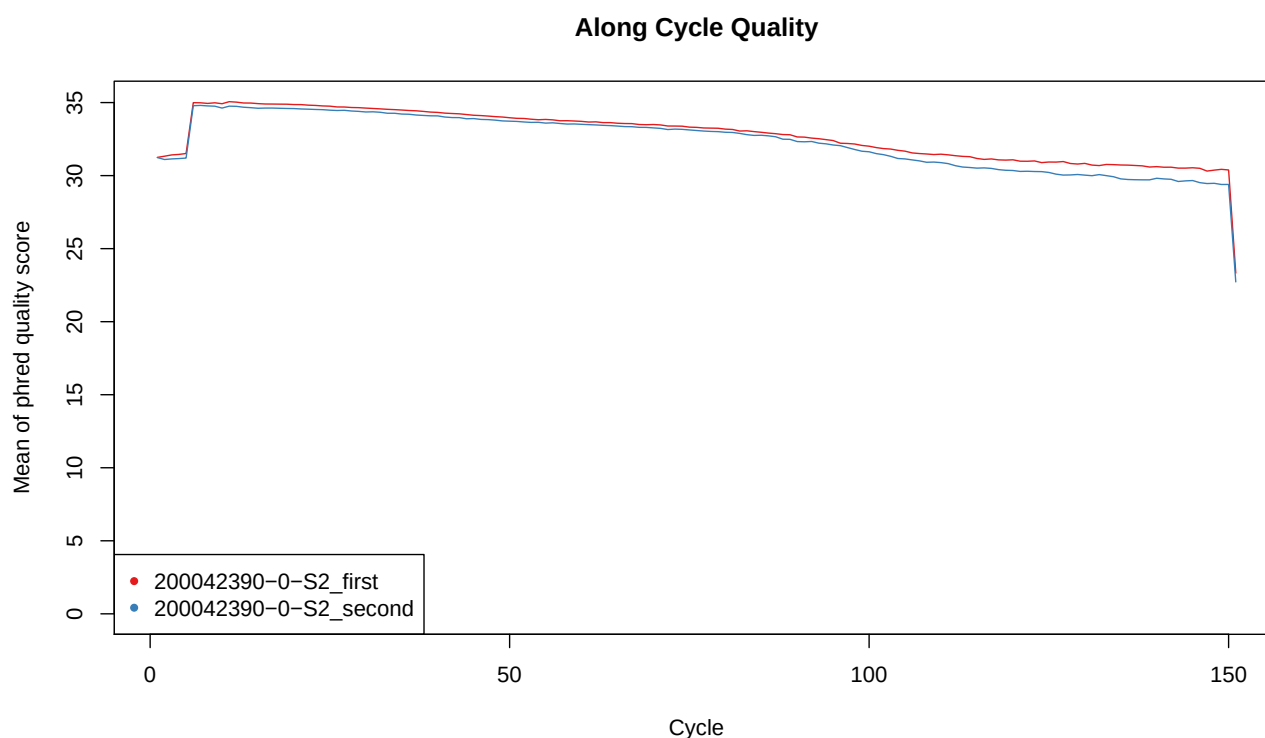


Figure 8.1: Mean base call quality by cycle.

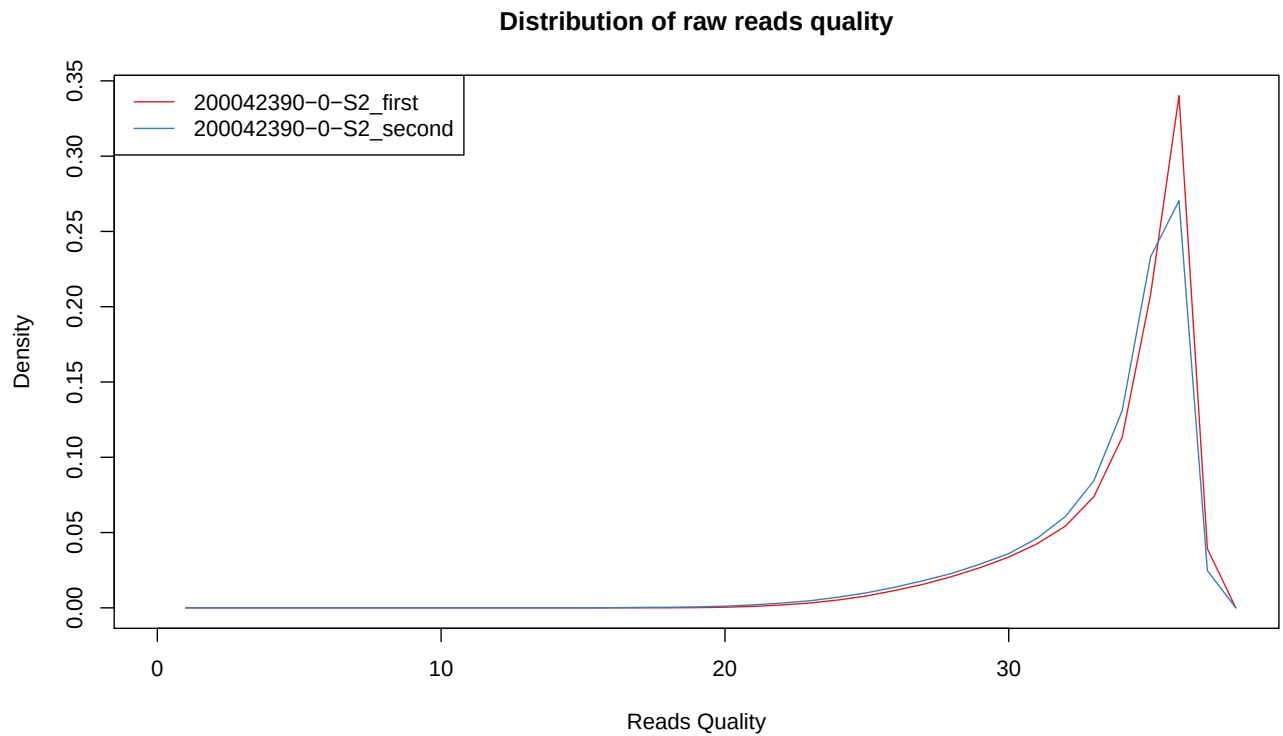


Figure 8.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

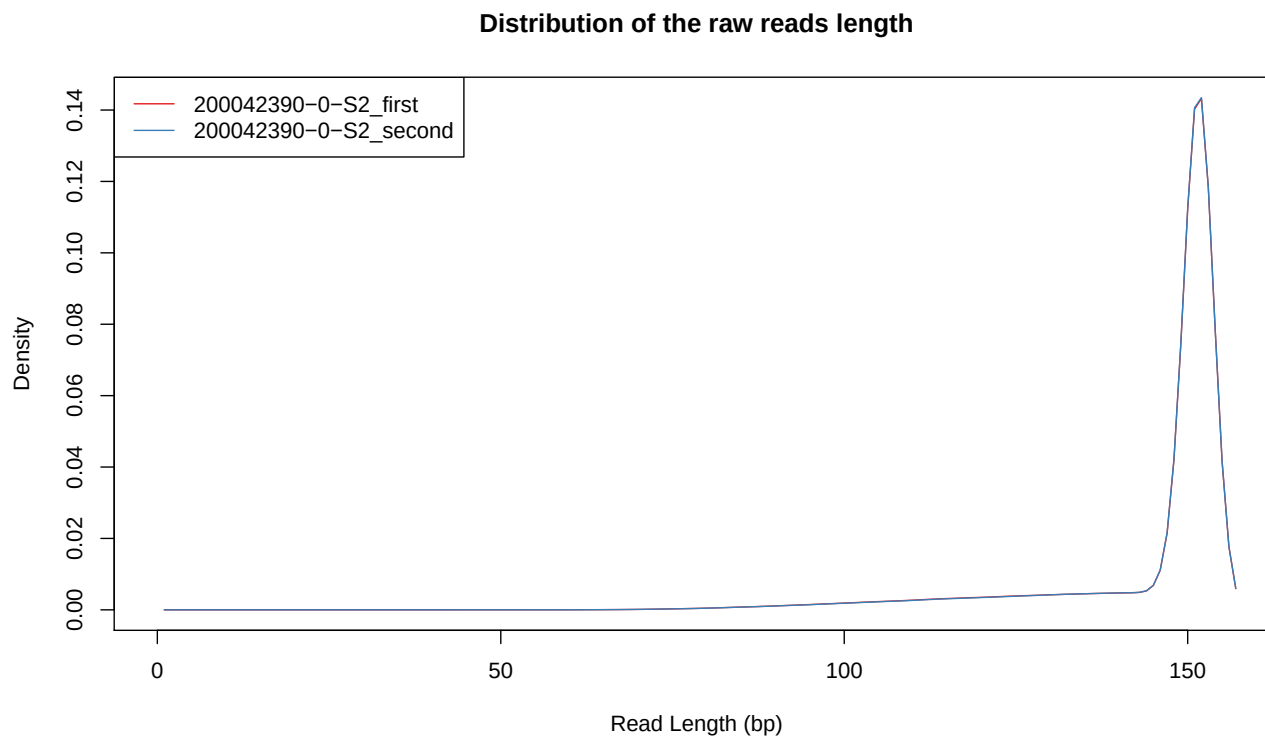


Figure 8.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

8.3 Alignment statistics

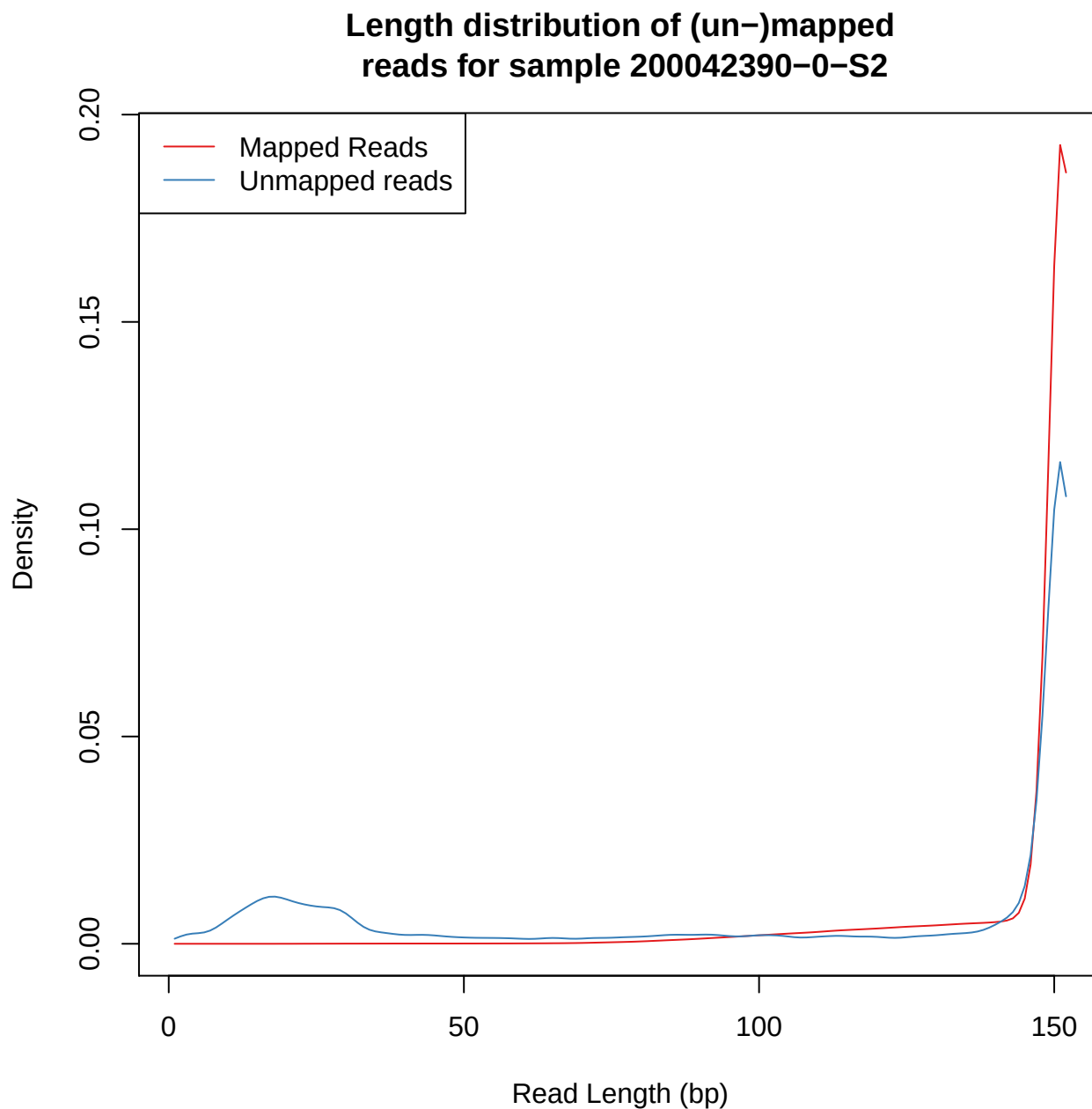


Figure 8.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

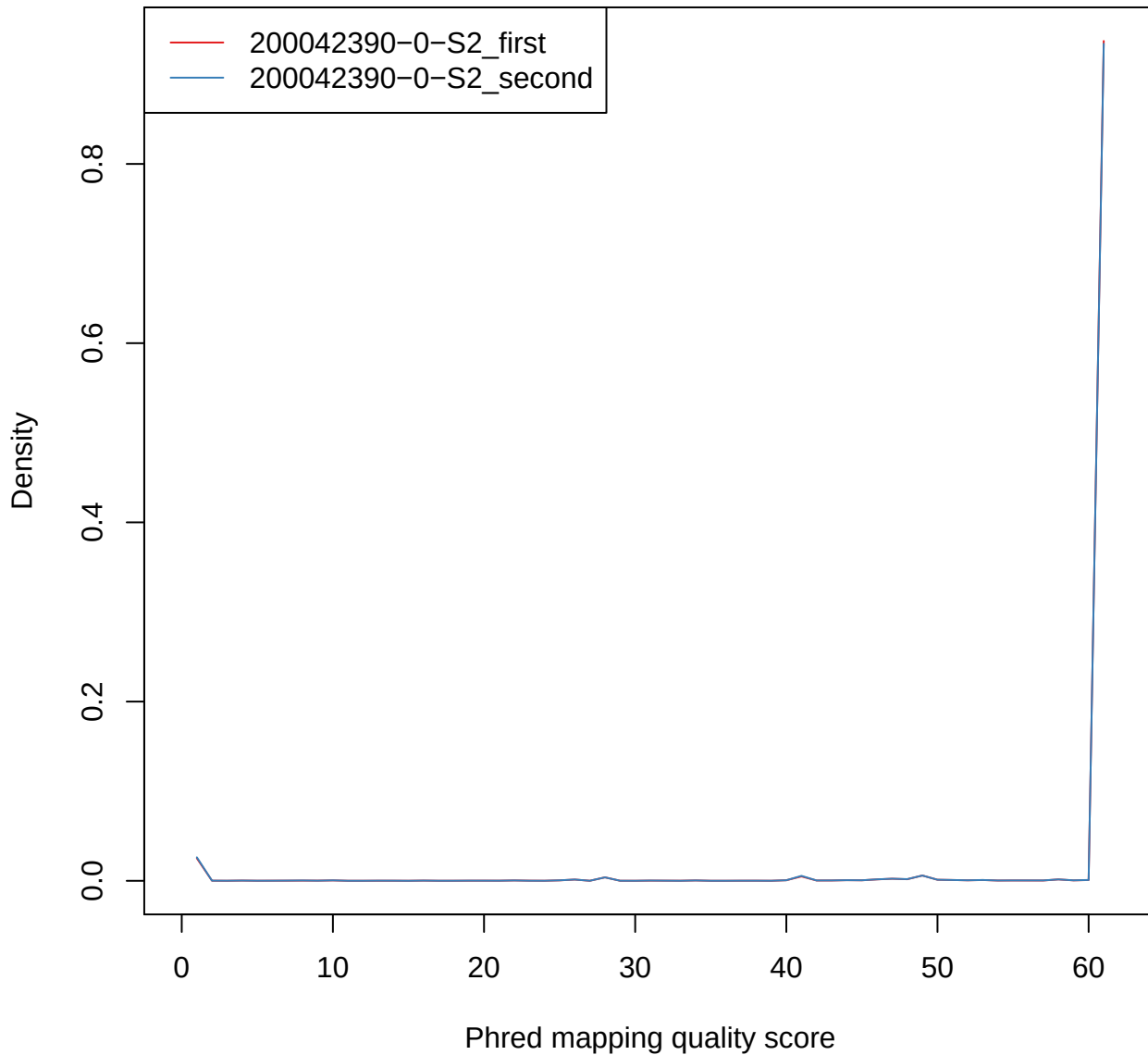


Figure 8.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

8.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042390-0-S2 (2,509,406 aligned reads)**

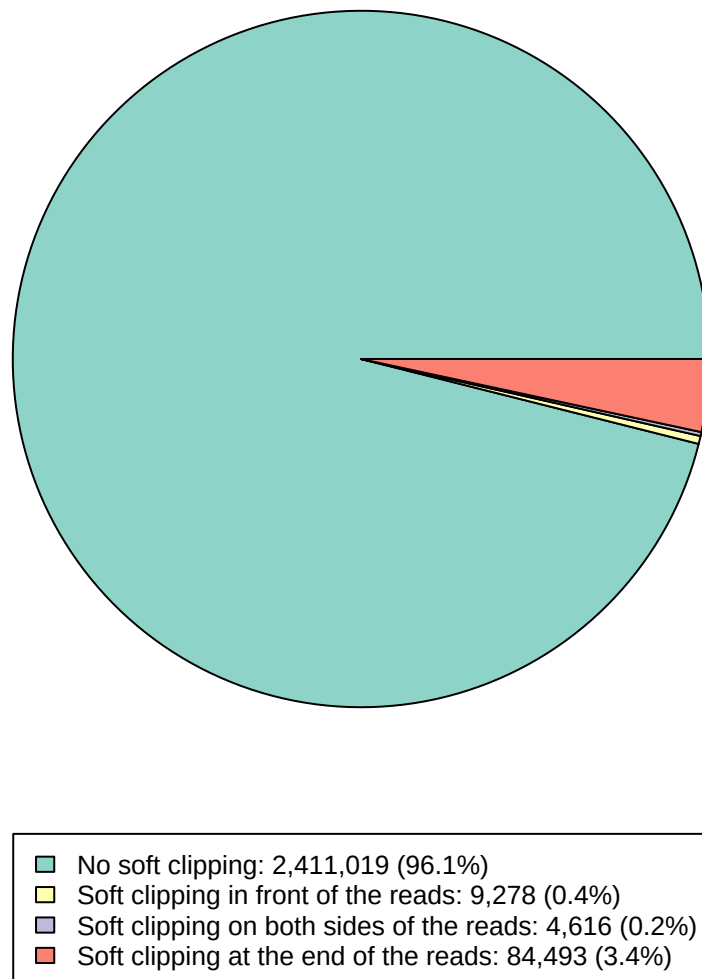


Figure 8.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042390-0-S2

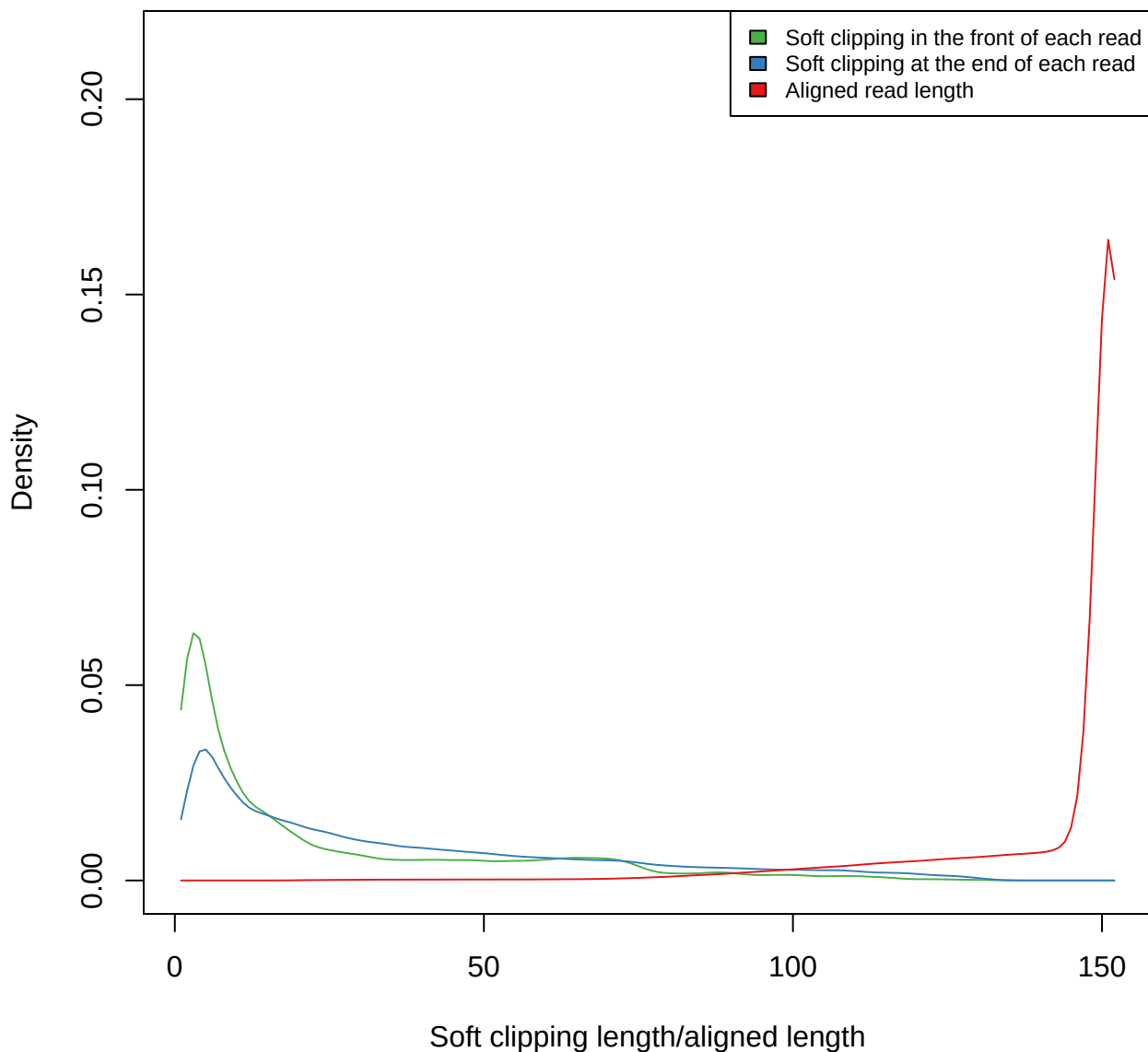


Figure 8.7: Density plots of soft clipping and aligned read length.

8.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 8.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042390-0-S2	74.28%	11.47%	7.42%	6.83%

Table 8.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042390-0-S2	5.28%	58456	3204	178	11	2	0	0

8.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 8.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042390-0-S2	1

Table 8.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	14	MSH2	NM_000251.3	8	c.1277-3297_1277-3284	44

Target region coverage distribution for sample 200042390-0-S2

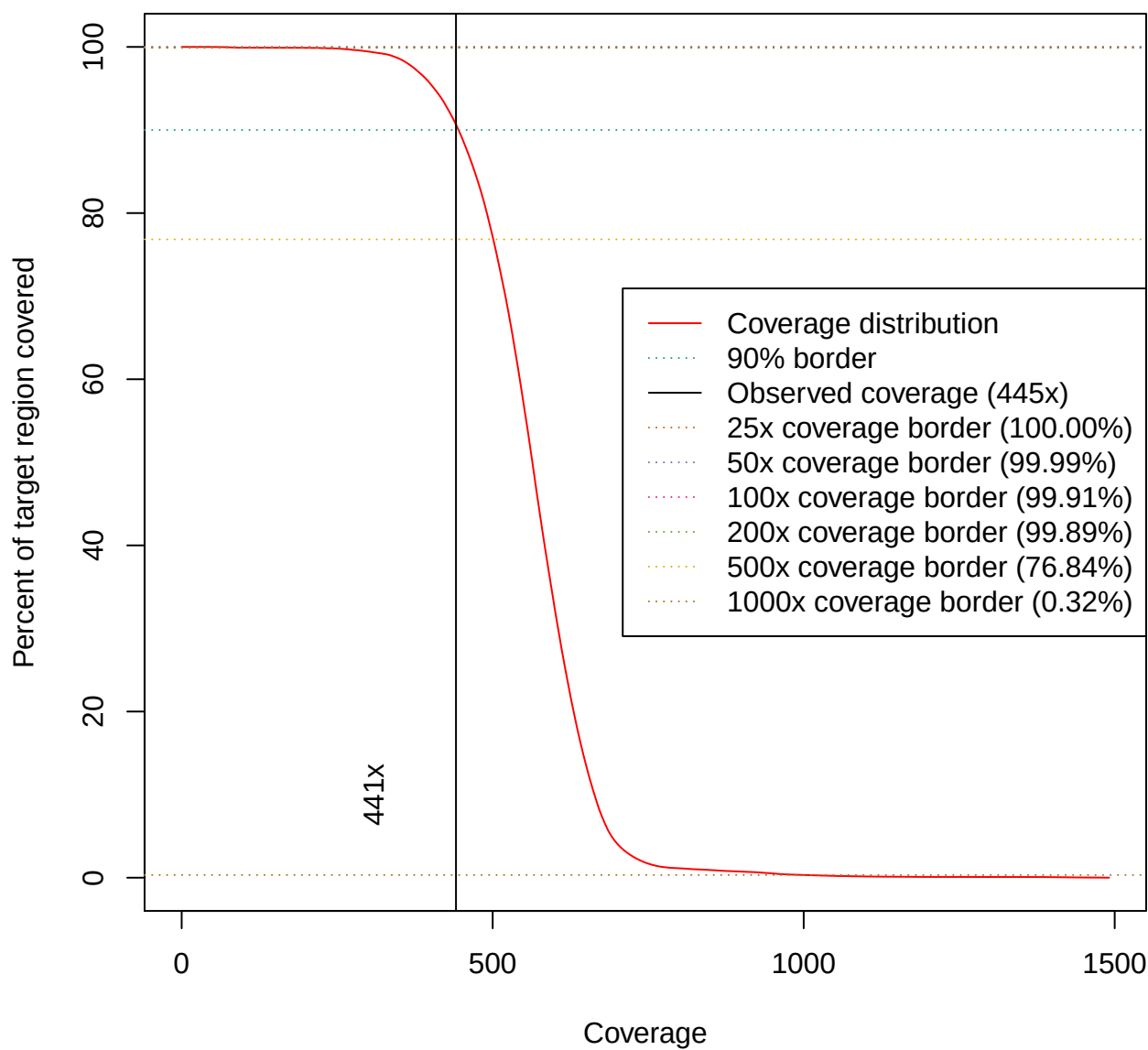


Figure 8.8: Target region coverage.

Table 8.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25x	50x	100x	200x	500x	1000x	Coverage 10% quantile	Coverage Heterogeneity
200042390-0-S2	100.00%	99.99%	99.91%	99.89%	76.84%	0.32%	445x	0.09%

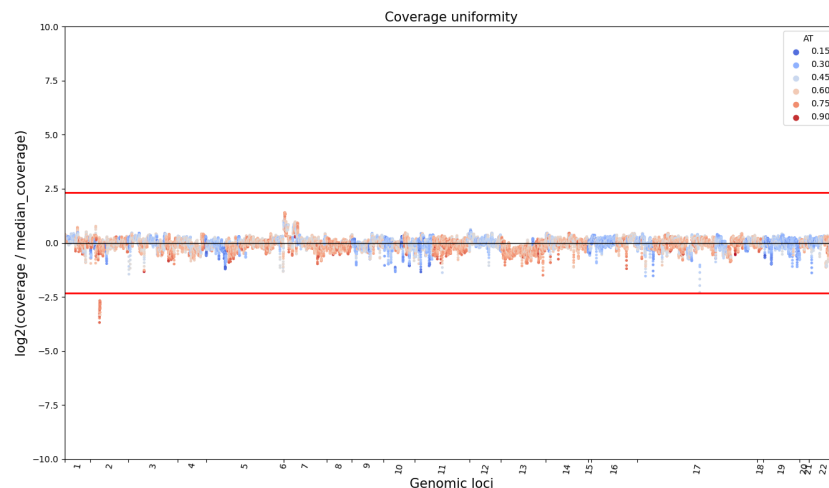


Figure 8.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.

Sample 200042391-0-S3

9.1 Dataset

For sample 200042391-0-S3 2,530,000 reads were sequenced, of these 2,511,050 were mapped corresponding to 99.25% of all reads.

9.2 Raw reads quality

This section focuses on the quality of raw reads. In Figure 9.1 the average base call quality at each sequencing cycle is plotted. Typically base call quality tends to decline slightly towards the end of the read. Figure 9.2 shows the distribution of average base call quality, while Figure 9.3 shows the distribution of read lengths.

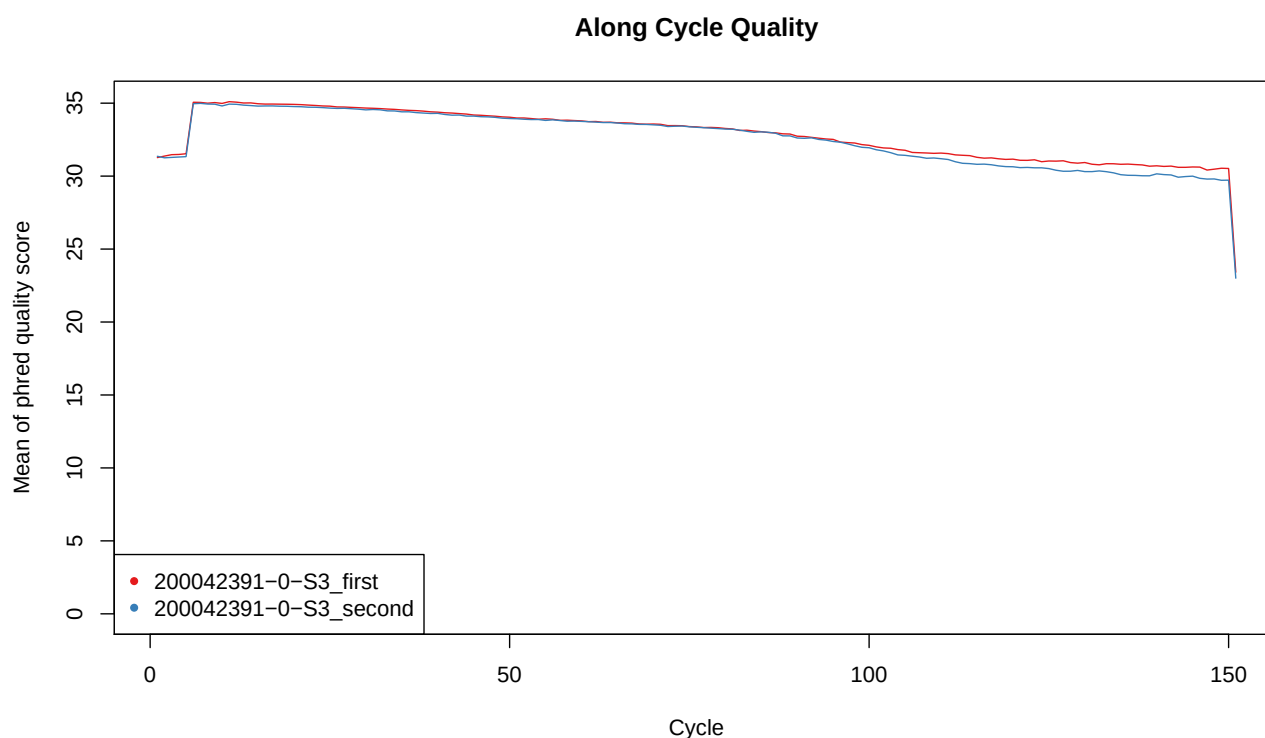


Figure 9.1: Mean base call quality by cycle.

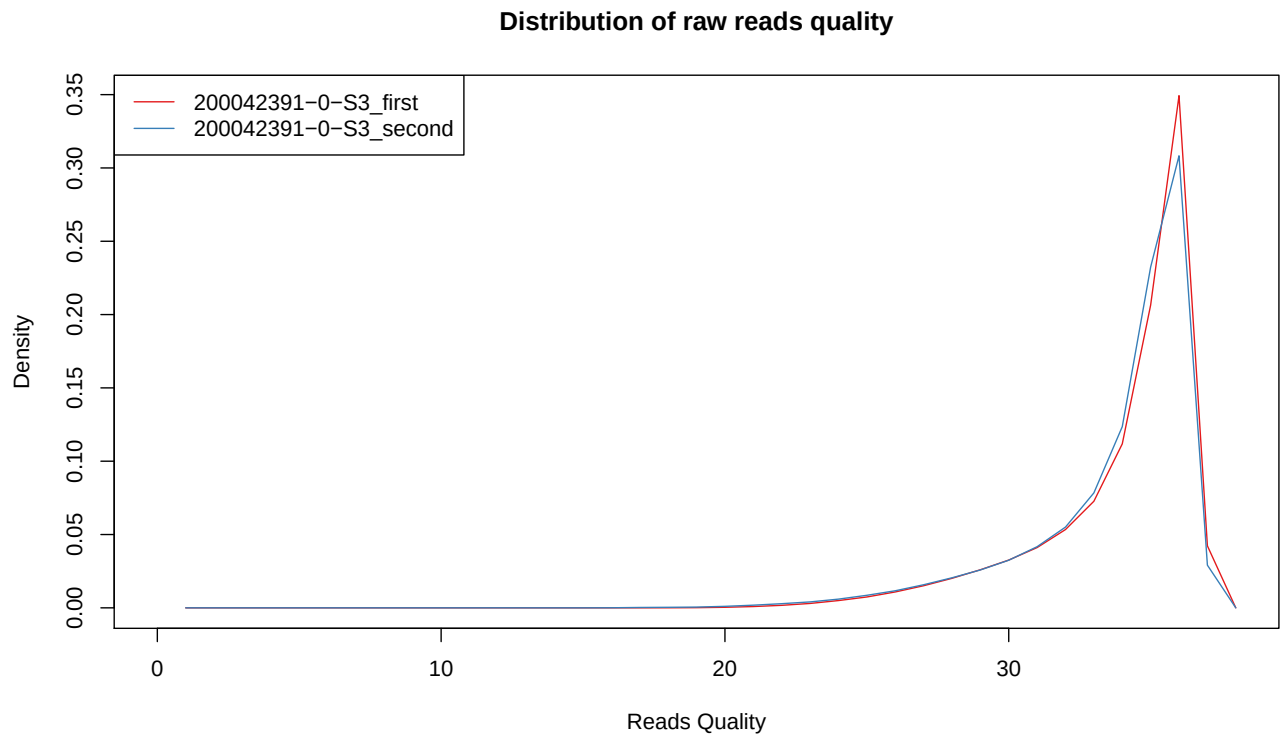


Figure 9.2: Distribution of average read qualities for each dataset. Ideally, most reads should be of high quality resulting in a peak at 35–40 (Illumina platforms) or 20–30 (Ion Torrent platforms).

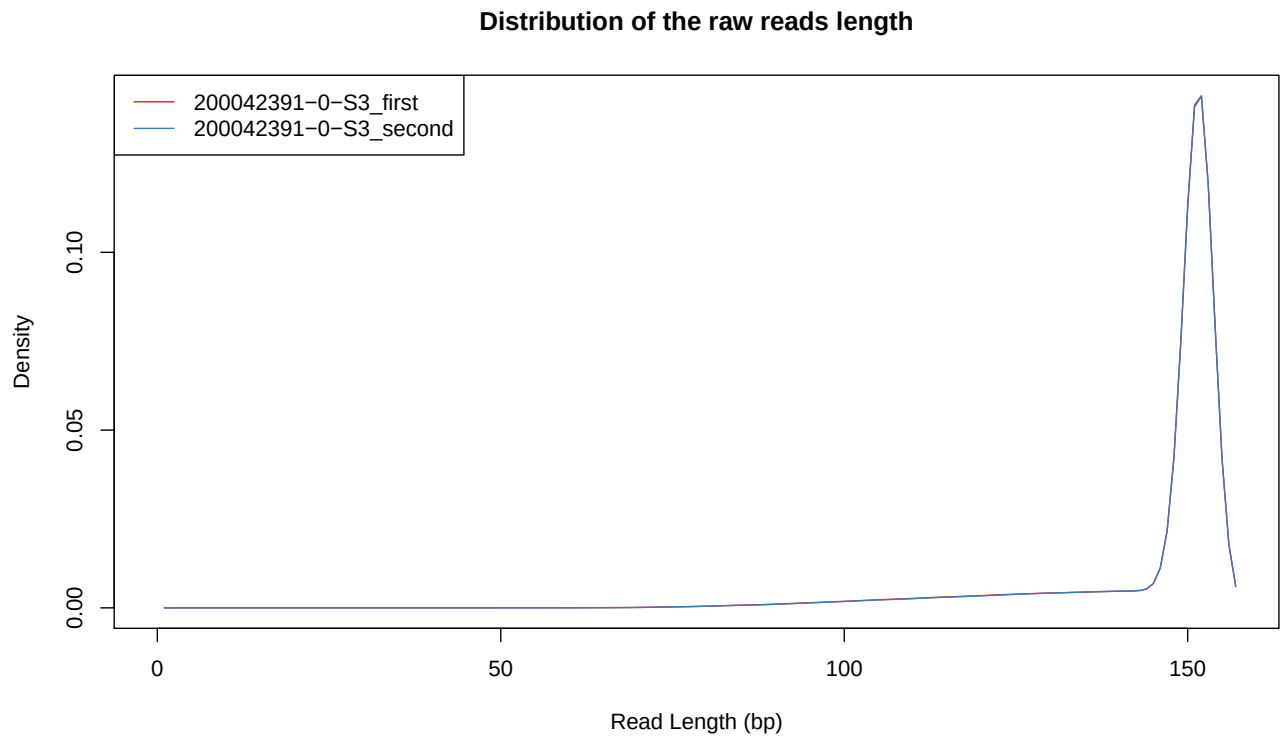


Figure 9.3: Distribution of raw read lengths. Optimally most reads should have a high read length resulting in one or more peaks at the end of the x-scale.

9.3 Alignment statistics

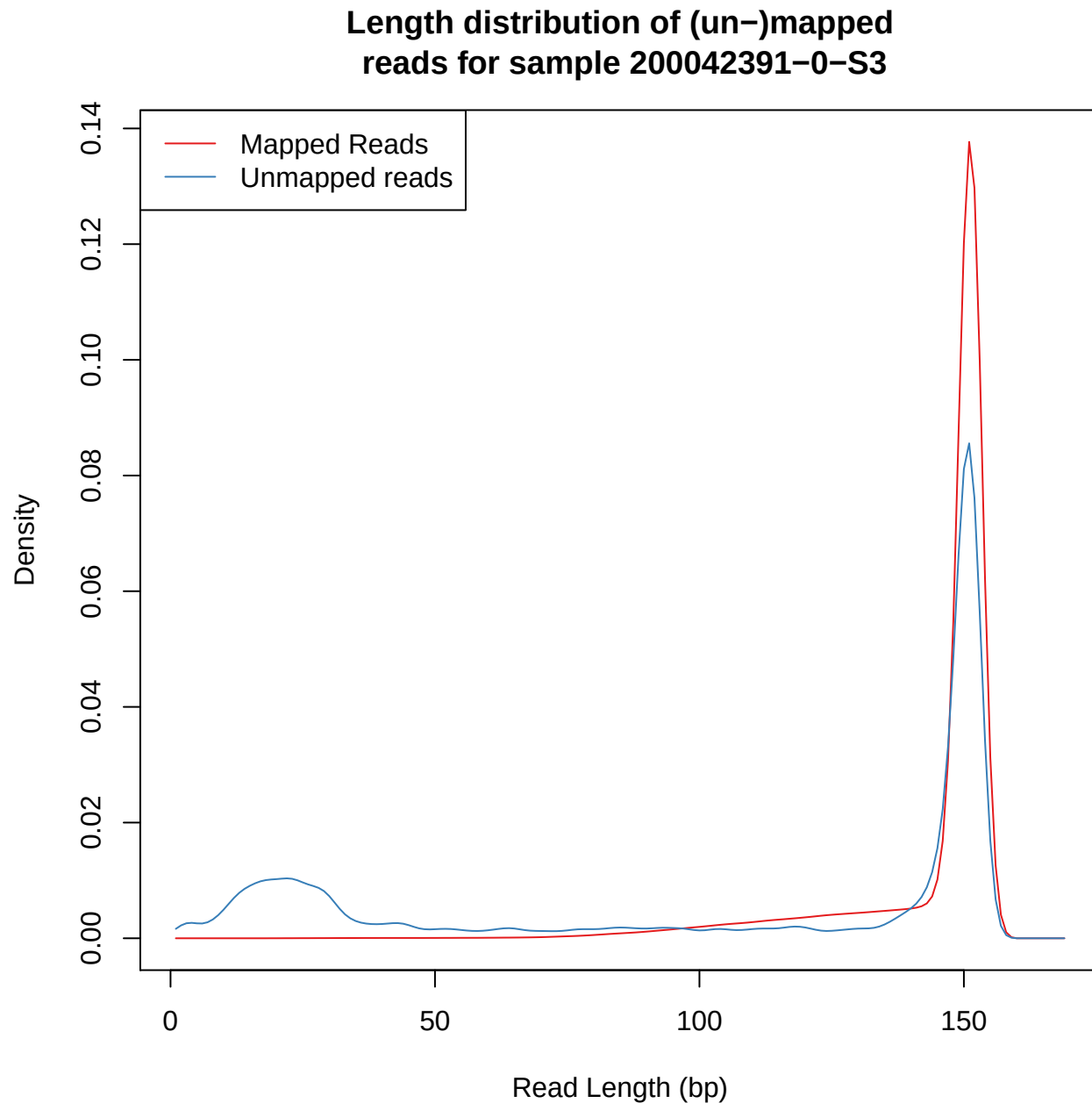


Figure 9.4: Read length distribution for mapped and unmapped reads. This plot can be used to explore why reads did not map. If the dataset contains shorter reads, these often tend to be unmapped.

Mapping quality distribution

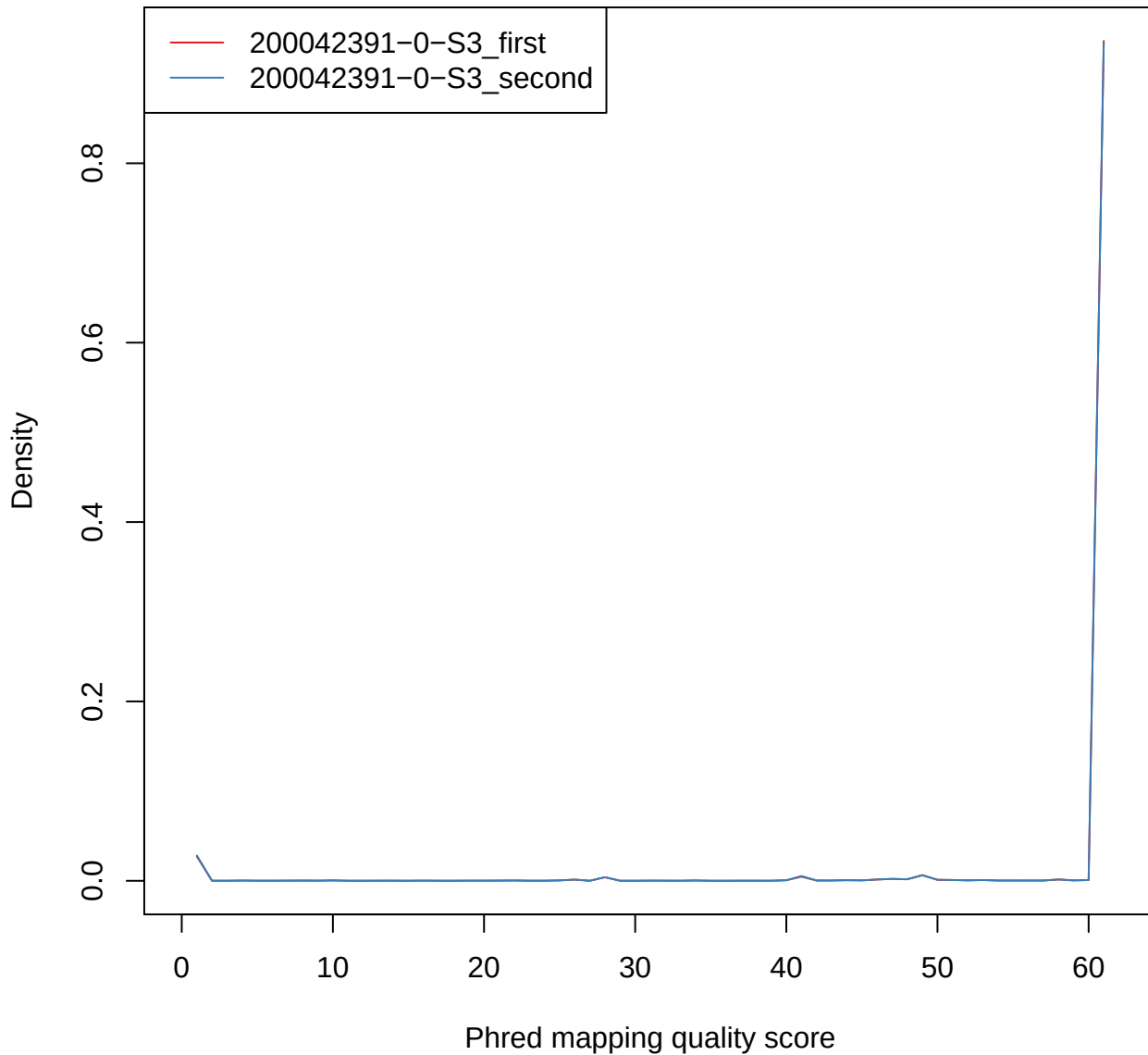


Figure 9.5: Histogram of mapping quality. The higher the better. Optimally most of the mappings should be of high quality resulting in one peak around 40.

9.4 Soft clipping

If a read cannot be aligned over its full length, some bases will be 'soft-clipped' at the borders of the read. Depending on the data there might be a lot of read alignments with soft-clipping, however, the number of bases that are clipped should be small. If for a lot of reads the number of soft-clipped bases is high, this indicates that there was a problem and that a lot of the data is discarded.

**Soft clip statistics for sample
200042391-0-S3 (2,511,050 aligned reads)**

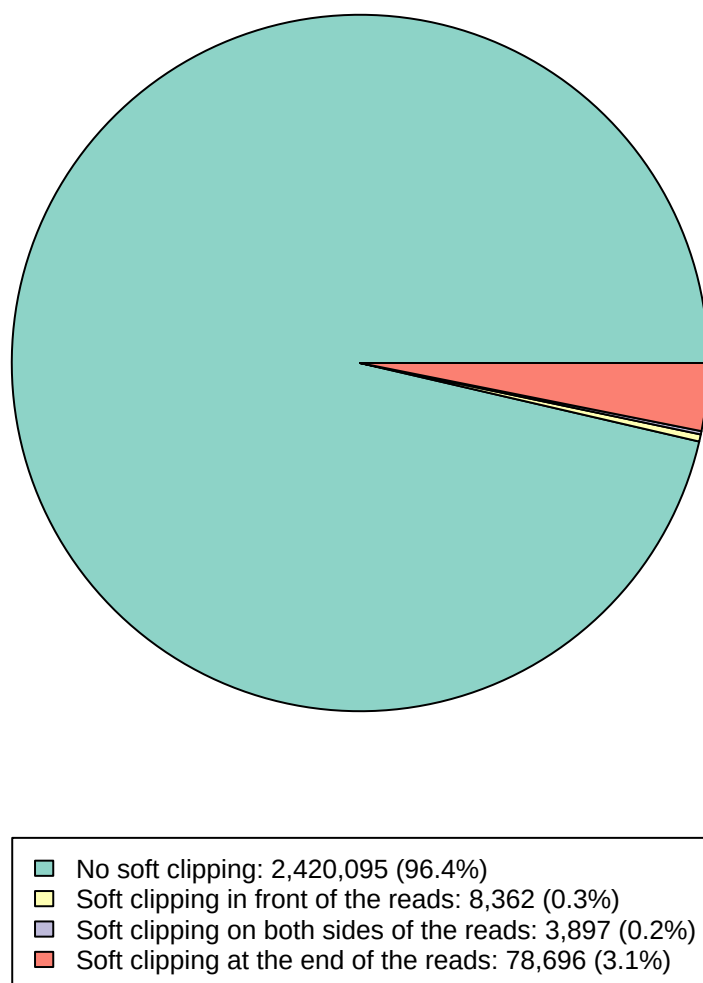


Figure 9.6: Pie charts illustrating the proportion of reads with soft clipping.

Soft clipping length distribution for sample 200042391-0-S3

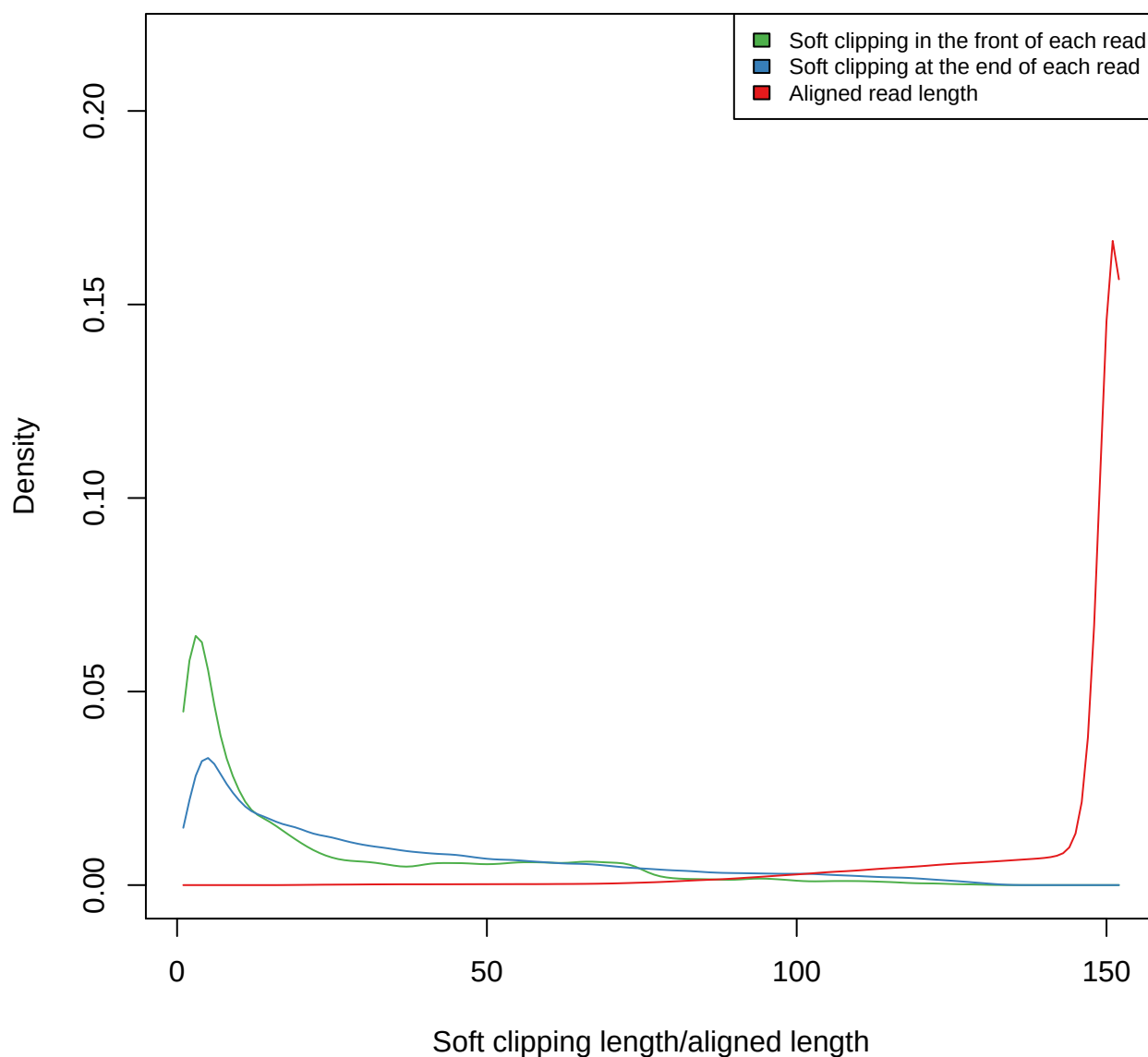


Figure 9.7: Density plots of soft clipping and aligned read length.

9.5 Mapping statistics for HCS_v2_hg38 gene panel

Table 9.1: Mapping statistics: Percentage of mapped bp's on-target (onTarget), within a fragment length of target regions (flankTarget) or completely off target. The off-target bp's are divided into reads that scatter across the genome with low coverage (offTarget_LowCov) and reads that accumulate in specific regions (off-Target_HighCov)

	onTarget	flankTarget	offTarget_HighCov	offTarget_LowCov
200042391-0-S3	74.19%	11.62%	7.48%	6.70%

Table 9.2: PCR duplicates: Displayed is the percentage of duplicate fragments per sample and the decomposition of all duplicated reads into the total number of reads that have two identical copies (2), three copies (3), etc.

	duplicate fraction	2	3	4	5	6-10	11-100	>100
200042391-0-S3	4.65%	51,972	2,614	141	4	0	0	0

9.6 Target region coverage for HCS_v2_hg38 gene panel

An important prerequisite for reliable variant detection from next generation sequencing data is the coverage of the targeted regions. The base coverage is defined as the number of reads covering a base. The coverage of a target region is defined as its average base coverage.

Table 9.3: Number of low coverage regions in datasets. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Sample ID	Low coverage regions
200042391-0-S3	1

Table 9.4: Low coverage regions in dataset. Low coverage regions are defined as regions that fall within target regions but show coverage less than 50.

Chr	Start	Width	Gene	Transcript	Exon id	cDNA	Mean coverage
2	47442251	96	MSH2	NM_000251.3	8	c.1277-3297_1277-3202	14

Target region coverage distribution for sample 200042391-0-S3

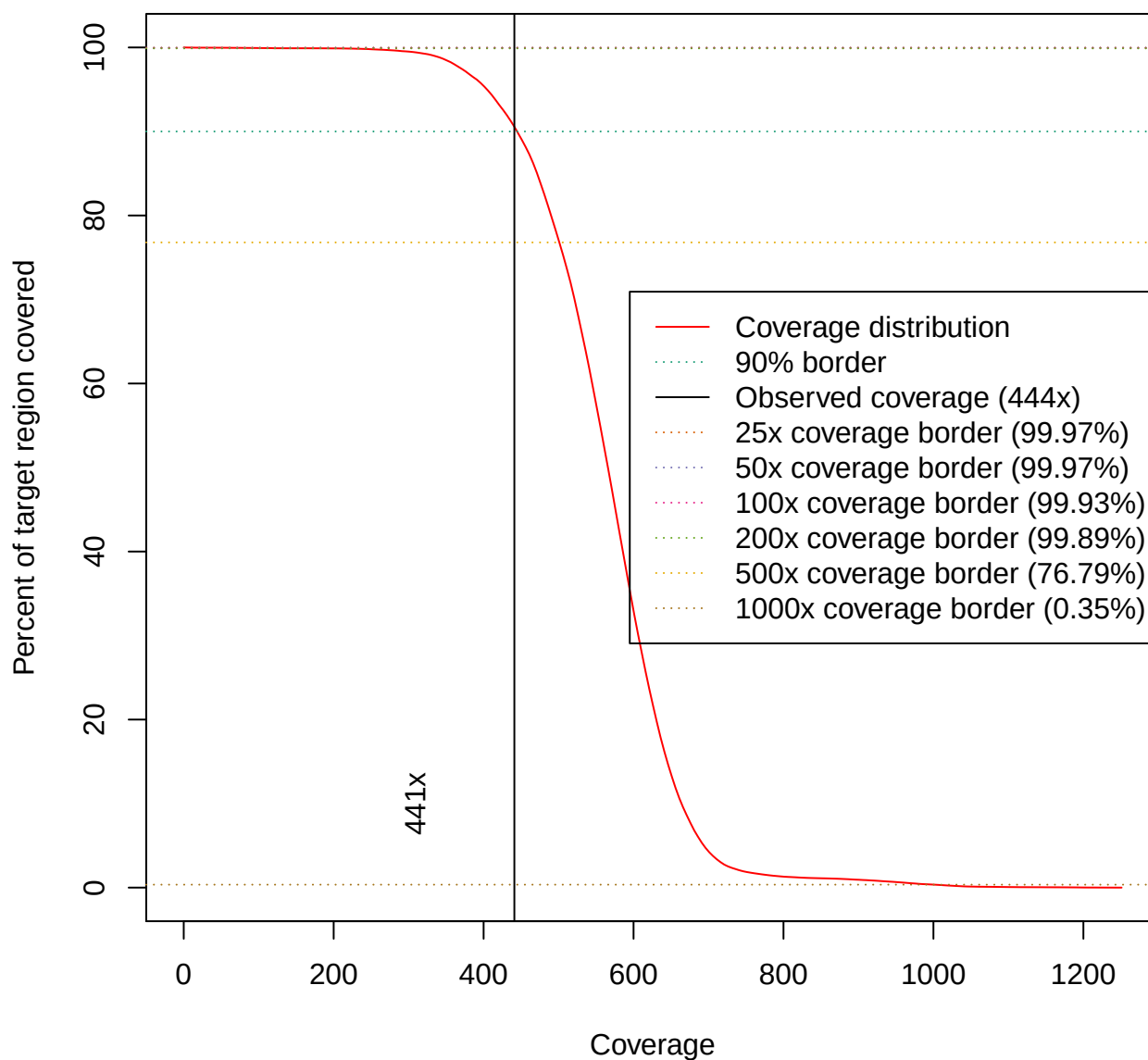


Figure 9.8: Target region coverage.

Table 9.5: Percentage of target regions with certain coverage and coverage heterogeneity across target region.

Sample ID	25×	50×	100×	200×	500×	1000×	Coverage 10% quantile	Coverage Heterogeneity
200042391-0-S3	99.97%	99.97%	99.93%	99.89%	76.79%	0.35%	444×	0.08%

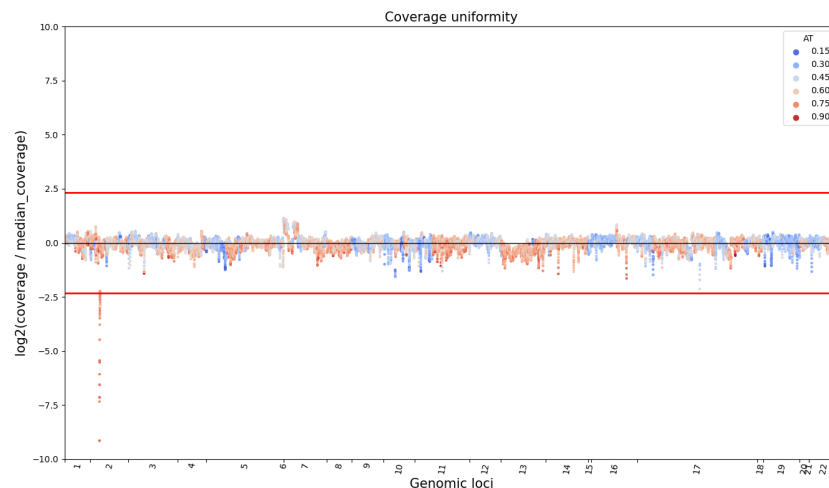


Figure 9.9: Coverage uniformity plot: Visualizing read coverage variability across the target region to support the identification of potential coverage biases. The x-axis corresponds to the target region and depending on the number of genes (with a 40 genes cutoff) it is organized by chromosomes or genes. Overlapping gene names are not displayed for readability. The y-axis indicates the coverage within a given target region normalized by the median and log2 transformed, the exact formula is $\log_2((\text{coverage} + 1) / (\text{median} + 1))$. The black horizontal line indicates the median coverage while the two red horizontal lines correspond to 0.2 and 5 times the median coverage. Everything between the two red lines is considered as homogeneously covered. Each dot represents a small window of the target region with the color indicating the AT-ratio of that window. Dark blue is used to represent regions with high GC content while dark red symbolizes high AT content.